

Canonical conformal limits of subcritical FK planar maps

OpenAI

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Abstract

For every fixed $0 < q < 4$, we prove joint convergence of spherical Fortuin–Kasteleyn planar maps in their flag-triangle uniformization to the corresponding unit-area Liouville quantum gravity sphere decorated by an independent conformal loop ensemble. The convergence includes the area measure, deterministically rescaled graph distances between all vertex pairs, and the full nested interface collection, with interfaces converging uniformly up to reparameterization.

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1 Introduction

Random planar maps give discrete models of random surfaces. A central problem is to identify their scaling limits in prescribed conformal coordinates, together with area, intrinsic distance and a statistical mechanical decoration. The continuum area measure of Liouville quantum gravity (LQG) is a Gaussian multiplicative chaos measure. Its construction belongs to Kahane’s theory [33]; Duplantier and Sheffield developed the circle-average construction for LQG and proved the probabilistic KPZ relation [15]. For maps decorated by the critical Fortuin–Kasteleyn (FK) random-cluster model [17], the interfaces are predicted to converge to a conformal loop ensemble (CLE), independent of the LQG surface in its conformal coordinate.

The discrete route to this prediction begins with tree-decorated maps. Mullin’s correspondence and Bernardi’s bijective developments encode such maps through pairs of tree contours [5, 38]. Sheffield’s inventory bijection combines this description with Bernardi’s subgraph-to-tree construction [6, 42]. It encodes an FK-decorated map by a word whose two resolved contours have a correlated Brownian scaling limit. Duplantier–Miller–Sheffield identify the continuum mating with an LQG surface carrying an independent space-filling Schramm–Loewner evolution (SLE); Miller–Sheffield give the finite-area sphere version [16, 36]. Gwynne, Mao and Sun, and Gwynne and Sun, strengthen the discrete convergence to retain cone times and encoded FK-loop information, including quantum areas and boundary lengths. Their finite-volume result applies to words conditioned to reduce to the empty word [21–23].

These results identify the limit through its encoding. Convergence in a prescribed discrete conformal embedding requires a further comparison with the embedded geometry. An intrinsic metric-space limit alone likewise does not specify where the discrete vertices lie in that coordinate [27, 36]. Gwynne, Miller and Sheffield proved convergence of the Tutte embedding of mated-CRT maps, jointly with quantum area, the space-filling curve and embedded random walk [26, Theorem 1.1]. Holden and Sun proved joint metric and measure convergence under the Cardy embedding for uniform triangulations of polygons at $\gamma = \sqrt{8/3}$, together with convergence of independent percolation decorations [29, Theorems 1.3 and 1.6]. Holden and Yu proved large-scale comparison results for circle packing and Riemann uniformization of infinite planar maps in ergodic scale-free environments under geometric moment and connectivity hypotheses [31, Theorems 1.6–1.9]. These results concern different map laws or embedding inputs from the finite spherical FK model studied here. A general conformal-embedding universality conjecture is recorded as Conjecture 3.5 of Holden and Sun [30].

Here we prove the joint statement for the flag-triangle uniformization of critical spherical FK maps, for every fixed $0 < q < 4$. This establishes the canonical conformal, area, intrinsic-metric and interface limit for this specified model and embedding within the broader universality prediction. Here “subcritical” refers to the LQG range $\gamma < 2$, equivalently $q < 4$; the FK model itself is at its self-dual critical point. The intrinsic step uses the distinct metric companion [39, Theorem 1.1]; we restate precisely its required unmarked Gromov–Hausdorff consequence in Theorem 1.2. The new comparison identifies all these objects in the same canonical coordinate, including atypical

vertices and the order along every macroscopic loop.

1.1 The result

Fix $q \in (0, 4)$. Let $\gamma \in (\sqrt{2}, 2)$ and $\kappa \in (4, 8)$ be determined by

$$q = 2 + 2 \cos\left(\frac{\pi\gamma^2}{2}\right), \quad \kappa = \frac{16}{\gamma^2}. \quad (1.1)$$

The finite law and all topologies are specified in [Section 2](#). In brief, a rooted n -edge decorated map (M_n, A_n) has weight $q^{\ell(M_n, A_n)/2}$, where ℓ counts FK interfaces. Its $4n$ equilateral flag triangles form a conformal sphere X_n with normalized face area m_n . Three independent m_n points determine the orientation-preserving uniformization $\phi_n : X_n \rightarrow \widehat{\mathbb{C}}$ by their images $0, 1, \infty$. Distances d_n use every original primal edge. Let Γ_n be the images of the prescribed polygonal interfaces, with all nested loops retained, and let

$$K_n(a) = \{(\phi_n(u), \phi_n(v), ad_n(u, v)) : u, v \in V(M_n)\}. \quad (1.2)$$

On the continuum side, take the ordinary unit-area γ -LQG sphere, forget its original marks, and normalize its conformal embedding using three independent quantum-area points at $0, 1, \infty$. Denote the resulting area measure and intrinsic metric by μ_h and D_h . Fix the deterministic normalization of D_h throughout. Conditional on this embedded surface, let Γ have the Möbius-invariant whole-plane nested CLE_κ law on $\widehat{\mathbb{C}}$.

Theorem 1.1 (Joint canonical limit). *There is a deterministic sequence $a_n = a_n(q) > 0$ with $a_n \rightarrow 0$ such that, through all positive integers,*

$$((\phi_n)_* m_n, K_n(a_n), \Gamma_n) \implies (\mu_h, \{(z, w, D_h(z, w)) : z, w \in \widehat{\mathbb{C}}\}, \Gamma). \quad (1.3)$$

The three topologies are weak convergence of probability measures, Hausdorff convergence of the full distance graphs, and uniform reparameterization matching of loop collections as defined in [Theorem 2.3](#). Moreover,

$$\max_{T \text{ a flag triangle of } X_n} \text{diam}_\rho \phi_n(T) \longrightarrow 0 \quad \text{in probability,} \quad (1.4)$$

where ρ is spherical distance. The conditional law of Γ is independent of the embedded marked quantum surface.

On a coupling with the asserted convergence, the distance-graph conclusion controls every pair of vertices whose embedded positions converge, including vertices with atypical degrees or near the exploration root. The loop topology retains the order along each curve and its multiplicity, as well as its trace. These conclusions are simultaneous in the prescribed coordinate. All assertions are for a fixed q ; the proof gives no endpoint assertion, convergence rate or explicit formula for a_n .

1.2 The intrinsic input

Theorem 1.2 (Intrinsic spherical scaling; companion consequence). *For the finite law of [Theorem 2.1](#) and γ in (1.1), there is a deterministic sequence $a_n > 0$ with $a_n \rightarrow 0$ such that*

$$(V(M_n), a_n d_n) \implies (S, D_h)$$

in the Gromov–Hausdorff topology on compact metric spaces up to isometry, through all positive integers. The limit is the ordinary unit-area γ -LQG sphere of Duplantier et al. [16, Definition 4.21(ii)], disintegrated at total quantum area one, with its marked points forgotten and with the fixed intrinsic metric normalization. The root and FK decoration are forgotten in this convergence.

Proof. Apply the fixed- q FK clause of OpenAI [39, Theorem 1.1] and forget the degree measure from its Gromov–Hausdorff–Prokhorov conclusion. The finite law, all-edge distance, ordinary unit-area sphere law and fixed metric normalization are exactly those above. $\square$

The companion *Metric-measure limits of subcritical FK and spanning-tree planar maps*, by OpenAI, September 24, 2026, proves the intrinsic result. We use only the displayed GH consequence; its stronger measured conclusion and its separate spanning-tree clause are not needed here. We take the same deterministic $a_n(q)$ throughout. In the companion the vertex measure is $\deg(v)/(2n)$, with loops counted twice. Each vertex belongs to $2\deg(v)$ of the $4n$ flag occurrences, so projection of uniform flag area to its vertex has exactly that degree distribution. The Brownian encoding uses $2n$ letters and an area clock of duration one; neither this clock normalization nor the flag geometry changes the all-edge metric. The role of Theorem 1.2 is to complete locally identified passage metrics at every vertex and to fix the deterministic scale.

1.3 Proof strategy

We first place the discrete and continuum objects on one topological sphere. The contour limit and its continuum sewing are recalled in Section 3. In Section 4 they give homeomorphisms H_n from the complete flag surface to the reference LQG sphere, with every flag image small. These homeomorphisms preserve separation and intersections but do not yet compare conformal moduli or lengths.

Here is the local information needed for those comparisons. Take a compact disk inside a larger open region of the reference sphere. Retain every exploration-time visit to the disk, together with a buffer around those visits and every incidence at their primal and dual vertices. There are finitely many retained time blocks. Their raw words, relative contour heights and exact minimum tests determine the corresponding part of the discrete flag surface. One can then read, for example, the minimum graph cost between two small endpoint sets, or the extremal length of curves joining two sides of a quadrilateral. Section 5 constructs these finite records; retaining all visits is what makes their conclusions apply to every local path and vertex.

The continuum contours need not determine a subsequential discrete readout. Section 6 instead identifies its conditional law: given the full continuum surface and exploration, the readout depends only on the retained local input, and disjoint protected regions have product conditional laws. An exact word-resampling identity shows that changes from the bilateral word to the finite conditioned word alter only the density of the continuum input. The same kernels then transfer to ordinary Gaussian fields and to the sphere at its actual area-normalized height. Section 7 turns local events with high probability into successful annuli at many radii around every point. This supplies the common probabilistic input for the conformal and metric comparisons.

Conformal coordinate and area. Section 8 rules out degeneration of flag-surface extremal length. A hypothetically small crossing modulus gives a probability flow of paths with low traffic energy. Annular lower bounds force its projected limit to consist of nonconstant rectifiable curves whose average arclength is absolutely continuous with respect to planar area. Locality and rotation covariance then give cheap crossings in two transverse directions at the same point, contradicting

quadrilateral reciprocity. The resulting annular bounds give a quasiconformal limit of $\phi_n \circ H_n^{-1}$. Its infinitesimal conformal structure is read from local moduli; triviality of the field germ and rotation covariance make it round. Three area marks fix the resulting conformal map, and the area and maximum-mesh conclusions follow in that coordinate. The analytic framework comes from Ahlfors–Beurling extremal length [1] and the ring estimates of Ivrii–Marković [32]. The elimination of deterministic anisotropy by rotation covariance has a related use in Holden and Yu [31, Section 4.4], where determinism comes from averaged Dirichlet energies.

Ordered nested interfaces. Section 9 identifies the loops independently of the metric comparison. Flexible-order matches form a noncrossing chord system on the exploration-time circle. Its faces encode the finite FK loops, including their cyclic traversal order. Chord convergence first gives ordered limits of these face contours. The alternating CLE exploration identifies their initial portions; winding tests on both sides exclude an omitted moving portion. This matches every macroscopic loop, with multiplicity, and retains the full nested collection. The exploration convention follows Miller and Sheffield [35], Sheffield [41] and the nested construction described in Aru et al. [3, Section 2.1.6].

Local passages and all-vertex distances. In Section 10, graph passages between small endpoint sets first acquire upper and lower comparisons with the reference LQG metric. Repeated annular opportunities and local finite-energy field changes bring a reference geodesic near a shortcut. A weighted change-of-measure argument forces the optimal comparison constants to agree. This adapts the optimal-constant method of Gwynne and Miller [19, Section 1.5 and Sections 3–6] to the local passage kernels. The changed laws are compared before a shortcut or a geodesic hit is selected, so restriction to a rare selected event preserves the required domination.

These passages have freely chosen endpoints in small sets; reaching a prescribed lattice vertex is a separate problem. Section 11 uses Theorem 1.2 at this point. Small surrounding circuits and the sphere topology of the GH limit exclude nontrivial spatial fibers. A possible collapse of all local passage costs would concentrate the remaining metric at one spatial point; the joint local laws exclude it by placing two separated positive escape-cost tests in the same sphere. The resulting all-vertex metric is a deterministic multiple of D_h , and the intrinsic diameter law fixes that multiple. All field-law arguments use the original oriented reference coordinate; only then are their geometric conclusions transported to the canonical coordinate. Section 12 assembles the three branches, including their common marks and possible common reflection. Figure 1 records these dependencies.

2 The model, normalization and topologies

2.1 The finite FK law

Definition 2.1 (Finite ensemble). Let $\mathcal{F}_n$ consist of pairs (M, A) , where M is a connected planar multigraph with n edges embedded in the oriented sphere and rooted at a distinguished oriented edge, and $A \subseteq E(M)$. Pairs are identified under orientation-preserving homeomorphisms preserving the root and A . Loops and multiple edges are allowed. Let M^* be the dual map, and let A^* consist of the dual edges crossing $E(M) \setminus A$. Write $k(A)$ and $k(A^*)$ for the numbers of connected components of the respective spanning subgraphs, including isolated vertices. Set

$$\ell(M, A) = k(A) + k(A^*) - 1 = 2k(A) + |A| - |V(M)|, \quad Z_n(q) = \sum_{(M, A) \in \mathcal{F}_n} q^{\ell(M, A)/2}.$$

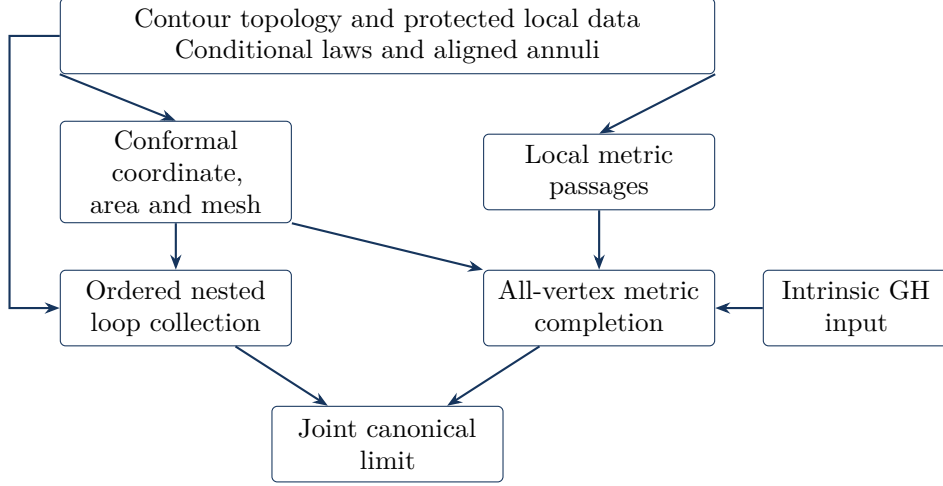


Figure 1: The conformal, loop and metric arguments share the contour correspondence. The intrinsic GH input enters only the all-vertex completion of the metric.

We sample (M_n, A_n) with probability $q^{\ell(M,A)/2}/Z_n(q)$. Its graph distance d_n is shortest-path distance on $V(M_n)$ using all edges of M_n , each of length one.

For completeness, thicken the vertices and occupied edges to a regular neighborhood $N(A)$. It has Euler characteristic $|V(M)| - |A|$. Every component has genus zero. If its total number of boundary components is b_A , then $|V(M)| - |A| = 2k(A) - b_A$, so $b_A = 2k(A) + |A| - |V(M)|$. The complementary components correspond to the components of A^* ; Euler's formula on the sphere gives $b_A = k(A) + k(A^*) - 1$. This proves the displayed identities and identifies ℓ with the number of FK interfaces, including configurations containing bridges or loops.

2.2 Flag triangles and the fixed conformal coordinate

For each occurrence of an endpoint and a side of an original edge, take an equilateral triangle with labels (v, e, f) : original vertex, edge midpoint and face center. Glue sides by their incidence identifications. There are $4n$ triangles. Repeated incidences are different occurrences, even when two labels refer to the same vertex, edge or face. In particular this construction is the incidence subdivision of the embedded sphere and produces a topological sphere X_n , also for loops and bridges.

The Euclidean metrics on the triangles define a conformal structure away from their conical vertices. At a cone of total angle $\theta > 0$, the coordinate obtained by raising the local cone coordinate to the power $2\pi/\theta$ extends this structure across the vertex. Give X_n normalized Euclidean face area m_n , so each triangle has mass $1/(4n)$. Conditional on X_n , take three independent samples x_n^1, x_n^2, x_n^3 from m_n . Since m_n has no atoms they are almost surely distinct. Genus-zero uniformization [28, Theorem II.2.15] and three-point normalization give a unique orientation-preserving conformal map

$$\phi_n : X_n \longrightarrow \widehat{\mathbb{C}}, \quad \phi_n(x_n^1) = 0, \quad \phi_n(x_n^2) = 1, \quad \phi_n(x_n^3) = \infty.$$

This is a function of the discrete surface and the three samples.

In a triangle write (t_v, t_e, t_f) for barycentric coordinates. Include $\{t_v \geq 2/3\}$ and, if the original edge belongs to A_n , also include $\{t_f \leq 1/3\}$. Their glued union is a regular neighborhood of the occupied primal subgraph. Its boundary components are the prescribed polygonal interfaces.

Write $\widehat{\Gamma}_n$ for their collection on X_n , taken as unrooted and unoriented loops, and $\Gamma_n = \phi_n(\widehat{\Gamma}_n)$ for the canonical images.

Lemma 2.2 (Flag area and a uniform root). *Let π_n send the interior of each flag triangle to its primal vertex, with an arbitrary convention on triangle boundaries. Then*

$$(\pi_n)_* m_n = \nu_n := \frac{1}{2n} \sum_{v \in V(M_n)} \deg_{M_n}(v) \delta_v. \quad (2.1)$$

Conditional on the unrooted decorated map, the tail of a uniform oriented root edge has law ν_n , and re-rooting uniformly preserves the finite decorated law. If $H_n : X_n \rightarrow S$ are homeomorphisms into a compact metric sphere with $\max_T \text{diam } H_n(T) \rightarrow 0$, then $(H_n)_ m_n$ and $(H_n)_* \nu_n$ have the same weak subsequential limits.*

Proof. A primal vertex v is incident to $2 \deg(v)$ flag occurrences; a loop contributes twice to $\deg(v)$ and all four of its occurrences are counted. Thus its mass is $2 \deg(v)/(4n)$, proving (2.1). There are $2n$ oriented edge occurrences, and exactly $\deg(v)$ have tail v . Uniform re-rooting preserves the weight $q^{\ell/2}$; equivalently, count rooted occurrences before quotienting by the automorphism group. This also proves the assertion in the presence of automorphisms. Couple $x \sim m_n$ to $\pi_n(x)$. Their H_n images lie in the same flag image, so their distance is at most $\max_T \text{diam } H_n(T)$. Testing against uniformly continuous functions on the compact sphere proves the last assertion. $\square$

The area samples used for ϕ_n are fresh independent samples. Theorem 2.2 describes the distribution of the forgotten discrete root; it does not replace those three samples by exploration marks.

2.3 The continuum object

The continuum surface is the ordinary γ -LQG sphere of Duplantier et al. [16, Definition 4.21(ii)], disintegrated at area one. Forget its original marked points. Given the surface, sample three independent points from its quantum-area probability measure and send them to $0, 1, \infty$ by an orientation-preserving conformal map. The area measure is diffuse with full support, so this normalization is almost surely defined. Write h for a representative field in that coordinate, μ_h for its area and D_h for its intrinsic metric. The metric is a continuous length metric inducing the topology of $\widehat{\mathbb{C}}$. Its deterministic multiplicative normalization remains fixed.

Given this embedded marked surface, sample the Möbius-invariant whole-plane nested CLE_κ ensemble on $\widehat{\mathbb{C}}$. This conditional-law specification is part of the joint limit. The original root and the original quantum-sphere marks are absent from the limiting object.

2.4 Distance graphs and loop matching

Use the product distance

$$d_\times((z, w, r), (z', w', r')) = \rho(z, z') + \rho(w, w') + |r - r'|$$

on $\widehat{\mathbb{C}}^2 \times [0, \infty)$. The distance graphs $K_n(a)$ in (1.2) are nonempty compact sets. Since D_h is continuous on the compact sphere, its full graph is also compact. We use the Hausdorff topology for $d_\times$.

Definition 2.3 (Loop and ensemble convergence). Our curve distance and locally finite multiset topology follow Gwynne et al. [25, Section 2.2], with loop orientation additionally forgotten. They retain the order within each loop and its multiplicity. For continuous loops $\alpha, \beta : \mathbb{S}^1 \rightarrow \widehat{\mathbb{C}}$, let

$$d_{\text{loop}}(\alpha, \beta) = \inf_{\sigma} \sup_{t \in \mathbb{S}^1} \rho(\alpha(t), \beta(\sigma(t))),$$

where σ ranges over circle homeomorphisms of either orientation. Identify loops at zero distance. A loop collection is locally finite in diameter if it has finitely many loops of diameter greater than ε for every $\varepsilon > 0$. An ε -matching of two such collections is a partial bijection covering every loop of diameter greater than ε on either side, with matched loops at distance at most ε ; every unmatched loop has diameter at most ε . Convergence means that such matchings exist with $\varepsilon \rightarrow 0$. All collections in this paper retain nested loops.

This definition also allows repetitions to be retained as separate elements when forming a matching. The loop identification proof will establish both local finiteness in diameter and eventual matching of every macroscopic loop.

2.5 Extraction conventions

Let d_γ denote the deterministic Hausdorff dimension of the γ -LQG metric [19]. We use $p = 1/d_\gamma$, $\xi = \gamma p$, and $Q = 2/\gamma + \gamma/2$. The parameter q always denotes the FK weight; the inventory parameter is b . A crossing scale is denoted by $B(n)$, whereas a_n denotes the deterministic normalization in Theorem 1.2. The exploration root is p_* .

All subsequential arguments retain the same contour encoding, continuum surface, local observations and independent area marks. Whenever a new finite family of tests is added, refine this joint extraction. The families used below admit countable determining atlases: rational time cuts, rational spatial boxes, rational height slacks and integer truncations. A diagonal refinement therefore retains all their conclusions on a common event of probability one. Statements involving all spatial points will be proved by explicit uniform estimates before passing from an atlas to arbitrary points.

3 The inventory encoding and its reference surfaces

The inventory word determines the primal map together with an auxiliary spanning tree and its complementary dual tree. Their contours will provide the reference sphere used in the topological comparison. We first fix the time and height conventions. One inventory letter represents one exploration triangle, which consists of two of the flag triangles of Section 2. A word of length $2n$ has duration one after division of time by $2n$. Multiplying both contour coordinates by one fixed constant makes their limiting marginal variances one. This constant is also used when contour heights are interpreted as quantum boundary lengths.

3.1 The exact inventory law

Write $\mathbf{h}, \mathbf{c}$ for the two burger types, $\mathbf{H}, \mathbf{C}$ for their rigid orders, and $\mathbf{F}$ for a flexible order. Reading from left to right, an order removes the most recently produced available burger of an allowed type. Reduction retains the unfulfilled orders, in chronological order, followed by the surviving stack. Set

$$b = \frac{\sqrt{q}}{2 + \sqrt{q}} \in (0, 1/2), \quad \mathbb{P}(\mathbf{h}) = \mathbb{P}(\mathbf{c}) = \frac{1}{4}, \quad \mathbb{P}(\mathbf{H}) = \mathbb{P}(\mathbf{C}) = \frac{1-b}{4}, \quad \mathbb{P}(\mathbf{F}) = \frac{b}{2}. \quad (3.1)$$

In the bilateral iid word every letter has a unique match almost surely [42, Proposition 2.2]. Resolve F as H or C according to its matched burger. The resolved contour $Z = (L, R)$ increases by the appropriate coordinate vector at a burger and decreases by that vector at its order. It is linearly interpolated between integer times and anchored at $Z(0) = 0$. Write N_F for the number of flexible orders in a finite word.

Proposition 3.1 (Inventory realization). *The iid word of length $2n$, conditioned to reduce to the empty word, encodes exactly the rooted FK-decorated map law of the theorem. Its underlying primal map, with every primal edge retained, is the map obtained by the tree/cotree sewing of its resolved contour. The FK interface count equals $1 + N_F$.*

Proof. The inventory bijection is the rooted map bijection of [42, Sections 4.1–4.2]. The conversion from a spanning subgraph to the auxiliary spanning tree is the construction of Bernardi [6, Section 4], as explained in Sheffield [42, Section 4.1]. Its elementary sewing objects are incidence occurrences, so loops, bridges, and repeated incidences are included. A matched burger/order pair forms the two exploration triangles in one quadrangle. The L -pairs give the primal spanning-tree edges; the R -pairs give the dual spanning-tree edges, whose quadrangles carry the remaining primal edges. Resolving a flexible order selects the auxiliary tree/cotree diagonal. Restoring its FK diagonal changes the decoration, not the quadrangle or the underlying all-edge primal map.

A fully matched word has n burgers and n orders. Its unconditioned probability is

$$\left(\frac{1-b}{16}\right)^n \left(\frac{2b}{1-b}\right)^{N_F}.$$

The bijection's loop-opening operation gives one initial interface and one additional interface for each flexible order. Since $2b/(1-b) = \sqrt{q}$, the probability is a constant depending only on n, q times $q^{\# \Gamma/2}$, which is the specified finite map weight. The rooted bijection is bijective, rather than a quotient weighted by automorphism counts. Uniform rerooting at an incidence preserves the law because every map under consideration has the same number of root incidences. These observations also show why deleting cotree-associated primal edges would change the metric in the theorem. $\square$

Figure 2 distinguishes the two exploration triangles from the four flags of an original edge. It also records the local interface pairings determined by the edge's occupation status.

Lemma 3.2 (Tree and incidence identities). *For a fully matched resolved word, let T_n be its primal spanning tree and T_n^* the complementary dual spanning tree. At the cut after letter k , the exploration has a primal corner vertex $v_n(k)$ and a dual corner face $f_n(k)$. In unscaled integer height units,*

$$\begin{aligned} d_{T_n}(v_n(k), v_n(l)) &= L_k + L_l - 2 \min_{j \in [k, l]} L_j, \\ d_{T_n^*}(f_n(k), f_n(l)) &= R_k + R_l - 2 \min_{j \in [k, l]} R_j, \quad 0 \leq k \leq l \leq 2n. \end{aligned}$$

Every primal vertex and dual face occurrence is represented by these corner sequences. Each quadrangle is the union of its two matched exploration triangles; each such triangle is incident to the corner states immediately before and after its letter. Thus the endpoints, faces, and flag incidences of a quadrangle have a uniformly bounded collection of tour representatives. The representatives for each primal vertex are identified by the L -tree rule, and those for each dual vertex by the R -tree rule; moving from a cut to its adjacent triangle changes a height by at most one.

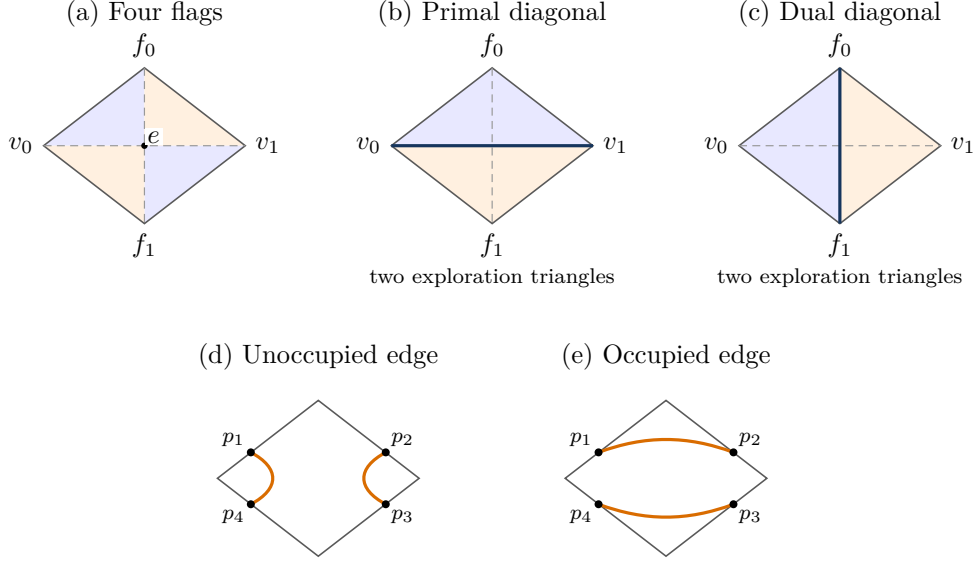


Figure 2: Incidences at one original edge. The four flags have labels (v_i, e, f_j) , $i, j \in \{0, 1\}$. Selecting the primal or dual auxiliary diagonal groups them into the two exploration triangles of the matched letters. The lower panels show the two pairings of the four interface ports: an unoccupied edge pairs ports about each primal endpoint, whereas an occupied edge pairs them about each incident face. This is a topological schematic, not an isometric drawing of the equilateral flags. Labels may repeat for a loop or bridge, but the four flag and port occurrences remain distinct.

Proof. After resolution this is the ordinary tree/cotree contour exploration in [42, Section 4.1, Figures 6–7]. A burger creates an edge to a child in its tree and its order returns along that edge. Letters of the other type hold this tree position fixed. The depth of the common ancestor of the positions at k, l is the minimum contour height between them, proving the two distance identities. The exploration crosses each tree-edge side and each quadrangle side occurrence, which proves coverage of the corners, including repeated occurrences. A letter triangle is bounded by the two successive exploration sides and the created or traversed tree edge. Pairing it with its matching letter triangle gives exactly the associated quadrangle. Subdivision into flags introduces an edge midpoint but no additional incidence choice. This proves the last assertions without any bound on vertex or face degrees. $\square$

Theorem 3.3 (Contour limits). *Let*

$$\varrho = \frac{b}{1-b} = -\cos\left(\frac{\pi\gamma^2}{4}\right), \quad \gamma \in (\sqrt{2}, 2), \quad \kappa = \frac{16}{\gamma^2}.$$

The diffusively rescaled bilateral resolved contours converge, uniformly on compact time intervals in law, to a bilateral Brownian motion whose marginal variance rates are one and whose correlation is ϱ . For a length- $2n$ word conditioned to reduce to the empty word, the same rescaling, with duration one, converges uniformly in law to the corresponding quadrant Brownian excursion from $(0, 0)$ to $(0, 0)$.

For an unconditioned raw block of $O(n)$ letters, the number of flexible orders left unidentified by reduction of the block is $o_{\mathbb{P}}(\sqrt{n})$. Consequently changing their resolutions changes the rescaled contour increments by $o_{\mathbb{P}}(1)$ uniformly on that block.

Proof. The bilateral assertion is [42, Theorem 2.5]. In the original units the covariance rates of the two net burger counts are $(1 - b)/2, (1 - b)/2, b/2$; multiplying heights by $\sqrt{2/(1 - b)}$ gives the stated convention. The conditioned assertion is [21, Theorem 1.8], followed by the deterministic change from duration two to duration one. The unidentified-order estimate is [42, Remark 3.17]; see also the reduced-word estimates in [23]. Each changed resolution changes a contour coordinate by at most one, so their total number bounds the supremum error. This last assertion concerns a raw unconditioned block. The transfer of discrete observations to conditioned extractions will be proved in Section 6. $\square$

3.2 The reference LQG surfaces

Theorem 3.4 (Peanosphere realization). *The bilateral Brownian contour is the boundary-length process of a γ -quantum cone explored by an independent, unparameterized whole-plane space-filling SLE_κ , subsequently parameterized by quantum area. The duration-one excursion is the boundary-length process of the ordinary unit-area γ -quantum sphere explored in the same way from an area-typical marked point p_* . The contour determines the curve-decorated quantum surface up to its marked conformal equivalence.*

Write η for the exploration parameterized by quantum area. In either realization, $\eta_(dt) = \mu_h$. Its spatial identifications are generated by finite chains of scalar horizontal chords*

$$s \sim_L t \iff L_s = L_t = \inf_{[s,t]} L, \quad s < t, \quad (3.2)$$

and the analogous R -chords, with the usual endpoint identification for the sphere. The resulting quotient carries the topology of the plane, or sphere, respectively. Frontier arcs are parameterized by quantum length, with running-minimum contour heights as their length coordinates.

Proof. For the cone, the surface and contour identification are the mating-of-trees and welding theorems [16, Theorems 1.9 and 1.11]. For the ordinary area-one sphere, [36, Theorem 1.1] supplies the Brownian-loop law and determination of the surface explored by an independent unclocked whole-plane space-filling SLE. In our range $\gamma^2 \in (2, 4)$ the covariance is the one in these theorems. The marked-point sampling rule is [16, Proposition A.13]. We use the common deterministic boundary-length convention specified above, with area-clock duration one.

Call a scalar chord *strict* if the contour is strictly above its endpoint height between the endpoints. The finite-chain assertion uses a further geometric part of the DMS construction. There are finitely many visits to each spatial point; ordering their left and right flow-line sectors links consecutive visits by scalar chords. This is the smallest generated equivalence relation in [16, Lemmas 8.9, 8.13–8.15 and Theorem 8.18], without taking its closure. A non-strict scalar chord can be split at its possible intermediate minimum visit into adjacent strict chords; scalar fibers have at most three visits. The same finite-visit and adjacent-sector argument applies to the independent whole-plane exploration of the sphere away from its exploration root: changing to quantum-area time preserves the visits, their order and their frontier incidences. At the root one additionally identifies 0 and 1. Thus the sphere statement combines its Brownian-loop realization with this geometric argument; it does not invoke a separately stated sphere finite-chain theorem. The quotient is topological because the resulting continuous bijection from the compact time quotient to the sphere is a homeomorphism. Running-minimum frontier coordinates are the quantum lengths supplied by the same welding construction. $\square$

We use actual field heights when comparing passage laws. The following description records explicitly the area normalization that couples different patches.

Proposition 3.5 (Three-mark sphere representation). *Let $h_{\mathbb{C}}$ be the whole-plane Gaussian free field (GFF) with circle average zero on $\partial\mathbb{D}$, and put*

$$|z|_+ = \max\{1, |z|\}, \quad G_{\mathbb{C}}(z, w) = \log \frac{|z|_+ |w|_+}{|z - w|}, \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2}.$$

Define

$$h_L = h_{\mathbb{C}} + \gamma G_{\mathbb{C}}(0, \cdot) + \gamma G_{\mathbb{C}}(1, \cdot) - (2Q - \gamma) \log |\cdot|_+, \quad A = \mu_{h_L}(\mathbb{C}). \quad (3.3)$$

Tilt the law of h_L by the integrable strictly positive density $A^{4/\gamma^2-2}/\mathbb{E}[A^{4/\gamma^2-2}]$, and then set

$$h = h_L - \gamma^{-1} \log A. \quad (3.4)$$

The result is the ordinary unit-area sphere with three independent area marks embedded at $\infty, 0, 1$. Resampling its three area marks and renormalizing gives the same law. In particular, on any bounded region avoiding $0, 1$, its field modulo constants is locally mutually absolutely continuous with a whole-plane GFF modulo constants. Near 0 , away from 1 , the corresponding comparison field is $h_{\mathbb{C}} - \gamma \log |\cdot|$.

Proof. This fixed-area field law originates in the sphere construction of David, Kupiainen, Rhodes and Vargas [11, Section 3.3, Proposition 3.8 and Section 3.4]. We use its form in Borga et al. [8, Definition 2.2]; Aru et al. [2, Theorem 1.1 and Remark 1.2] identify it with the ordinary DMS sphere. The mark resampling statement and local comparison, including the γ -insertion, are [8, Lemmas 2.3–2.4]. Subtracting the constant in (3.4) gives area one by the identity $\mu_{h+c} = e^{\gamma c} \mu_h$. $\square$

The proposition deliberately retains the common random constant $-\gamma^{-1} \log A$. Local equivalence modulo constants alone does not give a joint law for actual heights in several patches. The required joint density is derived from (3.3)–(3.4) in Section 6.

4 A topological correspondence for every flag

Contour convergence initially relates exploration times. We need a correspondence on the whole discrete surface which preserves separation and intersections. This section constructs it without estimating the length of any connecting walk. Throughout the finite-volume argument we work on a coupling on which the contours converge uniformly, write $Z_n = (L_n, R_n)$ for the rescaled contours, and write $\eta : [0, 1] \rightarrow S$ for the limiting exploration. Equip the reference topological sphere S with a fixed spherical metric ρ in any of its conformal coordinates. The choice of this coordinate is immaterial to the deterministic assertions below.

4.1 Scalar trees and the continuous projection

For a continuous real function c on a compact time interval, put

$$d_c(s, t) = c(s) + c(t) - 2 \min_{[s \wedge t, s \vee t]} c. \quad (4.1)$$

The equivalence classes for $d_c = 0$ are the vertices of its contour tree. We use the exact primal-tree and dual-tree incidence description in Lemma 3.2. Heights of a discrete tree edge in this section have the same rescaled units as L_n, R_n ; its graph length is still one whenever graph costs are subsequently considered.

Lemma 4.1 (Continuity of the scalar trees). *If $c_n \rightarrow c$ uniformly, then*

$$\sup_{s,t} |d_{c_n}(s,t) - d_c(s,t)| \leq 4\|c_n - c\|_\infty.$$

Suppose $s_n \rightarrow s$, $t_n \rightarrow t$, and $d_c(s,t) = 0$. Every contour representative of every vertex on the discrete c_n -tree path between the vertices at s_n, t_n has distance d_{c_n} tending to zero from the vertex at s_n , uniformly over those representatives. Consequently, if c is L or R , their images under η tend uniformly to $\eta(s)$.

For $c_n = R_n$, take the faces represented by the vertices of this dual-tree path. The union of their primal boundary walks is a connected primal subgraph. All its corner representatives have η -images tending uniformly to $\eta(s)$.

Proof. The displayed inequality follows by bounding the two endpoint errors and the error of the minimum. A vertex u on a tree geodesic from a to b satisfies $d(a,u) \leq d(a,b)$. The exact contour-tree identity therefore bounds all the asserted distances by $d_{c_n}(s_n, t_n)$, up to one rescaled edge if a cut lies inside a step. This tends to zero. If the assertion about η were false, choose a representative u_n violating it and then a convergent subsequence of u_n . Its limit u satisfies $d_c(s,u) = 0$, so $\eta(u) = \eta(s)$ by the scalar chord rule, contradicting continuity of η .

Consecutive faces of a dual-tree path share the primal edge crossed by that tree edge. Their entire boundary walks therefore have connected union, including when a boundary edge or vertex is visited repeatedly. A corner of any of these faces represents its dual-tree vertex, so the same distance bound applies to all its representatives. This proves the last assertion. No bound on a face degree is involved. $\square$

The tour supplies finitely many time representatives for each flag incidence, as specified in Lemma 3.2. We may use either endpoint of an exploration step; changing that choice changes the time by at most one mesh interval. A representative of a primal vertex means any tour occurrence of that vertex, and similarly for a dual vertex. The occurrences at a loop or bridge remain distinct.

Proposition 4.2 (Projection of the complete flag surface). *There are continuous maps $P_n : X_n \rightarrow S$ and numbers $\varepsilon_n \rightarrow 0$ such that the following hold simultaneously.*

- (i) *For every flag T , every $x \in T$, and every tour representative associated to any of the corners of its quadrangle,*

$$\rho(P_n(x), \eta(t)) \leq \varepsilon_n.$$

The same conclusion holds for any other occurrence of an incident primal or dual vertex.

- (ii) *The P_n -images of individual flags, original edges, original closed faces, and complete primal or dual incidence stars have maximum diameter tending to zero.*

- (iii) *The sets $P_n(V(M_n))$ become dense in S .*

These conclusions hold for the original flag complex, with all repeated incidence occurrences retained.

Proof. First choose one representative for every primal vertex and every dual vertex. Representatives of a given primal vertex have zero L_n -tree distance; representatives of a given dual vertex have zero R_n -tree distance. Uniform contour convergence, compactness of the time interval, and the chord rule show that the maximum disagreement between their η -images tends to zero. This is a uniform statement: a failing sequence of vertices and two of their occurrences would have convergent time representatives and would contradict the limiting chord rule.

At a tour cut the primal and dual corner states have that same time representative. In one quadrangle its four corner states are linked successively by equality of a primal or a dual vertex, together with the two consecutive-cut errors of its exploration triangles. The preceding uniform assertion and continuity of η show that all these representatives agree up to a uniform error tending to zero. It follows that the chosen representatives of the primal vertex and face center of every flag agree to this accuracy. Assign its edge midpoint the image of one fixed corner occurrence of that edge. The preceding quadrangle argument gives the same conclusion for this assignment.

Identify S with the unit sphere in $\mathbb{R}^3$. On every flag send a point with barycentric coordinates (a_1, a_2, a_3) to

$$\frac{a_1 y_1 + a_2 y_2 + a_3 y_3}{|a_1 y_1 + a_2 y_2 + a_3 y_3|},$$

where y_1, y_2, y_3 are the assigned images of its vertices. For all large n the three images lie in a common ball of radius less than $\pi/4$, so the denominator is nonzero. These formulas agree on glued sides and give a continuous map on the flag quotient. Their images stay in the same small spherical ball. This proves (i) and the flag assertion in (ii). Every flag in a primal or dual star shares the image of its central vertex, so its entire star has diameter at most twice the maximum flag diameter. The assertions for edges and faces follow as well. Finally the tour meshes become dense in $[0, 1]$, η is onto, and every tour position has a primal corner vertex with the stated small error. This proves (iii). $\square$

Lemma 4.3 (Protection by complete time preimages). *Let $K \subset U \subset S$, where K is compact and U is open. There are a neighborhood K' of K with $\overline{K'} \subset U$ and a finite union I of open time intervals such that*

$$\eta^{-1}(\overline{K'}) \subset I, \quad \bar{I} \subset \eta^{-1}(U). \quad (4.2)$$

Intervals are interpreted on the time circle when necessary. If $p_ \notin \bar{U}$, they are strict subintervals of $(0, 1)$. For all large n , every flag whose projection meets K , and the full incidence stars of its three vertices, have all their tour representatives in I .*

In the bilateral setting the same assertion holds for bounded compact spatial protections, with finitely many bounded time intervals.

Proof. Choose two successive compact neighborhoods of K in U . The inverse image of the smaller closed neighborhood is compact and lies inside the open inverse image of the larger one. Cover it by finitely many intervals whose closures lie in that open inverse image. This proves (4.2). If a sequence of omitted tour representatives belonged to stars of flags meeting K , Proposition 4.2 would force their η -images into an arbitrarily small neighborhood of K . They would eventually lie in $\eta^{-1}(\overline{K'}) \subset I$, a contradiction. Avoiding p_* keeps the inverse images away from 0, 1. In the plane, transience of η makes the inverse image of every compact set compact; the same finite-cover proof then applies. $\square$

4.2 Joining arbitrary lifts with spatial clearance

Lemma 4.4 (Local joining). *For every $\epsilon > 0$ there are $\delta > 0$ and n_0 such that, for $n \geq n_0$, $x \in S$, and primal vertices u, v satisfying*

$$P_n(u), P_n(v) \in B_\rho(x, \delta),$$

there is a primal graph walk from u to v whose entire projection is contained in $B_\rho(x, \epsilon)$. The analogous assertion holds for arbitrary prescribed points of X_n , using continuous paths. The choice

is uniform over the prescribed lifts. Its conclusion makes no assertion about the number of edges in the walk.

Proof. First suppose that representatives s_n, t_n of u_n, v_n converge to s, t with $s \sim_L t$. The primal-tree path joins these vertices and has image tending to the single point $\eta(s)$ by Lemma 4.1 and Proposition 4.2. If instead $s \sim_R t$, join the incident dual vertices by their dual-tree path. The union of the primal boundaries of its faces is connected, contains both primal corner vertices, and has the same vanishing image diameter. A walk in this connected subgraph supplies the required primal connection.

For arbitrary s, t with $\eta(s) = \eta(t)$, choose a finite chain of L - and R -chords joining them, using Theorem 3.4. For each intermediate time choose its nearest discrete cut and its primal corner vertex. The preceding construction for each link gives finitely many walks whose union tends to $\eta(s)$; concatenate them.

If the uniform assertion failed, there would be $n_j \rightarrow \infty$, $\delta_j \downarrow 0$, points x_j , and two prescribed vertices with images in $B(x_j, \delta_j)$ admitting no walk in $B(x_j, \epsilon)$. Choose time representatives and pass to convergent subsequences $x_j \rightarrow x$, $s_j \rightarrow s$, $t_j \rightarrow t$. Proposition 4.2 gives $\eta(s) = \eta(t) = x$. The finite-chain construction just proved produces the forbidden walks. For an arbitrary point of a flag, first join it inside that flag to its primal vertex. The extra pieces have vanishing projected diameter by Proposition 4.2. $\square$

To specify an ordered following test for a path $c : [0, 1] \rightarrow S$, choose $0 = t_0 < \dots < t_r = 1$, open ports U_j containing $c(t_j)$, and open tubes V_j containing $c([t_{j-1}, t_j])$. A follower visits these ports in order, with its intervening pieces in the corresponding tubes. The compact reference pieces and the chosen smaller port neighborhoods have positive clearance inside these open sets. A port specifies a visit; a transverse crossing will require the additional side and clearance conditions below.

Corollary 4.5 (Following a finite collection of paths). *Let $c_1, \dots, c_m$ be continuous paths on S and let prescribed lifts of their endpoints have the corresponding projected limits. For every finite ordered following test as above, all sufficiently large n admit lifted paths whose projections satisfy that test. Primal endpoint lifts admit primal walks. Endpoints which coincide in the reference collection may be assigned one common prescribed lift.*

Proof. Apply Lemma 4.4 with a spatial error smaller than the clearance of the chosen finite requirements. Uniform continuity gives a finite time partition for each c_i on whose pieces its oscillation is smaller than the resulting $\delta/3$. Choose primal representatives near the partition points using Proposition 4.2(iii), retaining the prescribed endpoints. Join successive representatives by the lemma. Subdivide at the intermediate ports before making these choices. Concatenation proves the ordered assertion, and the same construction works for all the finitely many paths at once. $\square$

4.3 From local joining to sphere homeomorphisms

Local joining initially supplies connected branch sets, rather than disjoint arcs meeting only at their endpoints. We use a cubic skeleton because a minimal tree joining three attachments has just one trivalent junction after its outgoing paths are attached. Pruning therefore turns the branch sets into a subdivision, to which Whitney's embedding theorem and disk extension can be applied.

Lemma 4.6 (Lifting a cubic skeleton). *Let $G \subset S$ be an embedded finite simple cubic graph with piecewise smooth edges. For every $\epsilon > 0$, all sufficiently large n contain a primal subgraph G_n which is a subdivision of G , and there is a graph homeomorphism $h_n : G_n \rightarrow G$ such that*

$$\sup_{x \in G_n} \rho(P_n(x), h_n(x)) < \epsilon.$$

If G is 3-connected, its facial cycles and those of G_n correspond under this homeomorphism, up to one common reversal of orientation.

Proof. Take pairwise disjoint small closed junction disks about the vertices of G . In each disk its three incident edges are three disjoint stubs except for their common endpoint. Choose three further intermediate points on the stubs, and narrow disjoint ribbons along the remaining edge portions. The ribbons of different edges are disjoint outside the junction disks. Inside a junction disk, choose the three ribbon ends disjoint, so each meets only its own stub neighborhood. Make all these neighborhoods small enough for the desired error.

Assign one primal lift to every graph vertex. By Corollary 4.5, lift its three initial stubs from that common lift to the three intermediate points. Their union is a connected branch set B_v within the chosen junction neighborhood. The branch sets belonging to distinct vertices are disjoint, since their projected neighborhoods are disjoint. Lift each remaining edge portion between the corresponding stub endpoints. The error may be chosen so small that different connecting walks can intersect only within branch sets at a common endpoint, and no connecting walk meets any other branch set. In particular, near a junction each connector meets only the end of its own stub; all such contacts occur before the prescribed marks delimiting the middle of that connector.

For an edge vw , take the portion from its last visit to B_v before its first subsequent visit to B_w . Erase loops from that finite walk. The resulting simple paths have interiors disjoint from every branch set and from each other. At a branch set retain a minimal finite tree joining its three attachment points. After the three outgoing paths are attached, this tree has exactly one vertex of degree three and every other vertex of degree two; this also covers coincident attachment points and the case where one attachment lies between the other two. Removing unused branches therefore gives an actual subdivision of G . The use of three attachments is the reason for choosing a cubic skeleton.

Here is the order control needed to define h_n . Give each edge ribbon a continuous longitudinal coordinate, equal to its edge parameter on the central arc. In the path-lifting construction use successive coordinate intervals of length less than σ , and choose the spatial errors so that each joining piece stays within σ of its prescribed interval. If a loop-erasure removes a portion between two occurrences of one vertex, the two original parameters differ by at most 3σ : their common projected point has a single longitudinal coordinate within the required error of both. Truncation at the branch sets changes only the initial or final stub. Assign increasing central-arc parameters to the surviving vertices, and interpolate strictly increasingly along the surviving edges. The skipped parameter intervals have oscillation tending to zero as $\sigma \downarrow 0$. Within a junction neighborhood map the three remaining tree arms to the three initial edge pieces. This gives a graph homeomorphism whose error is bounded by the ribbon widths, junction diameters, and those vanishing oscillations. Choose them in that order to make the total error smaller than ϵ .

Finally a subdivision induces the same embedding of the suppressed graph. Whitney's theorem says that two embeddings of a finite simple 3-connected planar graph in the sphere are equivalent up to a sphere homeomorphism [43]. Its hypotheses hold for G , and hence the identified facial cycles agree, with at most one global orientation reversal. $\square$

Theorem 4.7 (Homeomorphisms with vanishing projection error). *There are sphere homeomorphisms $H_n : X_n \rightarrow S$ such that*

$$\sup_{x \in X_n} \rho(H_n(x), P_n(x)) \longrightarrow 0. \quad (4.3)$$

In particular,

$$\max_T \text{diam}_\rho H_n(T) \longrightarrow 0, \quad (4.4)$$

and the images under H_n of every flag and all its tour representatives have vanishing maximum disagreement. The maps may preserve or reverse orientation, with a common sign on each sphere. They may be chosen measurably in the contour coupling. The assertion continues to hold when that coupling retains any additional random observations.

Proof. Fix a small $\epsilon > 0$. Triangulate the reference sphere with simplicial triangles of diameter less than $\epsilon/10$, and take its embedded dual graph G . Draw dual edges through adjacent triangle interiors. Its faces lie in stars of the original vertices, so their diameters are less than ϵ . This graph is cubic and 3-connected. For the latter assertion, delete any two dual vertices, or equivalently the corresponding two closed triangles. Their union has connected complement in the sphere, whether the triangles are disjoint, meet at a vertex, or share an edge. A path in that complement between two remaining triangle interiors can be perturbed away from triangulation vertices; the triangles it successively crosses give a dual path avoiding the two deleted vertices. Thus deleting fewer than three dual vertices does not disconnect the dual graph. The simplicial condition also ensures that the dual has neither loops nor multiple edges.

Apply Lemma 4.6 with an error much smaller than ϵ . Denote by F_n the face of G_n corresponding to a face F of G . Choose z_F so that $\overline{F} \subset B(z_F, \epsilon)$. We claim that, for all large n ,

$$P_n(\overline{F_n}) \subset B(z_F, 3\epsilon). \quad (4.5)$$

The boundaries already satisfy this assertion by the graph homeomorphism estimate. If the claim fails in the interiors, pass to a sequence $x_n \in F_n$ with $P_n(x_n) \rightarrow y \notin B(z_F, 3\epsilon)$. Choose a vertex v of G outside $B(z_F, 3\epsilon)$; such a vertex exists for sufficiently small ϵ , because the dual graph has small faces and covers the sphere. The complement of $\overline{B}(z_F, 2\epsilon)$ is connected, so join y to v by a path with clearance from $B(z_F, 3\epsilon/2)$. Lift this path from the prescribed point x_n to the junction lift of v , using Corollary 4.5. Its projection avoids $B(z_F, 3\epsilon/2)$ for all large n , whereas the projection of ∂F_n is contained in that ball. The lifted path therefore avoids ∂F_n . Its last point is outside $\overline{F_n}$, since v is not incident to the corresponding face. This contradicts the Jordan separation of ∂F_n . There are finitely many faces, so (4.5) holds for all of them simultaneously.

Each face of a 3-connected sphere embedding is a Jordan disk. The graph homeomorphism h_n therefore extends across every face to a homeomorphism $H_{n,\epsilon} : X_n \rightarrow S$. For completeness, a homeomorphism between the boundary circles of two closed disks extends by identifying each disk with the standard closed disk and extending the resulting circle homeomorphism radially. The Jordan–Schönflies theorem gives these disk identifications; here all boundaries are finite polygonal curves in triangulated surfaces. The facial correspondence makes the extensions agree on common edges. For $x \in \overline{F_n}$, both $H_{n,\epsilon}(x) \in \overline{F}$ and $P_n(x) \in B(z_F, 3\epsilon)$, and hence

$$\sup_{x \in X_n} \rho(H_{n,\epsilon}(x), P_n(x)) \leq 4\epsilon.$$

Take $\epsilon_k \downarrow 0$. At each k the construction works eventually; increase its threshold n_k so that the thresholds are strictly increasing, and use H_{n,ϵ_k} for $n_k \leq n < n_{k+1}$. This proves (4.3). Equation (4.4) follows from Proposition 4.2 and the triangle inequality.

All finite walk choices can be made first in a fixed enumeration. The graph and its disk extensions can likewise be chosen among piecewise linear maps after finite subdivisions, with rational coordinates in fixed spherical charts. These form a countable collection; the polygonal disk construction and an arbitrarily small perturbation give a member satisfying any strictly weakened error bound above. Choose the first such member. Its uniform error is measurable by taking a supremum over a countable dense set in each flag. Choosing the largest available accuracy among the first n accuracies gives a measurable version of the diagonal construction. Finally, the whole argument is deterministic on the event of contour convergence and the peanosphere assertions, so retaining other observations does not alter it. $\square$

4.4 Protected bilateral windows

The analogous plane statement is local. In particular, it does not require a conformal uniformization of an infinite discrete map, or the assertion that an arbitrarily capped limiting contour has a sphere quotient.

Proposition 4.8 (Closing a protected window). *Work in a bilateral contour coupling with locally uniform convergence to its peanosphere contour. Given finitely many compact spatial tests and open buffers about them, one may choose a bounded time window, and then a finite spherical tree/cotree sewing, with the following properties for all large n .*

- (i) *Every flag star needed by the tests is unchanged, including its cyclic order and all original primal incidences.*
- (ii) *On a neighborhood of these flags there are continuous projections with the conclusions of Proposition 4.2 and Lemma 4.3, and the local joining conclusion of Lemma 4.4 holds with arbitrary fixed clearance.*
- (iii) *The finite sewing admits homeomorphisms into the reference one-point compactification for which the projection error tends to zero over the tests. The inverse images of the tested compact sets are contained in the unchanged part. Consequently both directions of every strict crossing or separation test take place in the original discrete surface.*

The construction can be made simultaneously for a countable exhaustion of compact tests and their finite refinements, by diagonalization.

Proof. Transience makes the full time preimage of a spatial compact bounded. Choose a large disk containing the tests and all their buffers, then a larger disk for the finite joining routes below. Both Brownian coordinates are simultaneously below any prescribed pair of levels at arbitrarily distant times in either direction. Indeed, after an invertible linear change of coordinates this follows from planar Brownian recurrence applied to a disk contained in the corresponding open cone. Choose cuts $a < b$ beyond the complete time preimages in use, at which both coordinates are strictly below all their heights. Rounding the cuts to lattice times preserves these strict inequalities for all large n .

No scalar chord from a protected time can cross either cut: it would have to pass below its own endpoint height. Thus every protected primal vertex, dual vertex, matched quadrangle, and their full stars are determined inside (a, b) . Translate the resolved axial walk on $[a, b]$ into the strictly positive quadrant, precede it by a monotone axial path from the origin to its initial point, and follow it by a monotone axial path from its terminal point to the origin. This is a finite nonnegative resolved excursion and hence a finite spherical tree/cotree sewing. The appended steps

and all incidence classes which they affect are below the protected heights. The exact tree/cotree incidence rule therefore leaves the protected full stars unchanged. This proves (i).

The projection and scalar-path proofs apply in any compact time interval containing the complete preimages involved. Strict lower cuts confine all the relevant discrete representatives to that interval. A finite chain of limiting chords uses only finitely many times, and its discrete tree paths can be read in a bounded enlargement. The sequential compactness proof of local joining consequently gives (ii), uniformly on each compact spatial buffer. There is no need to bound the length of one of these paths.

We give the exterior argument for (iii), since it prevents the cap from entering a tested region. Choose a finite simple cubic 3-connected planar skeleton G whose bounded faces have diameter less than ϵ throughout the large disk, and whose outer face misses that disk. Such a graph is the dual of a fine triangulation of a disk completed on the sphere by a vertex at infinity; the dual edges adjacent to that vertex can be drawn in a collar of the disk. The proof of 3-connectivity in Theorem 4.7 applies without change. Lift its stubs and edge ribbons using (ii). All these finite walks and their complete incidence stars are in a bounded time protection.

Take T larger than every time in this protection and so large that both limiting tour tails $(-\infty, -T] \cup [T, \infty)$ avoid a disk containing G and its ribbons. Retain two tour positions near $-T, T$ as exterior sentinels. For each bounded face of G , join each sentinel to a graph vertex not incident to that face by a spatial path avoiding a slightly enlarged disk about that face. Retain also a path between the two sentinels outside the disk containing G . There are finitely many such routes. Enlarge the time protection to contain their lifted routes before choosing the final lower cuts $a < -T < T < b$.

The tour strip from T to b is connected to the appended terminal piece, and the strip from a to $-T$ is connected to the appended initial piece. These strips avoid the lifted skeleton: an intersection would be an incidence in one of its full stars, whose representatives have already been retained between $-T$ and T . All cap-affected flags are connected to these strips through the appended tour, including any old exploration triangle whose partner is supplied by an appended step. Thus the cap and both strips lie in one face of the lifted skeleton. The sentinel routes show that this face is not any of the bounded corresponding faces: a route to a vertex not incident to such a face avoids its boundary, by its projection clearance. Whitney's facial correspondence therefore identifies it as the outer face. Equivalently, the cap changes only that exterior component; every other complementary disk and all its incidence data are the original ones.

For a bounded face, the connected-exterior proof of (4.5) now applies. Its points have time representatives within the fixed window, so a violating sequence has a convergent representative. The required path away from its small boundary can be lifted in the original map by (ii); if that path reaches a cut-affected region, it has already reached the outer component, which gives the same separation contradiction. Hence the entire projected face lies in a 3ϵ disk. Extend the graph homeomorphism over all bounded faces and extend arbitrarily over the outer face. Since the tested compact sets avoid the outer face and its collar, their inverse images are in the unchanged part, with error at most 4ϵ . Let $\epsilon \downarrow 0$.

All windows, skeletons, and routes just used are finite at each stage. Enumerate the desired compact tests and accuracies, retain the first k at stage k , and enlarge their windows before choosing the cuts. Choosing the discrete index after that finite stage gives the asserted simultaneous diagonal version. $\square$

4.5 Strict intersections and primal separators

Corollary 4.9 (Transverse crossings). *Fix a Jordan quadrilateral in a reference chart. Prescribe one crossing between its first and third sides and another between its second and fourth sides, with endpoint ports beyond those sides, and with positive clearance from the inappropriate sides. Discrete surface paths satisfying sufficiently accurate versions of these requirements intersect. If both are primal graph walks, they have a common primal vertex. The assertion holds simultaneously for every fixed finite collection of such strict tests, on the sphere or in a protected bilateral window.*

Proof. Use H_n to transfer the two walks to the reference sphere. Its uniform error is smaller than all the fixed port clearances for large n . Each walk then has a subpath crossing the indicated opposite sides of a common quadrilateral. Two such transverse paths intersect: a simple subarc of the first joining its two sides separates the other two sides, by the Jordan theorem. Their inverse images intersect as well. The interiors of distinct edges in an embedded primal graph are disjoint, so an intersection of primal walks gives a common vertex, also when one of their edges is a loop or they share an edge. Apply Proposition 4.8 in the bilateral case. $\square$

Corollary 4.10 (Primal surrounding circuits). *Let $B_0 \subseteq B_1 \subseteq B_2 \subseteq B_3$ be concentric closed disks in a planar reference chart. Suppose a closed primal walk has projection in $\text{int}(B_2) \setminus B_1$ and nonzero winding about B_0 . For all sufficiently accurate correspondences it contains a simple primal circuit C_n with the following properties. The closed side Ω_n of C_n opposite the exterior of B_3 contains every lift whose projection lies in B_0 , is disjoint from every lift projecting outside B_3 , and satisfies*

$$P_n(\Omega_n) \subset B_3.$$

The circuit uses a subset of the walk's edges with multiplicities discarded; for nonnegative edge costs its cost is at most the cost of that walk. The conclusion holds in a protected bilateral window as well.

Proof. For large n , the H_n -image of the walk remains in a slightly enlarged annulus between B_0 and B_3 . The short pointwise homotopy from P_n to H_n avoids B_0 and an exterior point, so the winding number is unchanged. Decompose the finite closed walk at repeated vertices into simple circuits, deleting immediate backtracks. Winding numbers add in this decomposition. At least one simple circuit has nonzero winding about B_0 ; its embedded image is a Jordan curve, and therefore it separates all of B_0 from the exterior of B_3 . Its bounded side lies inside B_3 , because the connected exterior of B_3 avoids the circuit. Keeping a smaller fixed error than the clearances between the four disks gives precisely the assertions about all P_n -lifts and about $P_n(\Omega_n)$. The edge and cost assertion follows from the circuit decomposition. The protected-window homeomorphism proves the local version. $\square$

Lemma 4.11 (Finite separator meshes from open ports). *For every $\epsilon > 0$ and every finite set of excluded reference points, there is a finite list of reference paths avoiding those points, with strict ordered ports and tubes, with this property: any primal walks satisfying all the prescribed following tests contain a connected primal subgraph whose complementary components have H_n -image diameter at most ϵ . The subgraph uses only the supplied walks. In particular its total edge cost is at most the sum of their costs.*

If a geodesic metric D inducing the topology of S is specified, the reference paths can all be chosen to have finite D -length. There is no cost requirement on joins from prescribed discrete endpoints in this statement.

Proof. Choose a cubic 3-connected skeleton with face diameter much smaller than ϵ , perturbing it to avoid the excluded points. Put disjoint small junction disks at its vertices and narrow disjoint edge ribbons between them. In each junction disk prescribe four paths following the sides of a small square, extended past its corners so that adjacent paths have strict transverse crossings there. Prescribe edge paths which run from the central hole of one junction square through its edge ribbon to the central hole of the other.

Corollary 4.9 stitches the four square paths into a closed walk of nonzero winding about its central hole. To see the winding directly, cut consecutive paths at their forced intersections in the disjoint corner boxes and retain the intervening side portions. Their concatenation follows the square once, and a homotopy within its four thin side corridors preserves that winding. Corollary 4.10 then supplies a simple junction circuit. Every incident edge walk starts inside its central hole and exits its outer disk, so it meets that circuit. Thus all paths at a junction belong to one connected branch set, without specifying their lattice endpoints in advance.

Truncate each edge walk between its last visit to the first branch set and its first visit to the second, and erase loops. Different edge corridors are disjoint away from their assigned junctions. As in Lemma 4.6, minimal trees joining the three attachments produce a subdivision of the cubic skeleton. Its edges follow the reference edges in order. Under the actual homeomorphism H_n it is an embedded graph on the reference sphere with arbitrarily small tracking error. Whitney's facial correspondence and the connected exterior of each small face, exactly as in the proof of (4.5), imply that every complementary component has diameter at most ϵ . Keeping the entire connected union of the supplied walks only subdivides these components. No additional edges have been used, which proves the cost assertion.

Finally a continuous path in an open set can be followed through any fixed finite ordered ports by a finite-length D -path. To verify this, cover its compact image by finitely many D -balls whose doubled balls lie in the open set, subdivide the parameter interval so successive selected points lie in one such ball, and join them by short D -geodesics. Those geodesics remain in the doubled balls. Applying this to the finite path list, with smaller tubes and the same strict ports, gives finite D -length reference paths with the identical topological requirements. $\square$

5 Local reconstruction and its observations

We now use the topological correspondence to read a finite spatial observation from finitely many portions of the inventory word. The observation may be a graph passage, an extremal length on the flag surface, or a maximum over all vertices in a protected region. The word portions must retain every incidence needed by the observation; selecting one tour visit to each point would not suffice.

There are two sources of data. Local directed traversals and their side-labelled quantum boundary lengths determine the continuum contour increments and relative heights. The retained raw words determine exact discrete equalities and minima. We first construct instructions that combine these data without reading an omitted part of the word, then specify the observations retained under subsequential limits. The next section compares their conditional laws on different surfaces.

5.1 Local directed traversals and their lengths

Let $\chi = \sqrt{\kappa}/2 - 2/\sqrt{\kappa}$. The imaginary field $\tilde{h}$ is a whole-plane GFF modulo $2\pi\chi\mathbb{Z}$, with the independent uniform angular phase in $[0, 2\pi\chi)$ included in its definition. It is independent of

the real field before the area parameterization. This convention gives exact translation and positive-dilation symmetry of the unlocked directed exploration. Its local Markov decomposition is a Dirichlet GFF plus a harmonic function, interpreted modulo the same period [35, Section 2.2.1 and Propositions 2.8, 2.11].

Lemma 5.1 (Buffered local traversals). *Let $K \Subset U \Subset V$ be deterministic bounded planar regions. The restrictions of $\tilde{h}$ to V determine the directed entrance-to-exit traversals of U meeting K , as local curve pieces. Together with $h|_V$, they determine elapsed quantum time on these pieces and the quantum lengths of compactly retained frontier arcs. There are only finitely many such traversals. Given independent spatial Poisson marks with intensity $\varepsilon^{-1}\mu_h$, the marked curve cells contained in U are local functions of the two fields and marks there. The assertion does not assign an absolute time or an order between distinct traversals.*

Proof. Use a positive buffer between U and ∂V in [24, Lemma 2.4]. It determines the two frontier flow lines from a countable dense set, stopped on exiting U , and hence each directed traversal, with its elapsed Lebesgue-area clock. Quantum time is obtained by measuring successive images with $\mu_h|_U$. For an incoming or outgoing frontier arc compactly contained in V , restrict the field to its indicated side and use the LQG boundary measure in the welding construction of [16]. To make the locality explicit, truncate that side by crosscuts inside V and map it to a half-disk near a retained subarc. Apply boundary Gaussian multiplicative chaos to the transformed field there. Two such charts give the same measure by boundary coordinate covariance: their transition fixes a real boundary interval and extends by reflection across it. Exhaust the retained arc by these subarcs. Thus restriction to a smaller arc commutes with the measure construction, and lengths and outgoing minus incoming length increments are local once the sides and arcs are specified. This statement supplies no height shift between different traversals.

Properness of the plane exploration puts all visits to $\bar{U}$ in a compact time interval; on the sphere the time interval is already compact. Each traversal reaching K crosses the positive Euclidean distance $\text{dist}(K, U^c)$. Uniform continuity gives a positive lower bound on the duration of such a crossing, so only finitely many disjoint traversals can reach K . The marked-cell assertion is precisely [9, Lemma 3.7], or follows by applying the traversal determination to consecutive local marks. Its extension to the field laws used here is made on a fixed buffer by the local field comparisons above. Relative heights will be recovered from aligned frontiers below. The remaining traversal order and the conditional laws of discrete observations are treated in Section 6. $\square$

Lemma 5.2 (A single visit has a local time neighborhood). *For every deterministic planar point z , the whole-plane exploration visits z exactly once almost surely. The same holds at a point sampled conditionally on h from μ_h , independently of the unlocked curve. If t is the unique visit time and I is a time neighborhood of t , then some spatial neighborhood of z has all its visits in I .*

Proof. The deterministic-point assertion follows from the imaginary-geometry construction; see [8, Section 2.4, property (vi)]. Independence and integration against the random area measure give the second assertion. For the last, otherwise choose $z_j \rightarrow z$ visited at $t_j \notin I$. Properness, or compactness of the spherical time interval, gives a subsequence $t_j \rightarrow t' \notin I$. Continuity gives $\eta(t') = z$, contradicting the unique visit. $\square$

5.2 The two candidate suppliers of a flexible order

Immediately before a flexible order, the candidate supplier of each burger type is the most recently produced burger of that type still available, when it exists. The order takes the more recent of

these two candidates. If a flexible order is unresolved by reduction of an extended block, every candidate supplying it lies before that block. Thus an order in a strictly smaller inner interval has a positive time buffer from either external supplier. We will also retain a compact time window containing those suppliers.

For the limiting contour, a backward supplier of colour $c \in \{L, R\}$ at time t means an earlier time $s < t$ with $c(s) = c(t) = \min_{[s,t]} c$. We do not initially assume uniqueness. The next two lemmas show that, when both colours have suppliers with positive spans, each supplier is unique and their chronological order is stable on a compact protected family. This is the continuum fact needed to type the externally unresolved flexible orders.

Here all time windows lie strictly inside the excursion interval in finite volume. A *forward joint record with witness* $r > 0$ is a time t such that both coordinates on $[t, t+r]$ are at least their value at t . Backward joint records are defined on $[t-r, t]$. A scalar branch level is the value of a strict local minimum of one coordinate. Brownian scalar chord classes have at most three visits; a class with three visits has its middle visit at such a minimum.

Lemma 5.3 (Cone estimates). *Set*

$$\theta = \arccos(-\varrho) = \frac{\pi\gamma^2}{4}, \quad d_0 = 0, \quad d_1 = \frac{1}{2}, \quad d_2 = \frac{\pi}{2\theta} = \frac{2}{\gamma^2} > \frac{1}{2}.$$

For fixed positive duration, the probability that Brownian motion stays above $-\varepsilon$ in k specified coordinates is at most $C\varepsilon^{2d_k}$, for $k = 1, 2$. Constants may depend on a compact range of positive durations. A fixed-start killed transition probability ending with those k coordinates in an ε -neighborhood of the walls is at most $C\varepsilon^{k+2d_k}$, with start points in a compact set. Finally, on a compact time window and for fixed positive record witness, the image in $\mathbb{R}^2$ of the joint record times has upper box dimension at most

$$2(1 - d_2) < 1. \tag{5.1}$$

The same bound holds for either coordinate projection of this image.

Proof. Whitening the two-dimensional Brownian motion sends the quadrant to a wedge of opening θ . Its positive homogeneous harmonic function vanishing on the boundary is $r^{\pi/\theta} \sin(\pi\varphi/\theta)$. Brownian scaling and the killed wedge kernel give the survival power $\varepsilon^{\pi/\theta}$; in a half-plane it is ε . One can equivalently use [23, Lemma 3.2] for this survival estimate. For the endpoint assertion split the duration in two. Bound the first transition density by the free Gaussian density, reverse the second killed transition using kernel symmetry, and integrate the survival bound over the k wall coordinates, of total volume $O(\varepsilon^k)$. Gaussian tails make the unrestricted coordinate integral finite.

For completeness, the image-dimension conclusion needs a counting argument, not merely the dimension of the record-time set. Fix $\nu > 0$, a compact window, and a witness $r > 0$. On the almost-sure Brownian modulus event, the oscillation over every interval of length 2δ is at most $C\delta^{1/2-\nu}$ for all small dyadic δ . If a grid interval contains a joint record, Brownian motion starting at its appropriate grid endpoint stays in a quadrant enlarged by $C\delta^{1/2-\nu}$ for at least $r/2$. On truncating the random modulus constant by a deterministic bound, the survival estimate bounds the expected number of such grid intervals by $C\delta^{-1+d_2-2\nu d_2}$. Markov's inequality and Borel–Cantelli give, for every $\zeta > 0$, the eventual almost-sure bound $\delta^{-1+d_2-2\nu d_2-\zeta}$ for their number. Each image lies in a ball of radius $C\delta^{1/2-\nu}$. Letting $\nu, \zeta \downarrow 0$ through rational values gives (5.1). Coordinate projection cannot increase the number or size of covering balls. $\square$

Lemma 5.4 (Protected chord genericity). *Almost surely the bilateral contour, and the interior of the quadrant excursion, have the following properties simultaneously.*

- (i) *There is no positive-span common L - and R -chord.*
- (ii) *At a forward or backward joint record with positive witness, neither coordinate is at a scalar branch level anywhere in the time domain.*
- (iii) *If both backward supplier chords at a time have positive span, each supplier is unique, their times are distinct, and neither scalar chord class is a branch triple. More precisely, fix a compact time window J and $a > 0$. The admissible tuples $(t, s_L, s_R) \in J^3$ with $t - s_L, t - s_R \geq a$ form a compact set, and $|s_L - s_R|$ is bounded below on it whenever it is nonempty. For uniformly converging contour perturbations, admissible supplier tuples in J^3 with these span bounds converge along subsequences to such tuples. Their more-recent-supplier choice is therefore stable in a neighborhood of each limiting tuple, uniformly on this compact set.*

Proof. We first work with bilateral Brownian motion and use countably many compact windows and rational positive witnesses throughout. Record conditions can be shortened, so any two distinct times can be placed in disjoint rational windows with a positive gap, retaining their respective record witnesses inside those windows.

For two such windows, condition on their Brownian increments relative to their own starting heights. The independent increment across their gap gives their relative starting height a non-degenerate Gaussian density in $\mathbb{R}^2$. Thus a random set determined by the first window and one determined by the second are translated relative to each other by a vector with a density. If both sets have upper box dimension at most $a < 1$, their difference set has upper box dimension at most $2a < 2$ and hence zero planar Lebesgue measure. Lemma 5.3 applies to the image of forward joint records in the first window and backward joint records in the second. A common chord would give equal two-dimensional heights in these two images, an event of probability zero. This proves (i).

We give both parts of the branch-level argument. First, a scalar local minimum at a time t cannot also be a one-sided record of the other coordinate. Restrict to witnesses of fixed positive length on both sides of t . On a grid interval of length δ containing such a time, the increments in one direction must stay above $-C\delta^{1/2-\nu}$ in one coordinate, and those in the other direction in both coordinates. The past and future increments at the two grid endpoints are independent. The survival bounds therefore give $C\delta^{(1-2\nu)(d_1+d_2)}$ for a specified interval. Since $d_1 + d_2 > 1$, choose $\nu > 0$ small enough that this exponent exceeds one; a union bound over $O(\delta^{-1})$ intervals then tends to zero. Remove the modulus truncation and then the rational witness restriction. This proves the assertion when t itself is the local minimum.

Second, put a distinct scalar local minimum in a rational window separated from the joint record. The local-minimum heights of a scalar Brownian path in this window form a countable set: every strict local minimum is the unique minimum of some rational interval. The relevant coordinate projection of the joint-record image in the other window has upper box dimension less than one, hence Lebesgue measure zero. Condition on the increments in both windows. The relevant coordinate of the intervening Gaussian increment has a density; it almost surely does not translate this null set to any of the countably many local-minimum heights. Countable exhaustion proves (ii).

We recall why the scalar facts just used hold simultaneously. Brownian motion has a unique minimum on every fixed compact interval, and minima on disjoint rational intervals have distinct heights almost surely, by the transition density and independent increments. These countably many statements imply that every local minimum is strict, that different local minima have different

heights, and that a scalar chord class has at most three visits. Indeed, four visits at the same minimal height would give two distinct local minima at that height.

At a time in (iii), both coordinates are backward records. A branch triple would put the respective coordinate at a local-minimum level, contrary to (ii). In particular two earlier suppliers for one colour, together with the order time, would give that excluded triple. A supplier shared by both colours would contradict (i). The chord relation is closed on a compact time window, so the admissible tuples with spans at least a form a compact set. Their distinct supplier times consequently have a positive minimum separation. For the perturbation assertion, extract a convergent subsequence of any sequence of admissible tuples. Uniform contour convergence preserves the chord equalities and inequalities, and the lower span bound remains a . Uniqueness identifies the limiting suppliers; their positive separation preserves chronological order. A finite cover of the compact tuple set gives the asserted uniformity.

Finally, on every compact subinterval of $(0, 1)$ the quadrant excursion law is absolutely continuous with respect to a Brownian path with a starting-point density, after localization of its endpoint heights. This follows from its positive killed transition and entrance densities; see [21, Section 2]. The events proved above are local on a finite union of compact time windows. Countable exhaustion therefore transfers them to the excursion interior. Its common chord between the two global endpoints is the stipulated root convention. $\square$

5.3 Retained blocks and their scalar frontiers

We use Theorems 3.3 and 3.4, Lemma 4.1, Proposition 4.2, and Lemma 4.3. The exploration is proper in a plane chart. The preimage of a compact set is compact, every occurrence of a primal or dual vertex has the same limiting image, and its incident flags have vanishing projected diameter. Scalar horizontal identification is a closed relation on compact time windows; the quotient tree maps continuously to the surface. All restrictions have a compact protection $K \subseteq U$ and a further open margin before the chart boundary.

Definition 5.5 (A protected chart). A chart instruction comprises these finite data.

1. Disjoint used time intervals, each strictly inside an extended block. Overlapping extended blocks are merged. A subinterval of a merged block retains its actual internal increments, without a new free starting height.
2. Finite unions of rational subintervals specifying ports, confining cells, supplier cells, and protected stars. Rational endpoints may be fractions of a block duration. A full star retains every incidence occurrence and its cyclic orientation at its primal and dual vertices.
3. For each coordinate c , a forest F_c on the block indices, recording height differences at the *used* starts of blocks whose labelled scalar frontiers are compared.
4. Gluing bands, exact minima of specified retained subintervals, and a finite rational partition carrying types for externally unresolved flexible orders. Band endpoints have strict clearance, except at retained minimum heights, which are tested exactly on the lattice.

Its realization uses the raw extended words, the forest differences, and these instructions. It retains the actual flag shapes and oriented gluings. A readout is a measurable function of this finite incidence record.

Instruction types form a countable collection. Durations, forest differences, and contour paths are inputs. We use deterministic cuts and Poisson cuts, rounded to even lattice times. Parametrize

the marks by their absolute cut positions: deterministic coordinates are fixed parameters, and, conditional on the number of random marks, the free coordinates have smooth densities on their ordered instruction cells. Durations are differences of these positions. In the reference experiment of Section 6, every fixed coordinate carries the same point mass as in the actual cut law; smooth proposal densities and their ratios concern only the free coordinates. No joint density is asserted in a fixed coordinate or in a redundant list of durations. Spatial tests are implemented between inner and outer instructions with strict margins.

A *record frontier* of a block is its scalar ancestral arc from the right endpoint down to the block minimum (the suffix frontier), or from the left endpoint down to that minimum (the prefix frontier). Its height parameter is measured relative to the used start. The scalar-tree projection makes this a continuous parameterized curve, even when the corresponding record time jumps. Its image is contained in the block image. Lemma 5.1 gives this oriented, side-labelled arc and its boundary-length parameter from the local fields.

Two such frontiers have an *aligned overlap* if, on a nondegenerate height interval, they trace the same arc in the same ancestral direction, with compatible boundary-side labels under the contour welding, and their local height parameters differ by a constant. Compatibility retains the paired incident sides; it does not require the two incident domains to occupy the same geometric side. This is a local test on the parameterized arcs. Boundary length on an identified side is the local welding measure of Lemma 5.1; no identification is inferred from the coincidence of two isolated spatial points.

Lemma 5.6 (Local recognition of a scalar overlap). *Almost surely the following holds simultaneously for the countable block atlas. A positive aligned overlap of two frontiers of the same scalar colour identifies the corresponding points in that scalar tree. Its parameter translation is the difference of their actual used-start heights. The assertion extends to the closure of the overlap.*

Conversely, a scalar chord between visits lying strictly inside two different blocks is represented by the closure of a positive aligned overlap of the first block's suffix frontier and the second block's prefix frontier. At a three-visit scalar branch, the adjacent pairs have overlap above the branch level and the outer pair has overlap below it; if two visits belong to one block, its internal minimum performs that part of the identification.

Proof. We first distinguish a scalar class from its possibly larger spatial fibre. For a coordinate c , let E_c be the set of heights of visits which have a positive-span one-sided record in c and also a positive-span one-sided record in the other coordinate. This is a single pathwise Lebesgue-null set of heights. For witnesses in the same direction, the height-image estimate in Lemma 5.3 gives dimension at most $2(1 - 2/\gamma^2) < 1$. Here and below take a countable union over compact time windows and positive rational lower bounds for the witness spans.

For witnesses in opposite directions, cover a fixed window by intervals of length δ . On the Brownian modulus event with oscillation at most $\delta^{1/2-\varepsilon}$, an interval containing such a visit forces scalar survival for a fixed positive duration backward from its left endpoint and forward from its right endpoint, each with initial allowance at most $2\delta^{1/2-\varepsilon}$. These two Brownian increments lie on disjoint time intervals. The reflection principle therefore bounds the probability by $C\delta^{1-2\varepsilon}$. The expected number of occupied intervals is at most $C\delta^{-2\varepsilon}$. On a dyadic sequence, Markov's inequality and Borel–Cantelli bound the count eventually by $\delta^{-4\varepsilon}$; the excluded modulus events are summable. Letting $\varepsilon \downarrow 0$ and using any Brownian Hölder exponent below $1/2$ shows that the height image has dimension zero. Thus both directions contribute only a null set to E_c . Interior excursion absolute continuity gives the same assertion for the finite bridge.

An interior point of a genuine scalar ancestral segment belongs to a nontrivial scalar class.

Indeed its level is below the endpoint height: the last visit to that level before the endpoint and the first visit afterward exist by recurrence, or by the lower excursion endpoints, and the contour stays above the level between them. Every visit in that class has a positive-span one-sided c -record witness: join it to any different visit of the same class. If an opposite-colour chord is incident to one of its visits, its height consequently belongs to E_c . Outside E_c no such chord can leave the class. By the finite-chain identification rule of Theorem 3.4, its entire spatial fibre is then that one scalar class.

Consider an aligned overlap with local parameters r and $r + \Delta$. Almost every r in its nondegenerate interval has an actual height outside E_c . Equality of its spatial points therefore gives equality in the scalar tree and forces $\Delta = x_i - x_j$, where x_i, x_j are the actual used-start heights. The same statement holds on a dense set of parameters. The ancestral segments are continuous in the scalar-tree metric, so equality extends to the whole overlap and its closure. This also shows why an isolated coincidence arising from the other colour cannot give an accepted overlap.

For the converse, write the blocks as $[a_i, b_i]$, $[a_j, b_j]$, with $b_i < a_j$, and let their scalar minima be m_i, m_j . Their endpoint ancestral arcs intersect in the scalar tree on the height interval

$$[\max\{m_i, m_j\}, M], \quad M = \min_{[b_i, a_j]} c,$$

when this interval is nonempty. A chord at level h between used-interior visits has $m_i, m_j \leq h \leq M$, by its suffix and prefix tests. The interval cannot be a singleton. If, for example, $m_i = M = h$, the visit in the first block is an interior local minimum at level h , and a further visit at that level lies in (b_i, a_j) . There is a preceding visit before the first block, since the bilateral contour visits lower heights in its past, or since the interior excursion starts below the tested level. Its last return to level h before this block, the local minimum, the intervening visit, and the visit in the second block would be four visits of one scalar class. This is impossible: the middle two would be distinct local minima at the same height. The case $m_j = M$ is the time reversal. Cut endpoints are almost surely not scalar record times, since the atlas uses deterministic or independently marked cuts. They therefore do not create an endpoint exception in this argument.

The nondegenerate common ancestral interval projects to an aligned same-colour overlap, with its local boundary-length parameter. It contains the chord level in its closure. For three visits, the middle visit is the strict scalar minimum between the outer ones. The two adjacent pairs share the incident segment above that level; the outer pair shares the segment below it. All three visits of a protected spatial point are retained. Their branch boundary is thus a retained block minimum; if two visits are in one block, its exact internal minimum accounts for their identification. This proves the converse and the branch assertion. $\square$

We next control the contour minima omitted between retained blocks. The full-preimage protection of Lemma 4.3 is essential: the following statement concerns all times over a spatial protection, rather than a chosen visit to each point.

Lemma 5.7 (An omitted minimum has a strict gap). *Let c be one of the two limiting contours on a compact interval J . Let $I \subset J$ be open and contain $\eta^{-1}(K) \cap J$ for a compact set K , and let $O = J \setminus I$. Consider a compact family $\mathcal{C}$ of triples (s, t, h) , $s \leq t$, such that*

$$c(s) = c(t) = h, \quad \eta(s) \in K, \quad c(u) \geq h \quad (u \in [s, t] \cap I).$$

There is $\delta > 0$ such that, whenever $[s, t] \cap O \neq \emptyset$ and $(s, t, h) \in \mathcal{C}$,

$$\left| \min_{[s, t] \cap O} c - h \right| > \delta. \tag{5.2}$$

Consequently the omitted pieces cannot change an exact gluing decision at heights tending to a protected candidate height without either a retained obstruction or a strict height gap.

Proof. Suppose there are candidate triples (s_j, t_j, h_j) and minimizing times $u_j \in [s_j, t_j] \cap O$ for which the difference in (5.2) tends to zero. Pass to a limit $(s, t, h) \in \mathcal{C}$ and $u \in [s, t] \cap O$. We have $c(u) = h$. Every $v \in (s, t) \cap O$ belongs to $[s_j, t_j]$ eventually, and hence $c(v) \geq h$; the same inequality at the endpoints follows from $c(s) = c(t) = h$. By the candidate condition it also holds on $[s, t] \cap I$. Thus

$$c(s) = c(u) = c(t) = \min_{[s, t]} c = h.$$

The scalar chord rule gives $\eta(u) = \eta(s) \in K$, contradicting $u \notin I$. The same argument applies when $s = t$.

The strict gap is stable under sufficiently small uniform contour perturbations and endpoint perturbations within compact instruction cells. In a finite block cover, suffix minima, prefix minima, and the minima of the intervening retained pieces are included in the tested condition. The remaining minimum is separated as above. One can therefore put rational height bands strictly between the protected candidate heights and that remaining obstruction. Lemma 5.8 uses these bands together with the exact discrete retained minima; no approximation of an exact lattice equality is inferred from uniform convergence alone. $\square$

5.4 Exact reconstruction of the protected incidences

Lemma 5.8 (Finite protected reconstruction). *Fix $K \subseteq U$, finitely many local tests, and a contour realization on the common probability-one event of Lemmas 5.4 and 5.6. There exists a chart instruction retaining every time preimage and full star needed by the tests. It has a continuum input neighborhood on which the omitted-obstruction and supplier-order tests have fixed strict margins. For all sufficiently large discrete indices, its realization gives the exact required incidences by recomputing retained equalities and minima from the raw words and forest offsets. The neighborhood can be selected from a countable atlas of continuity sets. The instruction is measurable from local directed traversals, their labelled frontier lengths, and their relative order. Disjoint closed spatial buffers admit disjoint raw extended blocks; finitely many descriptions in one buffer admit a single union instruction.*

Proof. Choose $K \subseteq K_1^\circ \subseteq K_1 \subseteq U$. The compact set $\eta^{-1}(K_1)$ has a finite interval cover whose closure lies in $\eta^{-1}(U)$. Enlarge and merge its intervals to place $\eta^{-1}(K)$ in used interiors. By Lemma 4.3, all incidences of the required stars lie there eventually. Every visit to a supplier vertex is included, not only a selected path endpoint.

Within a block, scalar equalities and minimum tests are exact word operations. For each scalar colour, form the finite graph of blocks having a positive aligned frontier overlap. Lemma 5.6 proves that each edge gives the actual height difference, read from the two local length parameters. Choose a spanning forest by the fixed block enumeration; every height difference within a component is then a forest sum. Pairs in different components cannot have a protected scalar chord, by the converse part of that Lemma, so they are rejected locally.

Here is the local choice of the remaining bands. Fix an ordered pair of blocks in one component. In component height coordinates, let $\mathcal{C}$ be the compact family of candidate endpoints and heights after testing the suffix, prefix, and every intervening retained minimum in that component. One endpoint has its image in K . The closures of the positive aligned overlaps define a local subset $\mathcal{G} \subset \mathcal{C}$, and let $\mathcal{B} = \mathcal{C} \setminus \mathcal{G}$. By Lemma 5.6, $\mathcal{G}$ is exactly the set of true scalar chords; $\mathcal{B}$ consists of the rejected candidates. In particular this definition reads no omitted contour minimum.

We use the omitted minimum only to prove that these local sets admit a finite code. Its value M_{ij} , on the portions between the two blocks omitted from this component, is one fixed threshold:

$$(s, t, h) \in \mathcal{G} \iff h \leq M_{ij} \text{ for } (s, t, h) \in \mathcal{C}.$$

It has a strict gap from the candidate heights. Indeed, a convergent sequence of omitted minimizing visits approaching such a height would give an omitted visit in the same scalar class as a protected endpoint, as in Lemma 5.7. The full-preimage cover places that visit in a retained block interior. The converse of Lemma 5.6 then places this block in the same component, contradicting its omission. This also covers a retained block belonging provisionally to another component.

The gap separates the compact candidate family from the threshold M_{ij} . Thus its accepted and rejected parts are both closed in $\mathcal{C}$; compactness of $\mathcal{B}$ is a consequence of this gap, not merely of its being a complement. If $\mathcal{G}$ and $\mathcal{B}$ are both nonempty, their height projections are compact and satisfy

$$\max h(\mathcal{G}) < \min h(\mathcal{B}).$$

Choose the first rational threshold and positive rational clearance strictly separating these two *locally defined* sets, and allow exact candidate matches only below that threshold. If one set is empty, choose the corresponding all-or-none instruction. If there are no omitted portions, all candidates pass. These choices are measurable: extrema of the compact local candidate and overlap relations are measurable, and the rational search is countable. For the rejected set, the extrema can be obtained by first imposing positive rational distance from the closed overlap relation and then taking monotone limits. Aligned overlap itself is a local measurable test: on each rational parameter subinterval, with rational domain margins and a bounded offset range, test whether a constant translation makes the uniform distance between the two labelled arcs zero. The admissible offsets form a compact set, so its nonemptiness and least element are measurable. A positive overlap has only one translation, by Lemma 5.6. Thus even the rejection decisions use only local data. The unknown value M_{ij} is never an instruction input or a selection criterion.

A scalar branch boundary uses the exact minimum in its retained middle block, with no rational approximation to that equality. Additional supplier neighborhoods retain the adjacent overlap segments. To specify the discrete decision, write $c_{i,n}$ for the contour of block i relative to its used start and $x_{i,n}$ for that start height in its forest component. All heights here use the same rescaled units; equality is still tested exactly. For endpoints s, t in ordered blocks i, j , first require

$$c_{i,n}(s) + x_{i,n} = c_{j,n}(t) + x_{j,n} = h.$$

Test that the suffix after s , the prefix before t , and each intervening retained minimum in this component are at least h . These are raw-word tests. If there are omitted portions, also require that h lie below the selected rational threshold, with its prescribed clearance; use the all-or-none rule when appropriate. Endpoints in different forest components are rejected. This is the complete acceptance predicate for a scalar link.

There are finitely many block pairs and colours. Were their decisions wrong along arbitrarily close converging inputs, pass to limiting candidate endpoints and heights. A retained equality or minimum is decided by the exact tests just specified. Any remaining discrepancy must come from an omitted minimum crossing its rational band, contrary to the strict gap. Retained branch minima are never rounded to that band. This proves inclusion of genuine protected links and rejection of false ones without enumerating the Brownian branch points or inferring lattice equalities from uniform convergence.

Reduce each extended raw word internally. An unresolved flexible order in the used interior has both candidate suppliers before the left extension start. Otherwise it would have been internally resolved. Limits of those candidates give two positive-span backward chords. Their earlier endpoints are distinct by Lemma 5.4. A branch ambiguity would place the joint record at a scalar middle minimum or at the height of a separated scalar branch visit; the same genericity assertion excludes both. Their relative order is therefore stable. The positive left extension supplies the span lower bound a in Lemma 5.4(iii), and the protected supplier window supplies its compact interval J . A finite rational cover of the admissible tuples consequently specifies all needed external types, simultaneously for every unresolved order in the used interior.

Unresolved decisions in unused extension portions may be arbitrary: they do not change internal matches, and used-start heights are free inputs. A portion supplying a tested incidence is promoted to a used supplier portion first. All comparisons other than the retained equalities and minimum tests have strict margins and persist in a small continuum neighborhood; those excluded tests are computed exactly from the raw words. The neighborhood's boundaries can avoid the countably many atoms of the relevant distance-to-boundary variables. This gives the countable atlas.

Disjoint closed spatial buffers have disjoint compact time preimages, hence disjoint finite covers. Within one buffer, taking unions and merging overlapping intervals realizes all original readouts as restrictions of one incidence record. $\square$

Lemma 5.9 (Recovering the complete contour input). *Retain the directed traversals throughout every extended block of a chart, their elapsed quantum-time cuts, and their side-labelled frontier length functions. These data determine each extended contour path relative to its used start. Aligned overlaps determine the forest height differences. This assertion concerns the complete extensions as well as the used interiors. Each block has its own clock origin: its absolute placement and the elapsed times in omitted gaps are absent. The forest height differences are retained up to common height translation.*

Proof. For a coordinate c , a used start s , and a later local time t , the incoming and outgoing frontier lengths for that segment are

$$Z_c(s) - \min_{[s,t]} Z_c, \quad Z_c(t) - \min_{[s,t]} Z_c.$$

Their difference is $Z_c(t) - Z_c(s)$. For a point in the left extension, reverse the interval and subtract in the opposite order. The local length construction of Lemma 5.1 provides these functions on the indicated sides. Elapsed quantum time identifies their arguments. Rational local times and contour continuity therefore recover the whole relative path, not only its endpoint displacement. All cuts used here are within-block offsets; the recovery requires no elapsed time between distinct blocks. Lemma 5.6 recovers the forest differences from aligned boundary lengths. If several extended blocks belong to one spatial buffer, retain all of these data together, with their relative traversal order. No independence between blocks in this one aggregate input is asserted. $\square$

5.5 Compatible local observations

We compactify a nonnegative cost by $s \mapsto s/(1+s)$, with ∞ mapped to one. A finite path record lists its endpoints, range, finitely many ordered marked visits, and the ranges and costs of the intervening subpaths. Marks carry retained tour-occurrence labels; their list order records the order along the graph path. Different visits to the same vertex are retained separately, and

repeated marks at one visit are permitted. Time labels in a block are divided by its duration. For fixed numbers of marks these data lie in a compact product, including the hyperspace of compact subsets of the compact time-label space. We retain the closed set of all such records, for each finite number of marks. Records from the empty family have a separate isolated value. Spatial ranges are read from the cell labels after including the continuum exploration in the input. They are projections of the record, not substitutes for its order and cost coordinates.

All finite refinements are retained jointly on the same discrete paths. For a fixed family of paths, forgetting unrestricted additional marks merges adjacent subpath costs by addition and adjacent ranges by union, and leaves the endpoints and whole range unchanged. These are continuous maps on the compact record spaces, with the convention $s + \infty = \infty$. The ability to repeat marks makes the forgetting maps onto the corresponding coarser record sets. Additional port or cost constraints restrict that image to the subfamily satisfying those constraints. We also retain subrecords between any two marked visits, with their inherited order and cost.

In the first-exit example below, for each retained primal vertex u choose one retained protected tour occurrence $t_n(u)$, expressed in the time units of the reference chart, and write $z_n^{\text{ref}}(u) = \eta(t_n(u))$. The sets U, V, W lie in that chart. Proposition 4.2 makes different incident representatives uniformly indistinguishable in the limit. Spatial membership is read through inner and outer protected-cell instructions with fixed strict margins.

Lemma 5.10 (Attainable and universal readouts). *The chart readouts may simultaneously include:*

1. *the closed path-record sets just described, for all rational ports, confinement instructions, cost budgets, and finite ordered marks;*
2. *every numerical extremal length of a family of curves on the retained flag triangles, with cell-defined ports and confinement;*
3. *maxima over all retained starting vertices of minimum costs to specified cell-defined target sets, including*

$$E_n(U, V) = \max_{z_n^{\text{ref}}(u) \in U} \theta_n d_n(u, \{v : z_n^{\text{ref}}(v) \notin V\}), \quad U \Subset V \Subset W,$$

where $\theta_n > 0$ is arbitrary deterministic and full stars are protected in W . An empty maximum is zero.

These quantities have joint subsequential limits in compact spaces. Every retained finite path subrecord is approximable by actual finite paths with arbitrarily small open losses; every sequence of actual bounded-cost paths has a further finite-subrecord limit. Universal maximum–minimum readouts retain their universal quantifier. Spatial versions use inner and outer cell instructions with fixed strict margins before extraction. Forgetting marks and taking marked subrecords commute with this joint extraction. The lemma asserts finite-record attainability; existence of a continuous limiting path requires an additional modulus of continuity.

Proof. All quantities are functions of the exact incidence record. In particular, extremal length uses the actual triangle shapes and oriented side identifications. It is a numerical variational quantity in $[0, \infty]$; it need not be continuous in a combinatorial presentation to be retained. Nonnegative path costs and extremal lengths are compactified, while the hyperspace of closed subsets of a compact metric space is compact. A diagonal extraction over the countable instructions and numbers of marks gives joint tightness. The two assertions about finite records are the two directions of Hausdorff convergence. The continuous forgetting maps take the discrete refined

record sets onto their coarser sets; continuity on compact spaces gives the same identities after Hausdorff convergence. To retain a particular finite refinement of an approximating path, mark that path first and extract its refined record. Its coarse projection is the original limiting record. The same argument retains any finite collection of marked subpaths, since all are read from that one path before taking limits. A diagonal extraction can retain countably many such finite records, but by itself does not turn their spatial ranges into a parameterized continuous path. Later metric arguments supply the needed cost-based modulus. We make no converse inference about a maximum from a path hyperspace; its numerical maximum–minimum value is a separate coordinate.

For the first-exit test, stop a path on its first vertex outside V . Its initial part lies in V ; its final edge and endpoint lie in W for large n , by the full-star protection and vanishing projected mesh. The minimum in the displayed formula can therefore be computed in that retained incidence record. This includes every inner vertex, since every occurrence and full star has been retained.

Finally, for a compact set inside an open spatial confinement, η^{-1} of the compact set has a finite rational cover with closure inside the preimage of the confinement. Intermediate compact protections give both inner and outer such covers. Lemma 5.8 makes their discrete cell tests correct eventually. Apply this to ports, ranges, and target complements, with one more margin for stars. Countable refinement supplies arbitrarily small open losses. Every assertion used below is made with these margins fixed first. $\square$

6 Local kernels and changes of surface law

The protected incidence records of Theorems 5.5 and 5.8 allow us to retain graph passages and flag-surface extremal lengths in the same local observation array. We now identify their conditional laws. The key is an exact resampling identity: after independently sampling the protected words, ladder blocks in the intervening gaps supply a density for their relative heights. Its limit depends only on the continuum input. Consequently the local output kernel survives both finite-volume conditioning and the changes of field law used below.

A discrete cost may be divided by any deterministic positive number. In word resampling, n counts letters per unit of continuum time; for a sphere word of length $2N$ and duration one, use $n = 2N$. Put $s_b = \sqrt{2/(1-b)}$. Raw diffusive heights divide integer counts by $\sqrt{n}$; the variance-one heights of Section 3 multiply these by s_b . All intrinsic chart inputs use the latter convention.

6.1 The local input and the conditional law

Let $\mathcal{C}$ denote the full continuum input: the whole contour, its determined curve-decorated quantum surface, the chosen coordinate marks, and independent cut marks. In a plane-field experiment include also the entire real and imaginary fields and the area-clocked exploration. Any additional embedding choice or imaginary-field lift is sampled from its conditional law given the curve-decorated surface, independently of the discrete outputs.

For one union spatial buffer, the intrinsic local input I consists of the directed traversals of all its retained extended blocks, their elapsed quantum-time cuts, labelled frontier-length measures, forest height differences, and the finite relative traversal order used by the instruction. This includes the left extensions and supplier portions, not only the used interiors. Each block’s absolute time placement and the elapsed times in omitted gaps are excluded, as is common height translation. By Theorems 5.1 and 5.9, these data are measurable from the real field on the real buffer, the imaginary field on its possibly larger buffer, local marks, and that order.

It is useful to distinguish the input used in the raw-word reference experiment. Write $\mathbf{l}$ for the relative contour paths on every complete extended block, together with their durations, within-block cut offsets, forest differences and fixed instruction. The local input I recovers $\mathbf{l}$, by [Theorem 5.9](#). The compact readouts of [Theorem 5.10](#) are initially recorded with time labels; call this array O^0 . Relabelling its ranges and ports in the reference surface uses only the traversals retained in I . We denote the resulting spatial array by O .

Theorem 6.1 (Compatible local kernels). *Fix a countable atlas of protected charts and countably many observations from [Lemma 5.10](#). From every sequence of integer scales tending to infinity and arbitrary deterministic cost units one can extract jointly kernels $K(I, dO)$ with the following properties.*

1. *In the bilateral word and in the interior of the finite empty-word bridge, the conditional output law given $\mathcal{C}$ is $K(I, dO)$. The finite-bridge assertion refers to the actual joint finite-sphere observations on the chosen subsequence.*
2. *For disjoint closed real buffers, the conditional joint law given $\mathcal{C}$ is the product of their kernels. Each buffer may carry a whole countable array of observations and refinements. Overlapping descriptions of one observation agree through their common union chart.*
3. *Smooth positive changes of the reference input densities do not change these kernels on their common input law. Consequently a locally absolutely continuous change of continuum field law uses the same kernels; a dominated field law gives dominated joint laws.*
4. *The extraction may retain all fixed deterministic ratios of microscopic time scales in a countable list, together with their actual joint input laws. The output cost units may be arbitrary deterministic positive numbers.*

Fixed measurable continuum functionals, including the reference internal metrics of the LQG surface, may be included jointly. Auxiliary variables sampled conditionally on $\mathcal{C}$ preserve the assertions.

The proof follows the finite density calculation below. We first construct the reference kernels to which that calculation will apply. For each union spatial buffer, sample its complete extended raw words, all duration and cut variables, and its forest parameter under the reference laws described below. Retain the entire array O^0 for that buffer, including observations which use several of its time blocks. These constitute one aggregate reference package. Packages in disjoint spatial buffers are sampled independently; no product assertion is made between outputs from constituent time blocks of one buffer.

The contour invariance principle and [Theorem 5.10](#) give joint subsequential limits of each package. On a common diagonal over finite families of instructions, disintegrate these limits as

$$\mathcal{R}_\alpha(d\mathbf{l}_\alpha)K_\alpha^0(\mathbf{l}_\alpha, dO_\alpha^0), \quad (6.1)$$

where α indexes union spatial buffers. The spaces are standard Borel, so these conditional kernels exist. The later gap and noise variables must be appended through the input alone; proving that fact is the substantive reference-limit step in [Theorem 6.5](#). After spatial relabelling we write the kernels as $K(I, dO)$.

6.2 An input-dependent density preserves a kernel

Lemma 6.2 (Local density ratios). *Let I, O, A be Polish variables, with O compact: retained intrinsic input, output, and additional input, respectively. Let $\hat{e}_n$ be a frozen microscopic exterior*

with macroscopic record e_n . On a fixed good chart suppose that along every sequence $e_n \rightarrow e$,

$$R_n^{\hat{e}_n} \implies R_e(dI, dA)K(I, dO). \quad (6.2)$$

Suppose the actual conditional resampling law is exactly

$$Q_n^{\hat{e}_n} = \frac{W_n^{\hat{e}_n}}{R_n^{\hat{e}_n}[W_n^{\hat{e}_n}]} R_n^{\hat{e}_n}. \quad (6.3)$$

Factors depending only on $n, \hat{e}_n$ may be canceled. If the weights are bounded on this chart and converge jointly to $w(e, I, A)$, with positive limiting normalizer, the limiting conditional output kernel remains K . This holds for finite products of independent complete reference packages, and after integration of auxiliary variables absent from the finite readout.

Proof. Bounded joint convergence gives convergence of the normalizer and each bounded continuous test multiplied by the weight. Division by the positive normalizer gives the limit

$$\frac{w(e, I, A)}{\int w(e, I, A) R_e(dI, dA)} R_e(dI, dA) K(I, dO).$$

A monotone-class argument identifies the conditional kernel. No continuity of K in its input is required. Independent complete packages converge jointly to the product of their joint laws. A subsequent input-dependent weight changes the input law but preserves the conditional product of output kernels. An unused auxiliary variable may be integrated first. $\square$

6.3 Forest heights and exact lattice smoothing

Notation 6.3 (Forest lattice). For m used starts at even integer times put

$$\Lambda_m = \{x = (x_i^1, x_i^2)_{i=1}^m \in \mathbb{Z}^{2m} : x_i^1 + x_i^2 \in 2\mathbb{Z} \text{ for all } i\}.$$

Common height translation is immaterial. Let P_F take the integer-linear differences specified by F_1, F_2 , and set $\Lambda_F = P_F \Lambda_m$. If x is the vector of raw integer start heights, the normalized forest parameter is

$$z = s_b n^{-1/2} P_F x \in s_b n^{-1/2} \Lambda_F. \quad (6.4)$$

Sample it with probabilities proportional to a positive smooth density $\varpi_F(z)$. In raw diffusive coordinates $r = n^{-1/2} P_F x$, the corresponding continuum density is $s_b^{d_F} \varpi_F(s_b r)$, where $d_F = \text{rank } P_F$. The fixed factor $s_b^{d_F}$ cancels in the lattice normalizing sum.

A nonempty fiber is a translate of $\Lambda_m \cap \ker P_F$, so its coarea constant does not depend on the feasible value. Crucially, the finite readout depends on full heights only through $P_F x$. To integrate unused height variables, first include their reference sampling masses in the finite weight (6.12), and then sum in that weighted joint law, as in the proof of Lemma 6.5.

Read an iid word backward until the next type-1 burger survives to the search origin. Such a *ladder block* has no internally unresolved flexible order: that order would consume the surviving burger. Successive ladder blocks are independent marked raw words. Denote their durations and second displacements by T_j, Y_j , with backward sign giving first displacement -1 . Set

$$J_m = \sum_{j=1}^m T_j, \quad S_m = \sum_{j=1}^m Y_j, \quad p_m(k, \ell) = \mathbb{P}[J_m = k, S_m = \ell].$$

In this lattice calculation let $Z^\circ = (L^\circ, R^\circ)$ be the Brownian limit in raw diffusive height units. Its marginal variance rate is $(1-b)/2$ and its covariance rate is $b/2$. Thus the variance-one Brownian motion is $Z = s_b Z^\circ$. Write $\tau_a^\circ = \inf\{t : L_t^\circ = -a\}$. The quantities J_m/m^2 and S_m/m below converge in these raw units.

The backward-ladder and local-limit approach is developed in Gwynne and Sun [22, Section 2.1, Lemma 2.1, Proposition 2.2 and Lemma 2.3], whose argument invokes Doney's bivariate stable local-limit theorem [13]. We keep the exact parity lattice and prove the conditional bridge extension needed for the resampling identity.

Lemma 6.4 (Lattice smoothing and bridges). *Let g be the density of $(\tau_1^\circ, R_{\tau_1^\circ}^\circ)$. It is bounded and positive smooth for $t > 0$, and*

$$\sup_{\substack{k \geq 1, \ell \in \mathbb{Z} \\ k-m-\ell \in 2\mathbb{Z}}} \left| m^3 p_m(k, \ell) - 2g(k/m^2, \ell/m) \right| \rightarrow 0. \quad (6.5)$$

The probabilities vanish off that parity coset. For admissible endpoint sequences converging in these units to (t, y) , $t > 0$, the entire stopped contour converges conditionally to the Brownian first-passage bridge with that endpoint and duration. The convergence is uniform on compact interior sets of scaled depth, duration, and second displacement. Strictly admissible open uniform path tubes have positive limiting probability.

Proof. The iid ladder limit follows from the stopping rule and the contour invariance principle, as in [22, Section 2.1]. Let $\sigma^2 = \text{Var}(L_1^\circ) = (1-b)/2$, $\beta = \text{Cov}(L_1^\circ, R_1^\circ)/\sigma^2 = b/(1-b)$, and $\nu^2 = \text{Var}(R_1^\circ) - \beta^2\sigma^2 > 0$. Writing $R^\circ = \beta L^\circ + \nu B$ with independent standard B , the reflection principle gives

$$g(t, y) = \frac{1}{2\pi\sigma\nu t^2} \exp\left\{-\frac{1}{2\sigma^2 t} - \frac{(y+\beta)^2}{2\nu^2 t}\right\}. \quad (6.6)$$

It is bounded and extends continuously by zero for $t \leq 0$. If $\hat{g}_{s_b}$ denotes the hitting density at depth s_b for the variance-one process Z , then

$$\hat{g}_{s_b}(t, z) = s_b^{-1} g(t, z/s_b). \quad (6.7)$$

Indeed the hitting time is the same and the second endpoint is multiplied by s_b . This is the explicit change from the raw lattice density to normalized boundary-length coordinates.

We specify the lattice factor rather than suppress it. For $\varphi(s, u) = \mathbb{E}e^{i(sT_1 + uY_1)}$, domain-of-attraction convergence gives uniform compact convergence of $\varphi(s/m^2, u/m)^m$. On the compact set

$$\mathcal{A} = \{(s, u) : \tfrac{1}{2} \leq \sqrt{|s|} + |u| \leq 2\},$$

the limiting modulus has a maximum $\rho < 1$, since the limiting law has the density g . Uniform convergence bounds $\sup_{\mathcal{A}} |\varphi(s/m^2, u/m)|^m$ by some $\rho' < 1$ for all sufficiently large m . For small $r = \sqrt{|s|} + |u| > 0$, choose $m = \lfloor r^{-1} \rfloor$. Then $(m^2 s, mu) \in \mathcal{A}$, so $|\varphi(s, u)|^m \leq \rho'$. Taking the m -th root gives $1 - |\varphi(s, u)| \geq c/m \geq cr$, the required bound near zero. The forward raw words $\mathbf{h}, \mathbf{hC}, \mathbf{hc}, \mathbf{hhH}$ have pairs $(1, 0), (2, 1), (2, -1), (3, 0)$, respectively, in the backward displacement convention. In each word the leftmost $\mathbf{h}$ is the first surviving type-1 burger found by the backward search. Their differences generate the parity lattice. Thus the only unit-modulus points on the Fourier torus are zero and (π, π) ; away from them powers decay exponentially. Near them, the scaled Fourier integrands are dominated by $e^{-c(\sqrt{|s|} + |u|)}$, an integrable function. Fourier

inversion gives uniform convergence. On the feasible coset the two neighborhoods contribute equally, producing the factor two; on the other coset they cancel. This proves (6.5).

Fix $0 < \varepsilon < 1/2$, put $r = \lfloor \varepsilon m \rfloor$, and retain the first $m - r$ marked blocks. Their density under the specified total endpoint is

$$\frac{p_r(k - J_{m-r}, \ell - S_{m-r})}{p_m(k, \ell)}. \quad (6.8)$$

On compact target sets it is at most $C\varepsilon^{-3}$ and has the continuous limit given by (6.5). This proves convergence of the retained prefix. Under the total endpoint, complete marked blocks are exchangeable. The last r blocks have the same ordered marked law as the first r , whose conditional density is bounded by

$$\frac{p_{m-r}(k - J_r, \ell - S_r)}{p_m(k, \ell)} \leq C.$$

Their duration and full relative contour oscillation therefore vanish in the m^2, m units when first $m \rightarrow \infty$, then $\varepsilon \downarrow 0$. This uses unconditioned stopped-contour convergence; no internal path is reversed. Converging together proves the full bridge limit.

For parameter continuity, the first coordinate above its lower wall is a three-dimensional Bessel bridge from a to zero. Independent standard bridges b^3, b^1 on $[0, 1]$ give

$$A_u = \|(1 - u)ae_1 + \sigma\sqrt{t}b_u^3\|, \quad L_{tu}^\circ = A_u - a, \quad R_{tu}^\circ = \beta L_{tu}^\circ + u(y + \beta a) + \nu\sqrt{t}b_u^1.$$

The killed Brownian transition density verifies this representation. It is continuous for $a, t > 0$; applying the preceding convergence to each convergent parameter sequence yields compact uniformity. Positive killed transition densities and Brownian path support give the stated tube positivity. $\square$

In each positive gap choose, backward, a guard of $\lfloor a\sqrt{n} \rfloor$ ladder blocks followed by noise of H_n blocks, with $H_n/\sqrt{n}$ independently discretized uniform on $[a, 2a]$. Here a is in raw diffusive units; the normalized guard depth tends to $s_b a$, and the normalized noise depth ranges in $[s_b a, 2s_b a]$. Forward, the noise precedes its guard. The joint density of scaled depth, duration, and second displacement is positive smooth on interior compact sets, with the displayed parity. The randomized depth is not frozen in a height resampling. As $a \downarrow 0$, finitely many such pieces fit strictly in their prescribed gaps with probability tending to one, by the contour limit at their ordinary search origins.

6.4 The exact protected-word density

Lemma 6.5 (The protected-word ratio). *Lemma 6.2 applies to the protected charts of Lemma 5.8, in the bilateral iid word and strictly inside the finite empty-word bridge, jointly with every fixed finite family of bounded continuous whole-contour observations.*

Proof. Fix a chart instruction and a good continuity neighborhood. We give the finite identity before taking limits.

The marked sample space. For a raw word w , let $\mathbf{p}(w)$ be the product of its letter probabilities. Let $\mathcal{L}_H$ be the forward words for which the backward search finds its H -th surviving type-1 burger exactly at the left endpoint. Their ladder decomposition is unique. Such a word has no internally unresolved flexible or type-1 order: either would consume the surviving leftmost type-1 burger before seeking an earlier supplier. Unresolved rigid type-2 orders cause no ambiguity. Thus $\mathcal{L}_H$

and its internal types are intrinsic to the finite word, independently of its incoming inventory. If $d(w)$ is forward displacement, then

$$p_H(k, y) = \sum_{\substack{N \in \mathcal{L}_H, |N|=k \\ d(N)=(H, -y)}} \mathbf{p}(N). \quad (6.9)$$

At a feasible endpoint let $\mathbf{B}_{H,k,y}$ be the probability law whose summands in (6.9) are divided by $p_H(k, y)$.

Fix each deterministic absolute cut of the instruction at its prescribed lattice index. These cuts are parameters of both laws, with identical point masses, rather than resampled duration variables. For the random cuts, refine the instruction by fixing their numbers and assigning the remaining absolute cut coordinates of each protected package to its own small deterministic windows. Positive gaps allow disjoint windows, with no random cut shared by packages in disjoint spatial buffers. Shared blocks or cuts belong to one aggregate package. Durations and internal relative cuts are derived from these fixed and free absolute coordinates.

Search origins are fixed by the instruction or drawn independently of the free package cuts in separate deterministic gap windows. Independently of the word, draw the random cut and search-origin marks once, and draw each noise depth with its integer probability mass $\pi_n(H)$. Perform the backward guard and noise searches and restrict to fitting, disjoint configurations. There is no repeated sampling until marks fit a given word. The cut record retains all mark labels; if rounding or forgetting a label has several preimages, their masses are summed. Fix the exterior e , including the prefix, suffix, guard words, search origins, and discovered noise and guard endpoints. Discovered endpoints carry the stopping restrictions, not an additional independent cut probability. The free ordinary-cut coordinates remain unfrozen variables θ ; prescribed deterministic cuts remain fixed. The notation θ also records the durations and relative cuts derived from these coordinates. Discovered noise endpoints determine auxiliary gap remnants, not additional fixed coordinates of a protected package. Write $\chi_n(e, \theta)$ for the exact mass of the independent cut/search marks, with their rounding multiplicities; factors depending only on e may be suppressed. The random noise depths are not included in this frozen exterior.

Here is the two-block finite-bridge arrangement, in forward order:

$$E_L N_1 G_1 U_0 W_1 U_1 N_2 G_2 U_2 W_2 U_3 N_3 G_3 E_R. \quad (6.10)$$

The N_i are noises, the G_i are frozen guards, and the U_i are ordinary gap or extension pieces. For this display split an extended protected block at its used start, placing its left extension in the adjacent U_i ; W_i runs from the used start to the end of the extended block, including its right extension. Together these two pieces still form one extended protected block. The suffix $G_3 E_R$ is frozen. Let x_0, x_3 be the prescribed heights after E_L, N_3 , respectively, and let x_1, x_2 be the free used-start heights. Figure 3 displays these extended blocks.

The conditional experiment has four groups of variables:

- (i) the frozen exterior e , including guards, search origins and discovered noise endpoints;
- (ii) the ordinary raw words and their duration/cut variables θ , with every left extension kept in its protected block;
- (iii) the forest value and the remaining full-height coordinates;
- (iv) the noise words, sampled conditional on the endpoints determined by the preceding variables.

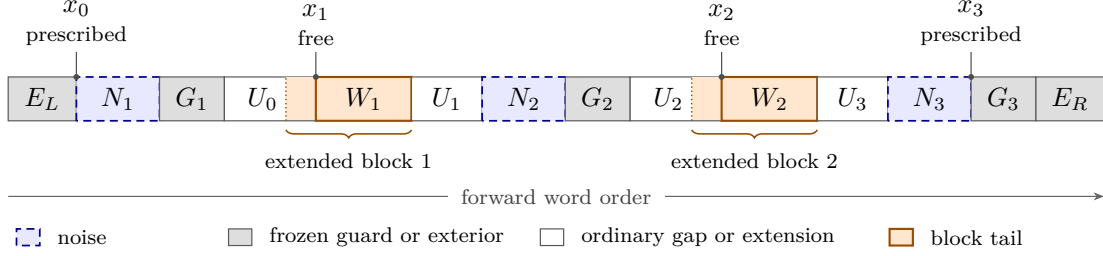


Figure 3: Forward word arrangement for two used blocks in (6.10). The shaded terminal portions of U_0 and U_2 are the left extensions: each and its adjacent W_i form one extended protected block, as indicated by the braces. Each W_i includes the block's right extension. The heights x_0, x_3 are prescribed; x_1, x_2 are free at the used starts. Segment widths are schematic.

The braces in the figure identify constituent time blocks. All blocks from one spatial buffer, together with their forest data and joint readouts, belong to the single package in (6.1).

For the fixed surrogate typing rule described below, set

$$u_1 = d(G_1 U_0), \quad u_2 = d(W_1 U_1 G_2 U_2), \quad u_3 = d(W_2 U_3).$$

All these are functions of the ordinary raw words w , the guards, and the fixed instruction, independent of the free heights and noise paths. The forward noise displacements are exactly

$$\Delta_1 = x_1 - x_0 - u_1, \quad \Delta_2 = x_2 - x_1 - u_2, \quad \Delta_3 = x_3 - x_2 - u_3. \quad (6.11)$$

The first two equations are an affine triangular integer bijection; the third imposes the terminal displacement. In particular $H_i = \Delta_i^1$, while the second *backward* displacement is $-\Delta_i^2$. The noise lengths k_i are fixed by their recorded endpoints. General finite collections give the same formula $\Delta_i = x_i - x_{i-1} - u_i$, with one final equation only when the terminal height is prescribed. All even-cut and displacement parities are retained. In particular the feasible condition is $k_i - \Delta_i^1 + \Delta_i^2 \in 2\mathbb{Z}$.

The augmented reference and its exact weight. Use the same point masses for the deterministic absolute cuts of the instruction. Within each aggregate package, choose positive smooth reference densities for its free absolute cut coordinates in a product of the windows just chosen, independently between disjoint spatial buffers. Compute all durations and relative cuts from these coordinates; do not independently sample durations linked by a fixed cut. After the exterior endpoints are frozen, compute genuine gap remnants from the remaining intervals. The boxes are small enough to leave these remnants positive on the good neighborhood. Draw the ordinary raw words with their iid letter weights. The additional gap/cut laws give an exact joint mass $r_n(e, \theta)$. Alternatively a proposal outside the fitting region may be completed by a dummy remnant and assigned weight zero; it is not conditioned on successful placement. Thus the aggregate packages retain their independent reference marginals across disjoint spatial buffers. Free cut coordinates remain integrated, and instruction-fixed cuts retain their point masses. Thus the reference and actual mark laws have the same deterministic support; the ratio $\chi_n(e, \theta)/r_n(e, \theta)$ contains only the free-coordinate mass ratio after their identical deterministic indicators cancel.

Draw the raw forest value $v_F = P_F x$ with its exact normalized mass $\beta_n(v_F | \theta)$, corresponding to the smooth density on $s_b n^{-1/2} \Lambda_F$. On each full-height fibre draw x with a positive smooth lattice mass $f_n(x | v_F, \theta)$. A concrete choice is a normalized Gaussian in $s_b(x - S v_F)/\sqrt{n}$, where S is an integer-linear right inverse of $P_F : \Lambda_m \rightarrow \Lambda_F$. Such an S exists because its image is a free

abelian group. Its fibres are translates of the fixed kernel lattice, so this choice has the required smooth limiting conditional density and constant coarea factor.

Compute (6.11). At feasible endpoints sample independent noise words with laws $\mathbf{B}_{H_i, k_i, -\Delta_i^2}$. At any impossible endpoint use a fixed dummy probability law and give it weight zero. This avoids conditioning the protected reference words on endpoint feasibility. The resulting probability measure is R_n^e .

Initially let A_n include all actual restrictions: fitting and cut instructions, the selected tube, admissible depths, true flexible resolutions, and, in the finite case, empty reduction. The exact weight for (6.10) is

$$W_n^e = \frac{\chi_n(e, \theta)}{r_n(e, \theta) \beta_n(v_F | \theta) f_n(x | v_F, \theta)} \prod_{i=1}^3 \left[\pi_n(\Delta_i^1) p_{\Delta_i^1}(k_i, -\Delta_i^2) \right] \mathbf{1}_{A_n}. \quad (6.12)$$

It is zero off actual support. The same expression has one factor per noise in the general case.

To verify the identity, multiply (6.12) by an atom of R_n^e . The cut, forest, and fibre reference masses cancel. Each noise endpoint mass cancels the denominator of its conditional word law, leaving $\mathbf{p}(N_i)$ with its intrinsic ladder restriction. The remaining ordinary-word product is already its iid weight. Frozen guards contribute the fixed factors $\mathbf{p}(G_i)$, which cancel upon conditioning. Their backward-discovered stopping locations are exactly their intrinsic $\mathcal{L}_H$ restrictions, so no further stop-selection factor is missing. What remains is the actual independent cut/depth marking mass times the raw-word product and $\mathbf{1}_{A_n}$. Therefore the actual marked conditional law is exactly

$$Q_n^e = \frac{W_n^e}{R_n^e[W_n^e]} R_n^e.$$

For the finite bridge its original empty-word normalizer also cancels. No asymptotic assertion is used in this finite calculation.

Removing microscopic admissibility from the weight. Internally reduce each guard together with its following ordinary stretch. Give every still-unresolved flexible order the type assigned by a fixed finite rational comparison partition. This is a deterministic rule of the ordinary raw letters, their time slots, and the instruction, fixed before sampling heights or noise words. The partitions are refined with those for the protected used portions. In particular the u_i in (6.11) are literal functions of w, e, θ , not functions of x or the noise paths.

An order still unresolved after that internal reduction has both candidate suppliers before the guard's start. A later available burger would instead have resolved it internally. The positive guard duration gives both candidates a positive backward span. On a compact supplier window, Lemma 5.4 excludes equal supplier endpoints and branch ambiguities; its compactness argument yields the finite comparison partition with strict chronological clearance. Choose the contour tube small enough to preserve those comparisons. If a proposed surrogate configuration passing that tube had a first wrong flexible type, its preceding contour would be the true contour. Its two actual supplier records would then have the prescribed strict chronological order, forcing the prescribed type, a contradiction. This induction applies to every admitted raw configuration, including independently sampled reference words. Internal matches agree automatically. The protected-used type lists obey the same buffered argument, so their readouts agree with the actual incidence record on the tube.

For the finite bridge, choose positive integer floors ℓ_1, ℓ_2 below the prefix counts and below all mutable contour heights. The bottom ℓ_c prefix burgers of each type, including their chronological

interleaving across types, are then preserved by every admitted mutable word. Impose the same exact endpoint counts at the last noise. Choose a fixed suffix cut $t_c < T$ where both original counts are below their respective floors. Such a cut exists by continuity at the excursion endpoint, and the portion up to t_c retains positive interior margins. Internally reduce the final guard together with the entire following suffix segment up to t_c , and apply the same fixed comparison partition to its unresolved flexible orders. Their candidate suppliers precede the guard's start, so its positive duration and the first-error argument give the correct suffix types up to this cut.

Here is the exact remaining inventory invariant. Two mutable words may leave different burgers above the protected floors, but have the same counts. Along the common typed suffix their counts stay equal. When a type's count first reaches its floor, all potentially different survivors above that floor have been removed by the last-in-first-out rule. Subsequent survivors of that type are common protected-prefix burgers and identically created suffix burgers. At t_c this has happened for both types. Their identities and creation order agree, so the entire chronological inventories agree. The rest of the raw suffix consequently evolves identically, including absent-candidate cases, and has the original empty reduction. Equality of high-rank burger identities before t_c is not asserted or needed.

Thus, on this good chart, the true-resolution and empty-reduction parts of A_n are automatic. Its remaining conditions are the specified cut, depth, positivity, fitting, and uniform-tube tests, chosen with continuum continuity boundaries. There is no residual microscopic freshness or success-normalization factor.

The reference limit and its explicit weight. Start with the aggregate package limits (6.1). Their input l contains the relative contour of every complete extended block, including the portions called left extensions in the display. Its cut coordinates are within-block offsets. Keep each block's remaining absolute placement, all omitted-gap durations, the scaled full heights $x/\sqrt{n}$, the genuine gap paths and the noise paths in an auxiliary variable A . Whole-contour tests will be read from the assembled paths and are retained in A as well.

Conditional on its duration and internal cut offsets, each iid raw block and its time-labelled readout are invariant under a common translation of that block's time indices. The aggregate readout uses only these relative block data, the forest parameter and the fixed instruction. Thus the remaining placements reveal no additional output information. Disintegrate the absolute-cut reference law into the retained relative cut data and those placements. On an interior good cell this is an affine change of the free cut coordinates, with the deterministic atoms retained. Its smooth lattice densities and fixed affine fibres give a limiting conditional placement law through the input alone; no continuity of the output kernel is required.

For a fresh ordinary word, two choices of its externally unresolved flexible types change its contour by at most the number of flexible orders unidentified by internal reduction. The reduced-word maximal and unidentified-order estimates [42, Lemma 3.13] [23, Lemma 3.7] make this error $o_{\mathbb{P}}(\sqrt{n})$, uniformly on the finitely many fresh pieces. The bound does not depend on the external inventory or on its chosen types. A ladder guard is intrinsically resolved. On the good tube the frozen typed pieces agree exactly with their actual contours by the first-error and protected-floor arguments above. Their rescaled paths are part of the frozen macroscopic record e . Consequently each $u_i/\sqrt{n}$ in (6.11) is, up to $o_{\mathbb{P}}(1)$, a sum of increments from l , fresh gap paths, and e . There is no residual dependence on a microscopic exterior inventory.

Fresh gap paths are appended through Brownian increment laws. The fibre Gaussian is an input-dependent kernel with a smooth limit. Conditional noise paths have the compact-uniform bridge limits of Theorem 6.4, at endpoints determined by the preceding inputs. A backward bridge

B of duration k gives forward contour $B(k - t) - B(k)$, a continuous path transformation. Thus, on each interior compact parameter set, these additions converge through a kernel $T_e(l, dA)$. The complete reference limit has form

$$\left[\prod_{\alpha} \mathcal{R}_{\alpha}(dl_{\alpha}) \right] T_e(l, dA) \prod_{\alpha} K_{\alpha}^0(l_{\alpha}, dO_{\alpha}^0). \quad (6.13)$$

This is the reference factorization required in [Theorem 6.2](#). The same construction retains each bounded continuous test of the assembled whole contour. It does not assert independence between the time blocks within a union buffer.

On interior compact parameter sets, $H_i/\sqrt{n} = a_i > 0$, $k_i/n = t_i > 0$, and $-\Delta_i^2/\sqrt{n} = y_i$ give

$$p_{H_i}(k_i, -\Delta_i^2) = 2n^{-3/2} a_i^{-3} g(t_i/a_i^2, y_i/a_i) + o(n^{-3/2})$$

uniformly on the feasible parity coset. The depth masses are $n^{-1/2}$ times their positive limiting density. The reference cut, forest, and fibre masses have their exact lattice normalizers. The identical deterministic-cut masses in the actual and reference laws have already canceled. Conditional on the fixed cut counts, the actual joint Poisson mark density and the positive reference density on the free-coordinate boxes have a bounded smooth ratio on interior compact sets; the actual mark density need not factor between packages. After their common powers of n are removed, these free-coordinate masses and the forest and fibre masses converge to their positive smooth densities. Fixed durations contribute no duration-density factor; the ladder estimate above is uniform also when such a duration is held at a prescribed positive value. The factors in (6.4)–(6.7) record every fixed height Jacobian. Thus (6.12), after canceling a common exterior-only factor, is bounded and converges jointly to a weight depending only on macroscopic inputs. The translations $u_i/\sqrt{n}$ have that same property by the surrogate contour estimate for fresh pieces and the retained macro paths for frozen ones. Lemma 6.4 gives compact-uniform convergence of the conditional noise paths. The aggregate fresh-package laws are independent of the microscopic exterior, the frozen rescaled paths enter only through e , and the noise limits are uniform on these compact parameter sets. Hence the factorization and weight convergence hold along every convergent sequence of good microscopic exteriors, not only at each fixed exterior value.

If an unused full height is integrated out, its denominator $f_n(x \mid v_F, \theta)$ in (6.12) must first be included. Fibre summation in the weighted joint measure then cancels that sampling density; its lattice Riemann sum has constant coarea. Whole-contour tests and their conditional bridge factors remain inside this integration. Equivalently keep the scaled height in A through Lemma 6.2 and integrate it afterward. The protected output depends on full heights only through v_F , so neither procedure introduces an output-dependent fibre factor. Random cut coordinates likewise remain integrated input variables throughout, while deterministic cuts remain fixed parameters.

The limiting normalizer is positive. Realized depths and durations are positive, and the strict tube contains a product subtest open in the free cut coordinates and in the free Brownian pieces and first-passage bridges. Deterministic cuts are held fixed in this subtest; if there are no free cuts, their factor is a point mass. The positive free-coordinate and endpoint densities and path support give positive weight, stably for nearby exterior records. Lemma 6.2 now applies. Finally exhaust under the actual contour law with its once-drawn independent marks: almost every realization has a finite protected cover, positive gaps, the proved local instructions, and noises fitting for sufficiently small raw depth a . Finite collections of good chart neighborhoods and interior compact cutoffs cover probability at least $1 - \varepsilon$. Their omitted contribution to a bounded conditional-identity test is at most its bound times ε ; let $\varepsilon \downarrow 0$. All whole-contour observations are retained in A , completing the proof. $\square$

6.5 Proof of the local-kernel theorem

Proof of Theorem 6.1. Use the aggregate kernels K^0 constructed in (6.1). By Theorem 6.5, their augmented reference law has form (6.13), and its limiting change of density depends only on the retained inputs. Applying Theorem 6.2 therefore preserves these kernels under the actual word law. The recovery $l = l(I)$ and the spatial relabelling described above give $K(I, dO)$. Multiplying a reference law by a positive smooth function of its input and normalizing leaves its conditional output law unchanged. Thus changing the auxiliary densities, and integrating parameters unused by the readout, identifies the kernels on overlapping input laws.

For a finite family of disjoint buffers, the aggregate reference packages, including their full inputs and output arrays, are independent. Their joint limits are products. Apply Lemma 6.5 to their union: the limiting resampling density depends only on the continuum input, so Lemma 6.2 preserves that conditional product. The same calculation applies in the finite bridge. Tests of the whole contour can be kept in the calculation; a monotone class of bounded continuous tests and the determination in Theorem 3.4 identify conditioning on $\mathcal{C}$. No additional information in $\mathcal{C}$ changes the kernel. Finite-dimensional products determine the assertion for countable arrays.

Compatibility is obtained before limiting. When two charts describe the same finite test, their readouts are restrictions of the same union incidence record and coincide on its good event for all large indices. The good-event exhaustion in Lemma 6.5 transfers this equality to the limits. Countably many such equalities hold simultaneously, and remain so under an absolutely continuous change of input law. This constructs one compatible array, not a collection of unrelated conditional versions.

For bounded changes of field density, multiply joint continuous tests by the density, first approximating that density in probability by bounded continuous functions of the input. Truncate an integrable density to remove boundedness. This proves the third assertion. Local marks can be retained first and then integrated with their conditional laws. Independent marks in disjoint regions preserve the conditional product.

For fixed offsets include their scaled versions in the same aggregate reference experiment before extraction. Its finite joint laws retain the actual correlations between these offsets; independence is used only between disjoint spatial buffers.

A measurable continuum functional has its fixed joint law with the input. On a set of input probability at least $1 - \varepsilon$, Lusin approximation replaces any bounded test of that functional by a continuous one. Include these approximations in the extraction and let ε tend to zero. Finally, variables sampled conditionally on $\mathcal{C}$, independently of the discrete outputs, can be appended by disintegration. This proves the final assertion. $\square$

6.6 Order and ordinary-field transfer

Lemma 6.6 (Finite traversal order). *For a compact protection K in a bounded open real buffer V , only finitely many traversals of V meet K , almost surely. After retaining their local directed pieces, their remaining global order is a finite permutation. Conditional on the local imaginary input J , write*

$$w_o(J) = \mathbb{P}[\text{order} = o \mid J].$$

Its feasible values are those with $w_o(J) > 0$; the actual order is feasible almost surely. These weights depend neither on the real field nor on the choice of quantum time unit.

For every finite protected test, the necessary order is determined by the imaginary field in a sufficiently large finite buffer, on events whose probabilities tend to one as that buffer increases.

The number of traversals can also be bounded deterministically on an exhaustion of probability tending to one.

The kernels can be augmented to include an output for every feasible order, simultaneously for all retained fixed scale offsets. Given the entire real and imaginary fields, the law of this augmented vector depends only on their respective local restrictions; such vectors have product laws in disjoint buffered regions. One component is the actual output at the actual order.

Proof. Properness places all visits to K in a compact time interval. Uniform continuity there supplies $\delta > 0$ such that moving between K and V^c takes at least δ units of time. Each distinct traversal meeting K contains such a move, so only finitely many occur. Lemma 5.1 reads their directed pieces in any strict imaginary enlargement. They can be enumerated measurably by taking the first representative in the fixed dense-point enumeration defining the local flow-line construction. The global curve order is determined by the imaginary field before it is parameterized by quantum area. Hence its conditional probabilities given J have the stated independence of real-field and clock choices. For a finite conditional distribution, $\mathbb{P}[w_O(J) = 0] = 0$, which proves feasibility.

For completeness, order determination can be witnessed in a finite region. For the finitely many representatives, trace their left and right frontier flow lines until the corresponding same-angle branches have merged. The whole-plane flow-line tree construction of space-filling SLE gives these mergers at finite times almost surely; the resulting finite trees and their cyclic sides determine the relative Peano order [35]. Their compact traces lie in a finite disk. Until exit from a slightly larger disk they are local sets determined by the imaginary field there, so the witnessed order is determined there as well. For increasing deterministic disks, the probability that all of these finitely many witnesses are contained tends to one. Intersecting with deterministic bounds on the number of representatives proves the exhaustion assertion.

By Theorem 6.1, conditional on the whole fields and order o , the actual local output array has a kernel $K_o(I, dO)$, where I now omits the order. This is defined for almost every local input with $w_o(J) > 0$. Keep the actual component and, conditionally on the fields and that component, sample each of the other feasible components independently from its kernel. The entire vector then has conditional law $\prod_{o:w_o>0} K_o(I, dO_o)$, which is independent of the actual order and depends only on local restrictions. The same construction uses the joint array at all retained fixed scale offsets, so it does not alter their within-order consistency. Conditional products between disjoint regions follow from Theorem 6.1, using independent auxiliary fills. This proves the last assertion. $\square$

Lemma 6.7 (A single traversal near a simple visit). *Suppose z has a unique exploration preimage. For every bounded neighborhood V of z , all visits to some smaller neighborhood U of z lie in one traversal of V . Protected observations therefore require no permutation input near z ; their kernels are measurable from the real and imaginary restrictions in every fixed sufficiently small surrounding buffer.*

Proof. Let t be the unique preimage, and let (a, b) be the component of $\eta^{-1}(V)$ containing t . If every neighborhood of z had a visit outside (a, b) , properness would give a convergent sequence of such times. Its limit would be a preimage of z outside (a, b) , contradicting uniqueness. Thus all visits to some U lie in that component. Within a single directed traversal, order is its own direction; labelled length differences are intrinsic to its local frontiers. The local measurability assertion follows from Lemma 5.1 and Theorem 6.1. $\square$

Proposition 6.8 (Field and unit changes). *On strict buffered compact subsets away from marked singularities, ordinary GFF restrictions, cone restrictions in an appropriate fixed chart, and their*

finite smooth shifts are locally mutually absolutely continuous. Restrictions on finitely many separated buffers are jointly equivalent to independent ordinary restrictions. The same statements hold for the independent imaginary field modulo its angular period, using its uniform phase. These changes transfer the kernels of Theorem 6.1, their products, and the all-order vectors of Lemma 6.6.

The kernels are intrinsically covariant under affine coordinate changes. For $\psi(z) = az + b$, $a \neq 0$, transform simultaneously

$$h \mapsto h \circ \psi + Q \log |a|, \quad \tilde{h} \mapsto \tilde{h} \circ \psi - \chi \arg a, \quad \chi = 2/\gamma - \gamma/2.$$

Transform ports and ranges by ψ^{-1} . Quantum durations, frontier lengths, raw graph costs, and flag-surface extremal lengths are unchanged. In particular translations preserve every chosen deterministic cost unit.

There is also an exact indexed change of quantum units. If the original time unit is n and the cost denominator is c_n , express the same map using time unit vn and cost denominator $c_n v^p$. Its field input, time labels, frontier lengths, and costs become, respectively,

$$h_v = h - \gamma^{-1} \log v, \quad t_v = v^{-1} t, \quad \ell_v = v^{-1/2} \ell, \quad O_v = v^{-p} O. \quad (6.14)$$

These identities hold jointly for the indexed kernel arrays. They also hold with $h + b$ in place of h . They assert a re-expression of the same array, not equality of two differently indexed kernels. For any other deterministic output denominator $d_{n,v} > 0$, the last factor is instead its literal ratio $c_n/d_{n,v}$; in particular extremal length divided by the same denominator in both conventions has no additional power of v .

Proof. Take larger disjoint disks around the compact buffers, with a pin outside them. Conditional on the exterior GFF, the fields in the disks are independent zero-boundary GFFs plus harmonic functions. Each harmonic function on a strict inner buffer has a smooth finite-energy extension supported in its disk. Conditional Cameron–Martin equivalence and then integration show that the joint restrictions are equivalent to independent zero-boundary restrictions. For the cone comparison, choose the normalization tests strictly inside its normalization circle and their buffers away from its marked center. There the field is an ordinary GFF plus the logarithmic drift, which has a finite-energy cutoff on the buffers. The same argument applies there and to its coordinate images. Finite smooth shifts are directly Cameron–Martin changes. For the imaginary field use its Markov decomposition modulo the angular period, with an independent uniform phase [35, Propositions 2.8, 2.9 and 2.11]. The argument applies to its harmonic part too. Densities can be truncated and integrated as in Theorem 6.1.

For affine covariance, express one intrinsic chart in its two coordinates. The quantum coordinate rules identify its area and labelled boundary lengths, and imaginary geometry identifies its directed frontiers. The retained raw incidence record and triangle shapes are literally the same; only their spatial labels change. Consequently their finite observations agree before extraction. Taking joint limits proves covariance of the kernels. The uniform phase convention makes translations and positive dilations exact symmetries of the unlocked whole-plane imaginary input. Absolute continuity alone would not prove this covariance; here it follows from the intrinsic finite readouts.

Finally, dividing quantum area units by v subtracts $\gamma^{-1} \log v$ from the field and divides boundary-length units by $\sqrt{v}$. Changing the denominator by v^p gives the last identity in (6.14) exactly on the same finite map. For integer time units use $v_n = \lfloor vn \rfloor / n$ throughout this identity and retain the same raw cuts. Then $v_n \rightarrow v$. If Poisson cuts are used, transform their intensity by the same clock change; include all required intensities in the countable atlas before extraction. Thus there is no endpoint-rounding error measured in an arbitrary graph-cost unit. Passing to the

joint indexed limit proves the formula. No relation between c_n and $c_{[vn]}$, or scaling rule for an unindexed passage limit under addition to h , is used. $\square$

6.7 A diagonal with varying volume labels

Proposition 6.9 (Fixed normalized inputs in a varying-label diagonal). *Let $v_r > 0$ be deterministic volume labels. For each r , take an unconditioned joint law supplied by Theorem 6.1, re-expressed in the units of (6.14). Suppose its base field in those units has one fixed ordinary normalized reference law, including the same imaginary-field law with uniform phase. Draw the observation marks using the same normalized convention at every row. A joint diagonal may retain every fixed relative offset in a countable list, provided every retained effective microscopic time unit tends to infinity. The output denominators may be arbitrary deterministic positive numbers. The resulting limits have the compatible kernels and the conditional product property of Theorem 6.1, and may retain fixed measurable continuum functionals and the all-order arrays of Theorem 6.6.*

Proof. Fix the ordinary normalized reference input h_* and its imaginary input with uniform phase. Choose common smooth auxiliary forest and free-cut laws for the reference experiments, keeping each instruction's deterministic cuts fixed. The density invariance in Theorem 6.1(3) allows this choice without changing the resulting actual-law kernels or prescribed observations. The observation marks themselves use the fixed normalized convention in the statement. Independent reference restrictions are used in the disjoint enlarged buffers. At label v_r , sample h_* in that label's unit convention. In the original convention this is $h = h_* + \gamma^{-1} \log v_r$; the indexed unit identity (6.14) then gives $h_{v_r} = h_*$. Thus the label changes the effective microscopic time and cost units, while the base normalized input has a fixed law. Write P_r for these unconditioned row laws and ν for their common marginal of continuum inputs and once-drawn marks. Thus $\pi_{\text{input}} P_r = \nu$ for every r .

At stage r retain the first r coordinates and relative offsets $w_1, \dots, w_r$. First hold r, v_r fixed. The fixed-label construction supplies microscopic approximants for that row's unconditioned joint law, including its protected reference experiment and reference metric observations. Choose an approximant along the sequence defining that row, with base integer scale n_r , so that the first r bounded joint tests have error at most 2^{-r} and

$$N_r = n_r v_r \longrightarrow \infty, \quad n_r v_r \min_{j \leq r} w_j \geq r.$$

Include the corresponding protected-density tests among those approximated. For integer-time conventions keep each raw cut and readout exactly and round only the unit used to scale its labels. Since the smallest unit tends to infinity, this changes the continuum labels by $o(1)$; it changes no edge or path cost.

Re-expressed in the effective unit N_r , these approximants form one raw-word sequence with a fixed continuum reference input law and arbitrary deterministic output denominators $c_{n_r} v_r^p$. For any fixed finite list of offsets, its joint field inputs are $h_* - \gamma^{-1} \log w_j$, a law independent of r . Apply Lemma 6.5 directly to this sequence. Its durations, forest variables, guards, noise depths, and ladder limits are all expressed in N_r units; in particular its raw depth window is proportional to $\sqrt{N_r}$. Its reference densities are fixed, and its compact good-chart normalizers have the positive limits proved there. The diagonal macro-input marginals converge to ν . Its actual-law good-chart exhaustion is therefore uniform along this sequence: first fix finitely many ν -continuity good cells of total mass at least $1 - \varepsilon$, then let r tend to infinity, and finally let ε decrease to zero. A finite union handles any fixed finite list of buffers and offsets. The exact

resampling identity and independent complete reference packages therefore prove the conditional product for this diagonal directly. Summable joint test approximation identifies the result with the desired limit of the unconditioned row laws. Subsequent fixed locally absolutely continuous field changes use [Theorem 6.1\(3\)](#). No assertion is made here for arbitrary varying field densities.

The fixed measurable functionals are retained by the Lusin argument in [Theorem 6.1](#), now under the common input marginal ν . The finite-order augmentation uses that same fixed imaginary marginal and the joint retained array. Thus it also applies to the diagonal, without changing any actual-order coordinate. $\square$

6.8 Normalized sphere heights and selected changes

Lemma 6.10 (Actual-height sphere densities). *Use the sphere chart with the exploration root at infinity and two further area marks at 0, 1. On a finite union K of separated closed disk patches avoiding the marks, the restriction of the normalized sphere field is dominated by the corresponding ordinary real-field restriction. Its density is strictly positive precisely on the ordinary restrictions for which $\mu_h(K) < 1$. Boundaries may be chosen to have zero mass. The joint law, including imaginary restrictions on separated strict buffers, is therefore equivalent on this mass event to independent ordinary real and imaginary restrictions on the patches.*

Together with their common conditional kernels, the same domination and positivity hold for any finite or countable protected output arrays in the actual finite-sphere subsequential law. Every patch keeps its actual additive height and its original deterministic cost and time units.

Proof. By [Proposition 3.5](#), before normalization write the field as $h_0 + d$, where h_0 is an ordinary whole-plane GFF with a pin, d is the deterministic marked drift, and the law has a strictly positive integrable tilt. Normalization gives

$$h = h_0 + d - \gamma^{-1} \log \mu_{h_0+d}(\widehat{\mathbb{C}}).$$

Repin outside K first. The change is an additive random constant, which cancels in this normalization; its effect on the original tilt is another positive integrable density. We may therefore use an exterior pin throughout the following Gaussian calculation.

Choose a smooth finite-energy g with $0 \leq g \leq 1$, equal to one precisely on K , less than one off K , and zero on an open set near the pin. It can be compactly supported away from the marks. Orthogonal Gaussian decomposition in the direction g gives $h_0 = h_\perp + Xg$, with X a nondegenerate Gaussian independent of $h_\perp$. Conditional on $h_\perp$, set $\nu = \mu_{h_\perp+d}$. This is a finite measure of full support, and its boundary mass on the chosen circles is zero. On K the normalized field is $h_\perp + d + c(X)$, where

$$c(x) = x - \gamma^{-1} \log \int e^{\gamma x g} d\nu.$$

Differentiation under the integral gives

$$c'(x) = 1 - \frac{\int g e^{\gamma x g} d\nu}{\int e^{\gamma x g} d\nu} > 0.$$

The inequality is strict because the exterior set where $g < 1$ has positive mass. Since $\nu(\{g = 0\}) > 0$, $c(x) \rightarrow -\infty$ as $x \rightarrow -\infty$. Also

$$\int e^{\gamma x (g-1)} d\nu \longrightarrow \nu(K) \quad (x \rightarrow \infty),$$

by dominated convergence, so $c(x) \rightarrow -\gamma^{-1} \log \nu(K)$. Thus c is a continuously differentiable increasing bijection onto that interval, with strictly positive derivative.

Compare the joint variables $(h_\perp, c(X))$ with $(h_\perp, X)$. Conditional change of variable gives a density which is finite and strictly positive exactly when $e^{\gamma y} \nu(K) < 1$. This event is the measurable condition $\mu_{h_\perp + d + y}(K) < 1$ on the resulting patch field. Pushing forward therefore proves equivalence of the normalized patch law with the unnormalized $(h_0 + d)|_K$ law restricted to that event. The initial positive tilt preserves its null sets and positivity. A finite-energy cutoff of d away from the marks, and Proposition 6.8, replace the unnormalized restrictions by ordinary ones, jointly on all patches. This proves the real-field statement at actual heights.

The unlocked imaginary field is independent of the sphere real field in this chart, by Proposition 3.5. Its separated restrictions are equivalent to independent ordinary restrictions by Proposition 6.8; enlarged imaginary buffers may be chosen before this comparison. Finally, Theorem 6.1 attaches precisely the same conditional output kernels on both sides. Multiplying a joint input law by a strictly positive density multiplies the input–output law by that same density. This proves the final assertion for the actual sphere array, including its products and all-vertex escape readouts. $\square$

Lemma 6.11 (Dominated transfer after finite selection). *Let the real input have one fixed ordinary pinned Gaussian law, and let f_j be a finite list of deterministic smooth functions in its Cameron–Martin space. In particular, for a whole-plane field whose circle-average pin is fixed outside a bounded region, any smooth recipe supported in that region is admissible. The pin and its law are chosen before the auxiliary draws or any event selection. For recipe j , suppose $f_j = b_j$ on a preservation buffer V_j and $f_j = 0$ outside a larger support buffer W_j . Before selecting a recipe, sample an auxiliary local output from the kernel with input $h + b_j$ in V_j , independently of the original outputs outside W_j , conditional on the fields. Use the same indexed cost and clock array, re-expressed by (6.14) when necessary.*

For each recipe, the retained auxiliary interior and original exterior can be completed to an output with the common conditional law for $h' = h + f_j$. Write

$$L_j(h') = \frac{d\text{Law}(h + f_j)}{d\text{Law}(h)}(h').$$

After restricting the source to $L_j(h + f_j) \leq L$, the image of any further source event, including a hit event and a rule selecting recipe j , is a submeasure bounded by L times the common output law. Summing over the finite list gives the bound $(\#j)L$. No independence after selection is required.

Proof. Condition on the whole fields. On V_j , the auxiliary output has exactly the kernel required by $h' = h + f_j$; outside W_j , the original exterior has exactly the unchanged kernel. Their product is the corresponding marginal of the common output array, by Theorem 6.1. Countable exterior restrictions are obtained by exhausting strict buffers outside W_j . Disintegrate the full common array given these retained restrictions and sample the remaining coordinates from that conditional law. This preserves the auxiliary interior and old exterior exactly. Local Poisson marks may be resampled with the correct conditional area law in the changed region and retained outside. The order is unchanged, since the imaginary field is unchanged.

Before restriction or selection, the resulting pair has law

$$\text{Law}(h', O') = L_j(h') \mathbb{P}_{\text{out}}(dh', dO').$$

The Cameron–Martin formula for this fixed pinned law gives

$$L_j(h') = \exp\{(h', f_j)_\nabla - \tfrac{1}{2} \|f_j\|_\nabla^2\}, \quad L_j(h + f_j) = \exp\{(h, f_j)_\nabla + \tfrac{1}{2} \|f_j\|_\nabla^2\}.$$

Here the pairing is the Gaussian linear functional. The second formula expresses the likelihood cutoff directly in the source field. Clipping this likelihood to L bounds the entire image measure by $L\mathbb{P}_{\text{out}}$. Restricting the source to any further event can only decrease its image measure. This remains true when that event depends on the original interior, auxiliary draws for other recipes, a minimizing path, or a selected hit. The discarded source coordinates are not required to be independent of the retained ones. Summing these measure inequalities proves the finite-list bound. For compactly supported smooth f_j , the Gaussian pairing is measurable in its support buffer and has variance $\|f_j\|_{\nabla}^2$. The clipping events can consequently be included among the fixed continuum tests before selection. $\square$

7 Simultaneous annuli at prescribed volume scales

A local estimate with probability close to one must work along every macroscopic route. Its volume label and its Euclidean radius are different parameters. This section matches them using circle averages and obtains a linear number of successful, distinct radii around every point. All selections of labels are made before revealing the fresh fields in the selected shells.

Write $A(a, b) = \{z : a < |z| < b\}$, and fix three shells

$$E = \overline{A(1/8, 4)}, \quad E^+ = A(1/16, 8), \quad U = A(1/32, 16). \quad (7.1)$$

The readouts below depend on field restrictions and local kernels in E ; U is the fresh-field buffer. Any fixed finite collection of strictly buffered shell geometries can be placed inside shells with these roles by changing their fixed radii. Constants are then allowed to depend on that collection.

The reference law consists of a zero-boundary real GFF on $B(0, 64)$ and an independent whole-plane imaginary GFF modulo its angular period, restricted to E , with the local kernels of Theorem 6.1. The logarithmic covariance normalization is $\text{Cov}(h(z), h(w)) = \log|z-w|^{-1} + \text{smooth}$. The Dirichlet inner product is $(f, g)_{\nabla} = (2\pi)^{-1} \int \nabla f \cdot \nabla g \, dz$. An imaginary representative has a circle-average phase in a fixed interval of length equal to the period. Subtracting this circle average leaves the usual centered Gaussian differences.

7.1 Uniform comparison on a strict interior

Lemma 7.1 (Conditional transfer). *Let $h^0, \tilde{h}^0$ be independent zero-boundary GFFs on U . Add functions $g, \tilde{g}$ harmonic on E^+ , with*

$$\|g\|_{L^2(E^+)} + \|\tilde{g}\|_{L^2(E^+)} \leq H,$$

and bounded real and imaginary constants. On E , retain any fixed family of the same local output kernels under both field laws. There is a function $\omega_H : [0, 1] \rightarrow [0, 1]$, with $\omega_H(\varepsilon) \rightarrow 0$ as $\varepsilon \downarrow 0$, such that every event of reference probability at most ε has probability at most $\omega_H(\varepsilon)$ under this shifted fresh law. Conversely, for each $\eta > 0$ there is $p(H, \eta) > 0$ such that every event of reference probability at least η has shifted fresh probability at least $p(H, \eta)$. The constants are uniform in the volume label and in the measurable event. They depend on a bound for the two added constants.

Proof. Choose a smooth cutoff supported in E^+ and equal to one near E . For a harmonic function g , its product with this cutoff is an extension in $H_0^1(U)$, equal to g on E , whose Dirichlet norm is at most $C\|g\|_{L^2(E^+)}$. Indeed, multiply $\Delta g = 0$ by $\chi^2 g$, integrate by parts, and use $2ab \leq a^2/2 + 2b^2$ to bound $\int \chi^2 |\nabla g|^2$ by $4 \int g^2 |\nabla \chi|^2$; use nested cutoffs for the product extension. Constants have

extensions of bounded norm by the same cutoff. Apply this to both fields in the product Dirichlet space.

If P_f is the fresh law shifted by such an extension f , and P_0 the unshifted fresh law, Cameron–Martin and Cauchy–Schwarz give, for every event A ,

$$P_f(A) \leq e^{\|f\|_{\nabla}^2/2} P_0(A)^{1/2}, \quad P_f(A) \geq e^{-\|f\|_{\nabla}^2} P_0(A)^2. \quad (7.2)$$

These inequalities still hold after adjoining the same conditional kernel: the likelihood ratio is a function of the fields alone.

Decompose the reference pair on U by the domain Markov property. Its harmonic parts, after separating the bounded imaginary phase, have almost surely finite $L^2(E^+)$ norms. Their extensions are independent of the two fresh parts. Choose a bound H_0 whose extension event has probability at least $1/2$. Integrating the second inequality in (7.2) gives

$$P_{\text{ref}}(A) \geq \frac{1}{2} e^{-H_0^2} P_0(A)^2.$$

The first inequality then bounds the target probability by a constant times $P_{\text{ref}}(A)^{1/4}$.

For the other direction, if $P_{\text{ref}}(A) \geq \eta$, choose H_η so that the reference extension exceeds H_η with probability at most $\eta/2$. Integrating the first inequality gives

$$P_0(A) \geq e^{-H_\eta^2} (\eta/2)^2.$$

The second inequality gives the asserted positive lower bound, for example $e^{-H_t^2 - 2H_\eta^2} (\eta/2)^4$, where H_t bounds the target extension norm. All bounds concern the field laws and therefore are independent of the output label. $\square$

There is one finite-order issue in applying this lemma. The unlocked imaginary input may allow several local traversal orders, whereas a marginal test uses the actual order.

Lemma 7.2 (All feasible orders). *For the local arrays of Lemma 6.6, events with actual-order failure probability at most ε , uniformly in their labels, admit augmented all-order events with failure probability at most $\omega(\varepsilon)$, where $\omega(\varepsilon) \rightarrow 0$. Success of the augmented event implies success for the actual order. Conditional on both whole fields, the augmented arrays in disjoint buffered regions have product laws.*

Proof. Let J be the local unlocked imaginary input. For each feasible order o let $w_o(J) > 0$ be its conditional probability and K_o its local output kernel. Lemma 6.6 identifies these weights and kernels: the weights and finite feasible set depend on J , independently of the real field and volume label. Adjoin the other order coordinates as in that lemma, retaining the actual coordinate, so that the conditional vector law is $\prod_o K_o$.

For any $\zeta > 0$, choose $J_0 < \infty$ and $\delta > 0$ so that, with reference probability at least $1 - \zeta$, there are at most J_0 feasible orders and all their weights are at least δ . This is an exhaustion of the full traversal list for the fixed protection, under one imaginary marginal law common to all labels. Neither the bound nor the minimum weight is asserted uniformly over all imaginary inputs. On this event, the conditional probability that some coordinate fails is at most

$$\sum_o K_o(\text{failure at } o) \leq \delta^{-1} \sum_o w_o(J) K_o(\text{failure at } o).$$

After integration, the all-order failure probability is at most $\zeta + \varepsilon/\delta$. Choose ζ first and then ε . The product assertion is the augmented product assertion of Lemma 6.6. The same argument works for a finite joint array of readouts in one buffer. $\square$

7.2 A Gaussian budget along displaced scale chains

Fix $\epsilon_0 = 2^{-13}$ and $s = \log 2^{16}$. At radius $r_j = e^{-t_j}$, $t_j = t_0 + js$, use the square grid $\epsilon_0 r_j \mathbb{Z}^2$. Given a finest-grid point y , let $c_j(y)$ be its nearest point of this grid, resolving ties deterministically. Thus $|c_j(y) - y| \leq \epsilon_0 r_j$. Set

$$U_j = c_j + r_j U, \quad H_j = h_{32r_j}(c_j), \quad (7.3)$$

where the subscript denotes circle average. These fixed choices ensure that the closures of U_j are disjoint: each smaller buffer, and its circle of radius $32r_j$, is strictly inside the hole of every larger buffer. The circle at the larger scale is outside its own buffer and all the smaller ones.

Work first with a zero-boundary GFF in a disk D , and with centers in a fixed compact subset of D . For sufficiently small largest radius, all displayed circles are inside D . Let $\mathcal{F}$ contain both fields outside $\bigcup_j U_j$. The domain Markov decompositions are

$$h = h_j^0 + \mathcal{H}_j, \quad \tilde{h} = \tilde{h}_j^0 + \tilde{\mathcal{H}}_j \quad \text{on } U_j.$$

The fresh fields are mutually independent and independent of $\mathcal{F}$; the harmonic functions and H_j are $\mathcal{F}$ -measurable. For an imaginary lift define $\tilde{H}_j$ by the same outer circle. Put

$$G_j(u) = \mathcal{H}_j(c_j + r_j u) - H_j, \quad \tilde{G}_j(u) = \tilde{\mathcal{H}}_j(c_j + r_j u) - \tilde{H}_j, \quad u \in E^+.$$

The imaginary differences are independent of the choice of lift.

Lemma 7.3 (Square budget). *There are constants C, Λ , depending only on the fixed shell geometry and s , such that the centered Gaussian vector*

$$Z = ((H_j - H_{j-1})_{1 \leq j \leq m}, (G_j, \tilde{G}_j)_{0 \leq j \leq m}) \quad \text{in} \quad \mathbb{R}^m \oplus \bigoplus_{j=0}^m L^2(E^+)^2$$

has covariance operator norm at most Λ and trace at most $C(m+1)$, once the largest radius is sufficiently small. The threshold for that radius may depend on the ambient compact set and disk. Consequently, when $m \leq C_0 N$, for each $R < \infty$ one can choose $A < \infty$ so that

$$\mathbb{P} \left[\sum_{j=1}^m (H_j - H_{j-1})^2 + \sum_{j=0}^m (\|G_j\|_2^2 + \|\tilde{G}_j\|_2^2) > AN \right] \leq e^{-RN}. \quad (7.4)$$

The bound is uniform in the finest-grid point y .

Proof. We verify the covariance assertion, including the effect of moving centers. Evaluation of $\mathcal{H}_j$ at $c_j + r_j u$ is the boundary pairing against harmonic measure $\nu_{j,u}$ on ∂U_j . For $u \in E^+$, its density on the two boundary circles, after scaling, is bounded uniformly, as are its derivatives on any slightly smaller fixed shell containing E . The charge representing $G_j(u)$ is $\nu_{j,u} - \sigma_j$, where σ_j is uniform measure on the circle of radius $32r_j$. It has total mass zero, bounded total variation, and support at distances comparable to r_j from y . The increment $H_j - H_{j-1}$ is represented by $\sigma_j - \sigma_{j-1}$, with the same properties at these two adjacent scales. Smooth circle densities make their logarithmic self-energies finite and uniformly bounded after scaling. The additive $\log r_j^{-1}$ cancels because the charges have mass zero. The ratio of the adjacent radii is fixed.

For two nonadjacent scales, the smaller charge has support of diameter $O(r_k)$, at distance comparable to r_j from the larger charge. Subtract the logarithmic kernel at one point of the

smaller support. Its gradient there is $O(r_j^{-1})$; the total-mass cancellation gives a covariance bound Cr_k/r_j . For increments, associate the charge to its larger of the two radii. Enlarging the constant to cover adjacent scales, every covariance block therefore has norm at most

$$Ce^{-s(|j-k|-2)_+}.$$

For the L^2 blocks, the same estimate follows from the uniform pointwise covariance bound and the fixed area of E^+ . For the smooth part of the Green function, cancellation in both variables bounds the block by $C_D r_j r_k$, with the larger adjacent radius used for an increment. Its operator norm is at most $C_D \sum_j r_j^2$; it is uniformly bounded once the largest radius is small. Pin terms for the imaginary whole-plane field are sums of a function of either variable and disappear against the zero-mass charges.

The row sums of the exponential block bound are uniformly finite. Cauchy–Schwarz, or the block Schur bound, gives the asserted operator norm. Integrating the diagonal covariance bound gives the trace estimate. This also constructs the profiles as Gaussian L^2 -valued variables.

If λ_ℓ are the covariance eigenvalues, for $0 < t < (4\Lambda)^{-1}$,

$$\mathbb{E}e^{t\|Z\|^2} = \prod_\ell (1 - 2t\lambda_\ell)^{-1/2} \leq \exp\left(2t \sum_\ell \lambda_\ell\right) \leq e^{C'(m+1)}.$$

Finite-dimensional approximation and monotone convergence justify the formula. Markov's inequality, followed by increasing A , proves (7.4). $\square$

7.3 Alignment and simultaneous coverage

A unit test at label $v > 0$, transplanted to center c and radius r , is evaluated with real field

$$h^{c,r,v}(u) = h(c + ru) + Q \log r - \gamma^{-1} \log v. \quad (7.5)$$

Its imaginary input is pulled back by the same affine map, with its phase modulo the period. Its local output is the corresponding array readout under the affine and relative-volume identities of Proposition 6.8. Thus its deterministic cost unit is part of the test. This definition allows arbitrary deterministic extremal-length units.

Theorem 7.4 (Aligned annuli). *Fix $\theta > 0$ and $M > 0$. For each integer N in any unbounded deterministic set, let*

$$I_N \subset [N, 2N] \cap \mathbb{Z}, \quad |I_N| \geq \theta N, \quad v_i = M4^{-i}.$$

Consider local unit events $\mathcal{G}_{N,i}$, $i \in I_N$, for the arrays of Theorem 6.1, on the fixed shell E . Their reference probabilities are evaluated under the single normalized ordinary field law specified above, at every label. The events may vary with N, i ; the local field law and full buffer geometry do not. There are constants

$$0 < a < B < \infty, \quad c > 0, \quad \varepsilon_* > 0$$

depending only on γ, θ and the fixed buffer geometries, with the following implication. If every $\mathcal{G}_{N,i}$ has reference probability at least $1 - \varepsilon_$, then in an ordinary field chart every compact set K has, almost surely for all sufficiently large admissible N , a collection of good transplanted tests such that:*

1. There are at most CN radii, all in $[e^{-BN}, e^{-aN}]$. At each radius the centers belong to a deterministic square grid of spacing $\epsilon_0 r$. At each fixed radius the buffers have overlap bounded by a fixed constant.
2. Every $x \in K$ has at least cN good tests at distinct radii r , with centers c_r satisfying $|x - c_r| < r/256$. In particular x lies with strict clearance in each central hole $B(c_r, r/128)$.
3. Each test has a label in I_N , is evaluated by (7.5), and succeeds for the actual traversal order. At most a fixed number of labels are used at any one center and radius; one may retain a single successful test there.

Under the ordinary Gaussian law, failure probabilities are bounded by $C_K e^{-c_K N}$ and hence are summable. The almost sure eventual conclusion persists under any fixed locally absolutely continuous change of the joint field law, with the same local kernels. It holds simultaneously over countably many compact sets and test families satisfying the stated hypotheses.

Proof. There are two different grid counts in the proof. We control coarse circle averages at the first and last radii on their own grids. The harmonic-profile and fresh-shell failures are then made to have arbitrarily large exponential rates, so that they can be summed over the much finer grid indexing every displaced scale chain.

Choose $2 < \beta < Q$ and set

$$a = \frac{\log 4}{4\gamma(Q + \beta)}, \quad B = \frac{4\log 4}{\gamma(Q - \beta)}, \quad t_j = aN + js, \quad m = \left\lfloor \frac{(B - a)N}{s} \right\rfloor.$$

All buffers are contained in a fixed compact neighborhood of K for large N . Use the grids and nearest-center chains above. The number of finest-grid points needed to cover K is at most $C_K e^{2BN}$.

First control the two endpoints on their own grids. The circle averages there have variance $t_j + O_K(1)$. There are at most $C_K e^{2t_j}$ centers on the grid at scale j . The Gaussian tail bound therefore gives, for $j = 0, m$,

$$\mathbb{P}[\text{some such center has } |H_j| > \beta t_j] \leq C_K e^{-(\beta^2/2-2)t_j} \leq C_K e^{-c_K N}. \quad (7.6)$$

Increasing C_K absorbs the bounded variance correction. On their complement, for every chain the exterior-measurable process

$$Y_j = \gamma(Q t_j - H_j) + \log M$$

satisfies $Y_0 < N \log 4$ and $Y_m > 2N \log 4$ for sufficiently large N . The fixed multiplier M changes only this last threshold.

Choose $R > 2B + 3$, and then the budget A in Lemma 7.3. On its good event,

$$\sum_{j=1}^m (Y_j - Y_{j-1})^2 \leq A_1 N \quad (7.7)$$

for a deterministic A_1 , since $t_j - t_{j-1} = s$ and $m = O(N)$.

Assign each $i \in I_N$ to the first j for which $Y_j \geq i \log 4$. Discard the assignment if $Y_j - i \log 4 > L$, where $L \geq \log 4$ will be fixed shortly. For one upward jump $\Delta = Y_j - Y_{j-1}$, the number of discarded lattice levels is zero unless $\Delta > L$, and in that case is at most

$$\Delta / \log 4 + 1 \leq 2\Delta / \log 4 \leq \frac{2\Delta^2}{L \log 4}.$$

The sum in (7.7) therefore bounds the number discarded by $2A_1N/(L \log 4)$. Choose L so large that this is at most $\theta N/8$. Every remaining assignment satisfies

$$|Y_j - i \log 4| \leq L, \quad (7.8)$$

and a single j has at most $K_L = \lceil 2L/\log 4 \rceil + 2$ such assignments.

Now choose H so large that $K_L A/H^2 \leq \theta/8$. There are at most AN/H^2 indices at which $\|G_j\|_2^2 + \|\tilde{G}_j\|_2^2 > H^2$. Deleting their labels leaves at least $\theta N/2$ labels at at least

$$\kappa_0 N, \quad \kappa_0 = \theta/(2K_L),$$

distinct radii. Select one label at each such radius, for example the smallest. Endpoint tests, budget tests, label assignments, and this selection are all $\mathcal{F}$ -measurable. This is why the coarse circles in (7.3) were placed outside every fresh annulus.

For a selected pair (j, i) , the conditional real field in (7.5) is a fresh field on U plus

$$G_j + \left(H_j - Qt_j - \gamma^{-1} \log M + i\gamma^{-1} \log 4 \right).$$

The constant in parentheses has absolute value at most L/γ . The imaginary conditional field is a fresh field plus $\tilde{G}_j$ and a phase in a bounded interval. Lemma 7.2 first makes the reference all-order failure probability small. Lemma 7.1 then makes its conditional target failure probability at most any prescribed $p_* > 0$, by choosing ε_* small enough. These choices are made after A, L, H and the geometric constants have been fixed.

The selected augmented tests are conditionally independent: integrate the fresh-field product law against the conditional product of local kernels in the disjoint buffers. The random selection causes no change, since it is $\mathcal{F}$ -measurable and takes values in a finite deterministic list. With $\ell \geq \kappa_0 N$ selected radii, a union bound over subsets shows

$$\mathbb{P}[\text{fewer than } \ell/2 \text{ tests succeed} \mid \mathcal{F}] \leq 2^\ell p_*^{\ell/2} \leq 2^{m+1} p_*^{\kappa_0 N/2}. \quad (7.9)$$

Choose p_* to make this at most e^{-RN} . Together with the square-budget bound this has an arbitrarily large exponential rate. Taking the union over the at most $C_K e^{2BN}$ finest-grid chains leaves a summable error. The endpoint errors are added using (7.6); they were union-bounded on their own grids and do not incur the finest-grid entropy.

For any $x \in K$, choose a finest-grid point y within $\epsilon_0 r_m$. At every scale of its chain,

$$|x - c_j(y)| \leq \epsilon_0 r_m + \epsilon_0 r_j \leq 2\epsilon_0 r_j < r_j/256.$$

Its at least $\kappa_0 N/2$ successes give the required distinct radii. Take the union of the successful tests produced by all chains. At a fixed center and radius, the alignment interval allows at most K_L labels. If several succeed, retaining any one preserves coverage, because their central holes are identical. The radius- r buffers lie in disks of radius $16r$ about grid points spaced by $\epsilon_0 r$, so their overlap is at most a constant depending only on ϵ_0 .

Borel–Cantelli proves eventual coverage. This event belongs to the restriction of the fields and kernels to any fixed neighborhood of K ; finitely many large initial radii can be discarded. A fixed absolutely continuous field-law change preserves this probability-one event by Proposition 6.8. It need not preserve the displayed summable numerical probability bound. Finally intersect the probability-one events for a countable compact exhaustion and countably many prescribed test families. $\square$

For transfer back to a jointly convergent discrete array, fix N and its finite collection of centers, labels and strict buffered tests first. On the event that these tests succeed, their strict inequalities and protected incidences hold in the approximating surfaces for all sufficiently large discrete indices, by Theorems 5.10 and 6.1. That threshold may depend on N , the finite configuration and its margins. Only afterward take a countable diagonal over increasing N . The radius mesh is therefore fixed before the discrete threshold is taken. This statement applies to extremal-length tests in any deterministic units, as well as to the other retained local readouts.

8 The canonical conformal structure

We identify the conformal structure obtained by gluing the fixed flag triangles. There are two steps. First, extremal length cannot degenerate: a degenerating annulus would produce a nonzero flow of rectifiable curves in the reference coordinate, and locality would then produce cheap crossings in two transverse directions. Second, the resulting quasiconformal comparison has a local infinitesimal conformal structure. The same locality argument makes this structure deterministic and round. Neither step uses the intrinsic metric comparison.

Throughout this section, a reference coordinate means the original oriented peanosphere coordinate of Section 3. On the finite sphere let $H_n : X_n \rightarrow \hat{\mathbb{C}}$ be the homeomorphisms of Theorem 4.7, chosen from the contour and surface data before drawing the three fresh normalization samples. On a compact planar test domain we use their protected-window counterparts from Proposition 4.8. In particular,

$$\max_T \text{diam } H_n(T) \longrightarrow 0 \quad (8.1)$$

in the reference coordinate. This assertion is topological and precedes the uniformization of X_n .

All extractions below retain the compatible local arrays of Theorem 6.1: actual flag-surface extremal lengths, strict port and confinement tests, and their joint refinements. A test whose projected confinement is compactly inside an open set can be specified by a finite union of protected cells with an intermediate margin. We always fix that finite test before passing to the discrete limit. The kernel theorem permits countably many such tests and arbitrary deterministic positive units for extremal length. Local changes of ordinary field law use Proposition 6.8. Transfer to the finite sphere at its actual area-normalized heights uses Lemma 6.10 together with the common kernels. For probability-one conclusions its domination direction suffices.

8.1 Extremal length and flows

We use the extremal-length and electrical-energy framework of Ahlfors and Beurling [1]. For a family $\mathcal{C}$ of curves on a Riemann surface, we use

$$\text{EL}(\mathcal{C}) = \sup_{\sigma} \frac{(\inf_{\gamma \in \mathcal{C}} \int_{\gamma} \sigma \, ds)^2}{\int \sigma^2 \, dA}. \quad (8.2)$$

Here ds, dA are any conformal background length and area elements and the supremum is over nonnegative measurable densities of finite positive energy. Equivalently, EL is the reciprocal of the infimal energy of a density charging every member of $\mathcal{C}$ by at least one. On X_n the background representative is the actual Euclidean flag geometry. The extended charts at the finitely many conical vertices give the same definition.

For an annulus A , write $\text{EL}_{\text{tr}}(A)$ for its through crossings and $\text{EL}_{\text{wind}}(A)$ for its nonzero-winding closed curves. Uniformization by a cylinder, and by a rectangle for a quadrilateral, gives

$$\text{EL}_{\text{tr}}(A) \text{EL}_{\text{wind}}(A) = 1, \quad \text{EL}(\mathcal{C}_{\leftrightarrow}(R)) \text{EL}(\mathcal{C}_{\downarrow}(R)) = 1. \quad (8.3)$$

For nested disjoint annuli crossed in series, through extremal lengths add as a lower bound. One can see this directly by adding admissible densities with disjoint supports and optimizing their relative weights.

A *unit flow* is a probability measure ν on rectifiable curves. Its traffic is the measure obtained by averaging arclength, with multiplicity. If this measure has density τ with respect to dA , then

$$\int \left(\int_{\gamma} \sigma \, ds \right) \nu(d\gamma) = \int \sigma \tau \, dA, \quad \text{EL}(\mathcal{C}) \leq \int \tau^2 \, dA \quad \text{if } \nu(\mathcal{C}) = 1. \quad (8.4)$$

Indeed, the first identity is Tonelli's theorem. For a density admissible for $\mathcal{C}$, the left-hand side is at least one; Cauchy–Schwarz and the reciprocal variational formula prove the second assertion. Restricting the curves to subpaths decreases traffic. Restricting to an event of flow probability $u > 0$ and normalizing multiplies the traffic energy by at most u^{-2} .

For a cylinder $[0, L] \times (\mathbb{R}/H\mathbb{Z})$, choose a horizontal segment with its second coordinate uniform. Its traffic density is H^{-1} , and its energy is $L/H = \text{EL}_{\text{tr}}$. Thus every conformal annulus has a unit through flow of energy equal to its through extremal length. Trimming between two intermediate separating curves gives compactly confined subpaths with no increase of energy. These observations also apply to piecewise smooth annuli in a flag surface.

8.2 Crossing quantiles and the two local tests

Fix a disk compactly inside an ordinary part of the cone coordinate, away from its marked singularity. Inside this disk choose a center z_0 and radii

$$0 < r_0 < r_1 < r_2 < R_2 < R_1 < R_0,$$

with additional strict collars around the six circles. These give a wide traversal band, a middle band $A(z_0, r_1, R_1)$, and a narrower band between radii r_2, R_2 . All radii and collars remain fixed.

Use the bilateral word with microscopic time unit 4^j , hence diffusive height unit 2^j . The protected-window construction of Proposition 4.8 gives a finite flag surface containing these tests. In it choose a piecewise smooth annulus $\mathcal{A}_j$ approximating the inverse image of the middle band under the topological correspondence, with its two boundary curves separating the corresponding inner and outer collars. Make this a fixed measurable choice: enumerate annuli with rational polygonal boundaries in finite subdivisions of the flag complex, and take the first satisfying these separation tests with boundary-image error less than 2^{-j} from the two middle-band circles. Polygonal approximation of their inverse Jordan curves gives such annuli; measurability of the correspondence makes the tests measurable. Put $Z_j = \text{EL}_{\text{tr}}(\mathcal{A}_j)$. If this temporary construction fails, put $Z_j = 1$; its failure probability tends to zero.

The chosen annulus need not itself be a local readout. Its two uses are captured by local curve families with strict ports and confinement. Let $\mathcal{W}_j$ be the wider crossing family, from the inner port near radius r_0 to the outer port near radius R_0 , confined to a strict annular collar of $A(z_0, r_0, R_0)$. Every member contains a through crossing of $\mathcal{A}_j$. A path from the central hole to the exterior of the outer buffer has such a wider crossing as a subpath: stop on its first outer visit and start at its last preceding inner visit. Conversely, trim each curve of the unit cylinder flow on

$\mathcal{A}_j$ between its last visit to radius r_2 before its first subsequent visit to radius R_2 . The trimmed pieces lie in the closed band between those circles. They belong to a narrower local family $\mathcal{N}_j$, whose ports and confinement include strict collars of that band. Their projected endpoints have a fixed positive separation. If τ_j is its traffic density with respect to Euclidean flag area dA_j , then

$$\text{EL}(\mathcal{W}_j) \geq Z_j, \quad \text{EL}(\mathcal{N}_j) \leq \int \tau_j^2 dA_j \leq Z_j. \quad (8.5)$$

For the first inequality, extend a density on $\mathcal{A}_j$ by zero; every wider crossing pays for a through subpath. The second uses (8.4) and the decrease of traffic under trimming.

Implement the two families by intermediate protected-cell covers of the ports and confinements. The extra collars allow the wider ports to remain on opposite sides of $\mathcal{A}_j$, and the narrower cover to contain every trimmed piece. The full-preimage protection and local readouts of Lemma 5.10 supply such finite covers, with margins fixed before taking the discrete limit. Their countable refinements are retained together. Thus the inequalities in (8.5), on the event that the construction succeeds, transfer its quantile bounds to local EL tests. They require no locality of the chosen topological homeomorphism or annulus.

Fix a sufficiently small $\varepsilon > 0$, and choose positive lower quantiles β_j satisfying

$$\mathbb{P}[Z_j \leq 2\beta_j] \geq \varepsilon/2, \quad \mathbb{P}[Z_j \geq \beta_j/2] \geq 1 - 2\varepsilon. \quad (8.6)$$

These inequalities allow atoms. The second gives high-probability barriers for the wider crossing family; the first gives a positive-probability unit flow in the narrower family with energy at most $2\beta_j$. The vanishing construction failures are absorbed in the probability slack. Choose ε so small that the lower test, after the fixed local changes of field law and band margins, meets the probability threshold of Theorem 7.4. The failure probabilities under these fixed changes of density tend to zero with ε .

We will rule out $\liminf_j \beta_j = 0$. On a hypothetical record-low sequence j_m , set $\beta_m = \beta_{j_m}$. For every fixed backward offset $i \geq 0$, $\beta_{j_m-i} \geq \beta_m$ eventually, so the wider-band tests still give barriers in β_m units. Their time units 4^{j_m-i} have relative labels $v_i = 4^{-i}$, exactly the labels used in annular alignment. Extremal length has no additional power of v_i . The next two lemmas turn these barriers and the small-energy flow at offset zero into a nonzero absolutely continuous average arclength measure in the reference coordinate. Local germs will then force cheap crossings in two transverse directions.

8.3 A density that charges all projected advances

We record precisely the consequence of the aligned-annulus construction that is needed here. Suppose that in an extraction, indexed by m , the high-probability shell tests give a lower through extremal length of order $\beta_m > 0$, at the backward labels required by Theorem 7.4. The units β_m are deterministic; the record-low experiment above is the first application. The aligned-annulus result of Theorem 7.4, applied to these local tests, supplies the following configuration for arbitrarily large integers N , with summable failure probabilities after subextraction. There are at most CN distinct geometric radii between e^{-BN} and e^{-aN} . At radius r the centers form a mesh of spacing comparable to r , and their buffered shells have bounded overlap. Every point of a fixed compact set lies in the central holes of at least cN good shells. For a good shell, every path from its central hole to the exterior of its outer buffer is charged by a density supported in that buffer, with charge at least r and energy at most Cr^2/β_m . The constants depend only on the fixed shell geometry and the probability cutoff. All these statements concern actual discrete-surface paths and supports under the homeomorphism H_m , with strict margins.

The density assertion follows from the extremal-length lower bound and (8.2), with a fixed slack in the constant. If a measurable choice is required, take a near-minimizing condenser potential and its gradient. Continuous piecewise affine potentials on successive finite refinements of the flag triangles, with rational values and fixed boundary values on the two port sets, form a countable energy-dense family. Selecting the first potential below the prescribed energy threshold gives a measurable choice. Its gradient, enlarged on triangle edges by the larger one-sided value, charges every crossing; this change on edges does not change energy.

Lemma 8.1 (Simultaneous density budget). *Let U be a finite union of rational boxes compactly inside an ordinary reference chart. In the preceding configuration put $R_N = e^{-a_N}$. There are nonnegative densities $g_{m,N,U}$, supported over U^{+CR_N} , such that*

$$\limsup_{m \rightarrow \infty} \beta_m \int g_{m,N,U}^2 dA_m \leq C \text{Area}(U^{+CR_N}), \quad (8.7)$$

$$\int_{\gamma|I} g_{m,N,U} ds_m \geq c \text{diam}(H_m \gamma(I)) - CR_N - o_m(1) \quad (8.8)$$

for every path interval I projected into U . For any fixed finite number of pairwise disjoint path intervals the charges add. The same density works for every path and every such collection. The constants are independent of U, N, m ; the threshold in m may depend on the finite configuration and margins.

Proof. At each center and radius retain one good shell density. Sum them over the centers whose buffers meet U , and then sum over the retained radii and divide by N . At radius r , bounded overlap and grid counting give total energy at most $C \text{Area}(U^{+CR_N})/\beta_m$. There are at most CN radii. Cauchy–Schwarz for their sum proves (8.7).

For the charge, choose a coordinate in which the path interval has oscillation at least its diameter divided by $\sqrt{2}$, and restrict to a subinterval between a minimum and a maximum of that coordinate. Discard coordinate values within CR_N of either endpoint. At each remaining value choose a visit of the path. This visit belongs to at least cN good central holes. A hole at radius r can account for a coordinate interval of length at most Cr , and the path must make a full crossing of its buffered shell. Integrating the number of covering holes over the coordinate interval gives

$$cN(\text{osc} - CR_N) \leq C \sum_{\text{shells crossed}} r.$$

The integral of the summed shell densities is at least the sum on the right before the harmless fixed factor. Divide by N . Strict inner and outer margins absorb the uniform projection errors, proving (8.8). Integrals over disjoint parameter intervals add even when their traces overlap. No choice of the density was made using the path. $\square$

8.4 Compact flows and average arclength

The density budget now controls how many disjoint advances a flow can make. This yields compactness without assuming that the projected discrete curves have bounded Euclidean length.

Lemma 8.2 (Flow extraction). *Suppose that the configurations of Lemma 8.1 hold along a sequence for all sufficiently large members of a deterministic sequence of N 's, simultaneously for the countable box library. Let ν_m be unit flows of compactly confined paths whose projected endpoints are separated by at least $d_0 > 0$, and whose traffic satisfies*

$$\int \tau_m^2 dA_m \leq C_0 \beta_m.$$

After reparameterization and subextraction, the projected flow laws converge weakly to a probability law ν on continuous paths. Its paths have endpoint separation at least d_0 , are rectifiable almost surely, and their average arclength measure Λ satisfies

$$\Lambda(U) \leq C\sqrt{\text{Area}(U)} \quad (8.9)$$

for every finite union of rational boxes compactly in the chart. In particular, $0 < \Lambda(K) < \infty$ on a sufficiently large compact confinement K , and Λ is absolutely continuous with respect to planar area.

Proof. Fix $h > 0$, and choose N with $CR_N < ch/4$. A collection of disjoint path intervals each of projected diameter at least h then pays at least $ch/2$ per interval for all sufficiently large m . By (8.4) and (8.7), its expected cardinality is bounded by C/h . The same bound holds for the maximal cardinality, by monotone convergence over finite collections.

Here is a measurable reparameterization implementing compactness. For a continuous projected path, mark its successive first moves by distance 2^{-k} , and denote their finite number by J_k . Give each such mark clock mass $2^{-k}/(1 + J_k)$, for every $k \geq 1$, and add normalized Lebesgue mass in its original parameter. The total clock mass lies in $[1, 2]$. Use its inverse, inserting a pause at each atom, and normalize the new parameter to $[0, 1]$. This is a representative of the same path up to zero uniform reparameterization distance. On a set where $J_k \leq M_k$ for every k , a new parameter interval shorter than $2^{-k-1}/(1 + M_k)$ contains no complete move at scale 2^{-k} , and the image oscillation there is at most 2^{2-k} .

The expected crossing-count bounds allow $M_k < \infty$ to be chosen with total exceptional probability as small as desired. For each fixed k , the finitely many indices preceding its eventual bound can be included by increasing M_k : their crossing counts are finite almost surely. Thus the reparameterized laws are tight in $C([0, 1], K)$, by Arzelà–Ascoli. Prokhorov’s theorem gives ν , and endpoint separation passes to the limit.

To prove the length assertion without exchanging infinite suprema, fix J . Consider at most J disjoint closed parameter intervals with rational endpoints whose images have positive clearance inside U , and sum the distances between their endpoint images. The supremum of these sums is a nonnegative lower semicontinuous path functional: each strict confinement test is open, and each finite sum is continuous on that test. For every such collection, (8.8) and Cauchy–Schwarz bound its flow expectation by

$$C\sqrt{\text{Area}(U^{+CR_N})} + CJR_N + o_m(1).$$

The bound also holds for the supremum, because the density charge holds simultaneously for all collections. Pass first to the weak flow limit, then let $N \rightarrow \infty$ with J fixed, and finally let $J \rightarrow \infty$. These finite interval sums exhaust variation in the open set U . Taking one larger confinement first proves finite total variation for almost every path; the same argument in each U then proves (8.9) for its arclength measure, with multiplicity.

Every limit path has length at least d_0 , so $\Lambda \neq 0$. The constant in (8.9) is uniform over finite unions of boxes. Cover a planar null set by countably many rational boxes of arbitrarily small total area, apply the estimate to finite subunions, and increase to the full union. This proves $\Lambda \ll dz$. $\square$

The lemma applies pathwise on a positive-probability event in a joint extraction. One may retain the flow laws as additional random variables and use the preceding tightness argument after restricting to compact bounds. The local kernel theorem is not conditioned on these flow laws: the flows will only witness events in the already retained local extremal-length arrays.

8.5 Local germs

We next isolate the pointwise use of locality. Write $\mathcal{H}$ for the sigma field of the real and imaginary fields in an ordinary chart. A *local germ readout* $Y(z)$ is a jointly measurable function of the retained local arrays with the following property: for every neighborhood V of z , it is readable from strict inner and outer tests in arbitrarily small subneighborhoods of z contained in V . Countable finite port covers are used in this definition.

At almost every deterministic z , Lemma 6.7 says that all visits to a sufficiently small neighborhood of z belong to one traversal in any prescribed larger neighborhood. The needed success event is local: fix two surrounding buffers inside that larger neighborhood. By Lemma 5.1, the imaginary field there determines the entire finite list of directed outer-buffer traversals meeting a prescribed compact inner disk. The condition that this list has cardinality one is therefore measurable there. Shrinking the inner disk exhausts a probability-one event by unique visitation and properness, as in Lemma 6.7. Consequently the germ requires no exterior ordering choice. The kernel theorem and its restriction compatibility imply that $\mathbb{E}[\varphi(Y(z)) \mid \mathcal{H}]$ is measurable with respect to the restrictions of the fields to every such larger neighborhood. For two distinct ordinary points these readouts have independent conditional kernels. These conclusions include local auxiliary marks, which may be integrated out. The single-traversal exhaustion has probability one; its success in an inner neighborhood is testable in the larger neighborhood, as stipulated in Lemma 6.7.

Lemma 8.3 (Deterministic germs and rotations). *A bounded local germ readout with values in a separable metric space is, for area-almost every deterministic point, almost surely deterministic. Its deterministic value respects every coordinate rotation under which the intrinsic readout transforms. The statements hold jointly for countably many readouts and rotations.*

Proof. First take a bounded real readout and set $p(z) = \mathbb{E}[Y(z) \mid \mathcal{H}]$, with jointly measurable versions. Conditional independence at distinct points and Fubini give, for every rational disk V ,

$$\mathbb{E} \left[\left(\int_V (Y(z) - p(z)) \, dz \right)^2 \middle| \mathcal{H} \right] = 0.$$

The diagonal in $V \times V$ has area zero. A countable disk exhaustion and Lebesgue differentiation imply $Y(z) = p(z)$ for almost every z , almost surely. A second application of Fubini gives this identity almost surely at almost every deterministic point.

For completeness, the point germ of a Dirichlet GFF is trivial. Realize the field as an isonormal Gaussian process over $H_0^1(V)$, with its Dirichlet inner product. The Gaussian subspace generated by its restriction to $B(z, r)$ is the closed span of the Green potentials of test functions supported there. An element in the intersection of these subspaces is orthogonal to every smooth function vanishing near z . Such functions are dense in $H_0^1(V)$: a logarithmic cutoff around z has Dirichlet energy tending to zero, so a point has zero capacity. The subspace intersection is therefore zero. Decreasing orthogonal projections, first on each finite Gaussian chaos and then by L^2 approximation, show that the intersection of the completed Gaussian sigma fields is trivial. The same argument applies to the direct sum of two independent fields. Taking the imaginary field modulo its period only decreases the observable sigma fields. Local absolute continuity in Proposition 6.8 transfers triviality to the actual field restrictions and their phase conventions.

For almost every deterministic z , the conditional expectation $p(z)$ is measurable with respect to each shrinking field restriction, hence to this trivial germ. It is deterministic, and so is $Y(z)$. A countable family of bounded continuous functions separating points proves the assertion for a separable target space.

Finally rotate a local coordinate about z , making the corresponding imaginary-field phase adjustment. The intrinsic time/height ports and flag-surface variational outputs are unchanged; only their spatial coordinate descriptions transform. This is the affine covariance in Proposition 6.8. The original and transformed field laws are locally equivalent. Thus every probability-zero or probability-one assertion about a germ transports to the transformed germ. Since its value is deterministic, that value must obey the claimed rotation rule. Taking countable intersections completes the proof. $\square$

8.6 Extremal length does not degenerate

Proposition 8.4 (No degeneration of crossing quantiles). *There exists $\beta_* > 0$ such that $\liminf_{j \rightarrow \infty} \beta_j \geq \beta_*$.*

Proof. Suppose otherwise, and take the record-low extraction described in Section 8.2: successive record lows j_m with $\beta_{j_m} \rightarrow 0$. For every fixed backward offset i , $\beta_{j_m-i} \geq \beta_{j_m}$ eventually. Retain all fixed offset laws and all local EL arrays in units $\beta_m = \beta_{j_m}$, using the diagonal extraction in Theorem 6.1. The second inequality of (8.6), through (8.5), gives the wider-band shell barriers in these units at every fixed backward label $v_i = 4^{-i}$. The aligned-annulus result and Lemma 8.1 therefore apply.

The first inequality of (8.6) retains a positive-probability small-EL event. Retain the trimmed cylinder flows from (8.5) as witnesses in the joint extraction. On a fixed buffer containing their confinement and the local tests, Proposition 6.8 gives a strictly positive Radon–Nikodym derivative of the ordinary field-pair restrictions relative to the cone restrictions. Reweight the entire coupled extraction by this field-dependent derivative. The local field and array marginal then has the ordinary law with the same kernels; the positive-probability flow event and its pathwise inequalities persist under this equivalent change of law. No kernel is conditioned on the flows. Their paths have fixed positive advance in a compact set and traffic energy $O(\beta_m)$. Lemma 8.2 gives, on that event, a probability flow ν of nonconstant rectifiable paths with nonzero absolutely continuous average arclength Λ .

We describe the local event witnessed by these paths. Take a finite sufficiently fine set of unoriented directions, closed under a quarter-turn. For a direction v , a center z , and $s > 0$, put

$$R_v(z, s) = \{z + tv + uv^\perp : |t| < 3s/2, |u| < s/4\}.$$

A strict crossing runs from $t < -s$ to $t > s$, has compact confinement in this rectangle, and is represented by intermediate port covers. Let $E_v(z)$ be the event that at every sufficiently small dyadic s some such cover has finite limiting EL in units β_m . Explicitly, this is a countable union over the first dyadic scale, followed by a countable intersection over smaller scales, and a union over admissible finite covers and finite integer EL bounds. Whether a cover fits strictly inside the indicated rectangle is a measurable open geometric test. Thus $E_v(z)$ is jointly measurable in z and the local arrays. Discarding finitely many scales leaves the event unchanged, so it is a local germ readout. Its finite EL bound may depend on the scale.

At arclength-almost every interior point of almost every path of ν , arclength parameterization has a unit tangent. Choose v within the angular slack of this tangent. Differentiability at that parameter value gives strict crossings of $R_v(z, s)$ at every sufficiently small s . Almost every path belongs to the support of ν . Each fixed strict tube crossing is an open path event containing that path, so has positive ν -mass. Weak convergence gives positive mass in the discrete flows for all sufficiently large m . Restrict and normalize that mass in (8.4). Its energy is still $O(\beta_m)$, with a finite constant depending on this radius and tube. A finite intermediate cover of positive mass

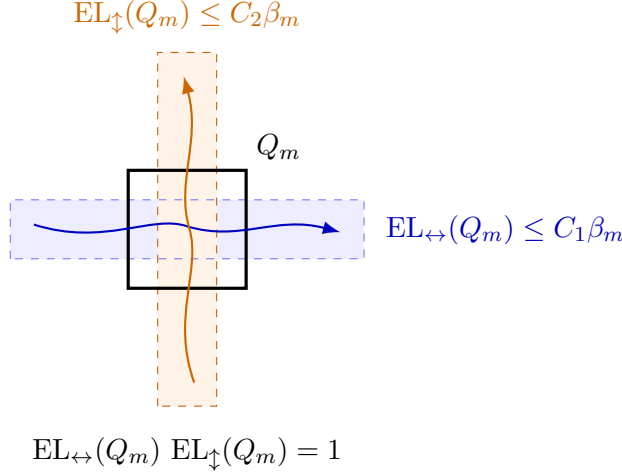


Figure 4: The dashed strips represent $R_v(z, s)$ and $R_{v^\perp}(z, s)$. Their strict lateral clearance forces crossings of the corresponding sides of the common quadrilateral Q_m . The picture uses the reference coordinate; the extremal lengths are those of the actual flag surface. Finite bounds at one common radius contradict duality when $\beta_m \rightarrow 0$.

supplies the required local EL witness. This proves $E_v(z)$ at Λ -almost every such point for some direction v .

Since $\Lambda \neq 0$ and $\Lambda \ll dz$, on the positive-probability flow event at least one of the finitely many events E_v holds on a set of positive planar area. Lemma 8.3 makes its indicator deterministic at almost every deterministic point and transports it under a quarter-turn. By Fubini, on a probability-one event these equalities hold outside one planar null set, simultaneously for the finite set of directions. The positive-area flow witness therefore meets that full-area set, where $E_v(z)$ and $E_{v^\perp}(z)$ have the same truth value. Choose a point satisfying both, and then one fixed dyadic radius below their two starting scales.

Let $\mathcal{F}_{v,m}$ and $\mathcal{F}_{v^\perp,m}$ be the two discrete curve families specified by the selected covers. Their extremal lengths are at most $C_1\beta_m$ and $C_2\beta_m$, respectively, for all sufficiently large m . Choose a common central quadrilateral Q_m , with projected sides near those of $\{|t| < s/2, |u| < s/2\}$, and smooth it on the discrete surface within the available margins. As shown in Figure 4, the narrower strips have lateral clearance from the inappropriate sides of this quadrilateral. Every member of either family therefore contains a subpath crossing the corresponding pair of sides of Q_m . For any density σ on Q_m , its extension by zero to the surface has the same energy, and

$$\inf_{\gamma \in \mathcal{F}_{v,m}} \int_{\gamma} \sigma \, ds_m \geq \inf_{\gamma \in \mathcal{C}_{\leftrightarrow}(Q_m)} \int_{\gamma} \sigma \, ds_m.$$

Taking the supremum in (8.2), and arguing identically in the perpendicular direction, yields

$$\text{EL}(\mathcal{C}_{\leftrightarrow}(Q_m)) \leq \text{EL}(\mathcal{F}_{v,m}) \leq C_1\beta_m, \quad \text{EL}(\mathcal{C}_{\updownarrow}(Q_m)) \leq \text{EL}(\mathcal{F}_{v^\perp,m}) \leq C_2\beta_m.$$

The constants are finite because the radius and the two covers have now been fixed. Their product tends to zero, contradicting (8.3). This excludes the record-low extraction. $\square$

Proposition 8.5 (Macroscopic extremal-length bounds). *In every diverging-scale extraction, including the finite sphere, the following hold in the reference coordinate.*

1. For every $d > 0$, the family of all discrete-surface paths whose projected diameter is at least d has a positive limiting lower extremal-length bound.
2. There is a deterministic $c_* > 0$ such that every fixed round annulus $A(z, r, 2r)$ in an ordinary coordinate chart has

$$\liminf_n \text{EL}_{X_n}(H_n^{-1} \mathcal{C}_{\text{tr}}(A(z, r, 2r))) \geq c_*.$$

The assertion holds simultaneously for a countable dense annulus library, with strict margins. It includes annuli surrounding an isolated exceptional mark whose closures avoid the marks.

Proof. Proposition 8.4 makes the high-probability barriers uniformly positive in raw EL units. To pass from powers of four to an arbitrary diverging integer sequence, write $n = 4^{j_n} v_n$, where $j_n = \lfloor \log_4 n \rfloor$ and $1 \leq v_n < 4$, and extract $v_n \rightarrow v \in [1, 4]$ jointly with every fixed offset array. Reexpress the same raw cuts and incidence readouts in the new units: the area clock is divided by v_n , frontier heights by $\sqrt{v_n}$, and the real field is shifted by $-\gamma^{-1} \log v_n$. The actual flag-surface extremal length is unchanged. Strict chart margins allow the converging factors to be retained through this indexed-array identity. On each fixed buffered domain, the constant shifts lie in a bounded family and admit cutoff extensions of uniformly bounded Dirichlet norm. The two-sided Cameron–Martin comparison therefore chooses a common high-probability barrier threshold before annular alignment. This argument does not identify differently indexed kernels at an unshifted field. Apply the aligned-annulus result and Lemma 8.1 with $\beta_m = 1$.

For a curve making advance d compactly in U , choose N with $CR_N < cd/4$. The density charges the entire family by at least $cd/2$, for large m , and has bounded energy. Its extremal length is therefore bounded below by a constant times $d^2 / \text{Area}(U^{+CR_N})$. A finite collection of ordinary patches catches a fixed positive subadvance of every curve of spherical diameter at least d . To obtain such a collection, first remove disjoint sufficiently small neighborhoods of the finitely many exceptional marks; a curve of diameter d either makes an advance away from them or crosses one of their surrounding compact annuli. Cover the resulting compact regions by finitely many ordinary boxes and sum the densities. This proves the first claim for the entire family at once.

For a through crossing of $A(z, r, 2r)$, take a slightly larger annular union of boxes of area at most Cr^2 . Each crossing makes projected advance at least a fixed multiple of r there. The preceding quotient is bounded below by a universal constant. The union can avoid the center and all other exceptional marks; in particular this proves the assertion for fixed shells surrounding an exceptional center. Changes of ordinary field law preserve the eventual statement and its constant. Countable extraction handles all rational tests and margins simultaneously. Lemma 6.10 transfers these probability-one conclusions to the finite sphere at its actual area-normalized heights; Theorem 6.1 retains the same incidence and extremal-length readouts. $\square$

8.7 An all-center ring criterion

We give the analytic compactness statement in a form requiring no estimate at a radius depending on n . In a finite conformal atlas of the round sphere, fix the countable library of annuli $A(z, r, 2r)$ with rational centers and radii and compact closure in a chart. Annuli around finitely many omitted centers are also allowed, as in Proposition 8.5.

Lemma 8.6 (Ring criterion). *Let $f_n : \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ be homeomorphisms. Suppose that $f_n(z_n^i) = w_n^i$, where both triples converge to triples of distinct points. Suppose that, for every annulus A in the preceding library,*

$$\liminf_n \text{EL}_{\text{tr}}(f_n(A)) \geq c_* > 0. \tag{8.10}$$

Then the maps and their inverses are equicontinuous. Every subsequential uniform limit is a $K(c_)$ -quasiconformal sphere homeomorphism, with either orientation.*

Proof. The two ring estimates below give equicontinuity and exclude collapsed continua. Annuli with shifted centers will then bound the linear distortion of each limit.

Two ring estimates. Let E, F be the two complementary continua of a sphere annulus. If both have spherical diameter at least u , every essential closed curve in the annulus has spherical length at least $u/4$. Otherwise its trace lies in a disk of radius $u/4$; each continuum has a point outside that disk, so the curve is null-homotopic in the sphere minus those two points, a contradiction. Constant spherical density and (8.3) give

$$\text{EL}_{\text{tr}}(\widehat{\mathbb{C}} \setminus (E \cup F)) \leq 64\pi/u^2. \quad (8.11)$$

If additionally $d = \text{dist}(E, F) \rightarrow 0$ while both diameters stay above u , choose nearest points and center geodesic circles at the point in E . For $2d < t < u/4$, both continua meet the circle of radius t . There is a circle arc joining them whose interior is in the annulus. Every admissible crossing density therefore has integral at least one on that circle. Cauchy–Schwarz and polar integration imply

$$\text{EL}_{\text{tr}}(\widehat{\mathbb{C}} \setminus (E \cup F)) \leq \frac{2\pi}{\log(u/(8d))} \rightarrow 0. \quad (8.12)$$

The density is extended by zero on the complementary continua when performing this integration.

Equicontinuity and exclusion of collapsed continua. Fix $\delta > 0$ smaller than the pairwise limiting distances of the target marks, and fix an image tolerance $u > 0$. Choose an integer k so large that $kc_*/2 > 64\pi/\min(u, \delta)^2$. Around each source point choose an outer coordinate disk omitting two limiting source marks with clearance. Inside it choose a stack of k rational concentric annuli, with disjoint closures and a smaller central disk. Rational centers and radii can be chosen so that these central disks cover the sphere; retain a finite subcover. For this finite collection, (8.10) gives a common eventual index with all image moduli at least $c_*/2$. Series addition and (8.11) imply that every retained central disk has image diameter less than u : its outer complementary continuum contains two target marks separated by at least δ . A Lebesgue number for the finite source cover proves equicontinuity. Only finitely many fixed annuli were used for each u .

Arzelà–Ascoli supplies a uniform limit f , which is onto: for each target point extract a convergent sequence of its preimages. Its fibers are connected. Indeed, if a fiber split into two disjoint nonempty compact sets, choose disjoint open neighborhoods of these sets. Their complement has image bounded away from the fiber value. For large n , the connected preimage under f_n of a sufficiently small disk about that value would lie in the two neighborhoods and meet both, which is impossible.

Suppose a fiber C is nontrivial. Choose $a \in \partial C$, $c \in C \setminus \{a\}$, and $d \notin C$. Take $b \notin C$ sufficiently close to a . A fixed rational annulus can be chosen with a, b strictly inside its inner disk and c, d strictly outside its outer disk. The complementary continua of its f_n -image have diameters bounded below by the positive limiting distances from $f(b)$ and $f(d)$ to $f(a) = f(c)$, but their mutual distance tends to zero. This contradicts (8.12) and the lower bound for this single fixed annulus. Thus f is a homeomorphism.

Quasiconformality and inverse equicontinuity. We give the quantitative local criterion as well. Work in finite planar coordinates on a disk whose image avoids the target pole. For a bounded planar annulus with through EL at least $m > 0$, let e_0, e_1 belong to its bounded complementary component and w to the unbounded component. The same circle integration as above, with the unbounded continuum crossing every sufficiently large circle, gives

$$|w - e_i| \geq \theta(m)|e_1 - e_0| \quad (i = 0, 1), \quad \theta(m) = \frac{1}{2}e^{-2\pi/m}. \quad (8.13)$$

For a fixed rational source annulus apply this with $m = c_*/2$ to f_n , then pass to f . Approximation of centers, radii, and points extends the inequalities to every sufficiently small source annulus. There is no common discrete threshold in this approximation.

Put $L(z, r) = \max_{|v-z|=r} |f(v) - f(z)|$ and define $l(z, r)$ by the corresponding minimum. With $\theta = \theta(c_*/2)$, (8.13) yields

$$l(z, 2r) \geq \theta L(z, r).$$

For $|w - z| = 2r$, let $v = (z + w)/2$ and apply the same inequality to the annulus centered at w , where v lies on its inner circle and z on its outer circle. Provided $B(z, 4r)$ is in the chart, this gives

$$|f(z) - f(v)| \geq \theta |f(v) - f(w)|, \quad L(z, 2r) \leq (1 + \theta^{-1})L(z, r).$$

Consequently

$$\frac{L(z, 2r)}{l(z, 2r)} \leq \theta^{-1}(1 + \theta^{-1}). \quad (8.14)$$

This is the metric criterion for planar quasiconformality [10, Theorem 1]; see also [4, Theorem 2.5.3]. The argument is precisely the lower-bound part of the proof of [32, Lemma 3.1, equations (3.3)–(3.7)]; the shifted centers explain why one needs the all-center hypothesis. One can also see the required Sobolev regularity directly: (8.14) bounds $L(z, r)^2$ by a constant times $\text{Area}(f(B(z, r)))$. Integrating squared coordinate difference quotients, and using Fubini for the locally finite measure $V \mapsto \text{Area}(f(V))$, gives bounded L^2 difference quotients on every compact subdomain. Hence $f \in W_{\text{loc}}^{1,2}$, and the metric criterion gives bounded distortion. Charts at the pole, or point removability, finish the sphere assertion; a reflection handles the other orientation.

Finally, failure of inverse equicontinuity would give target points approaching one another whose source preimages stay separated. Extract a uniform limit of the corresponding f_n 's and convergent source preimages. The limit would identify two distinct points, contradicting the homeomorphism just proved. $\square$

8.8 The infinitesimal structure is round

The elimination of deterministic anisotropy by rotations has a parallel in [31, Section 4.4], where determinism comes from energy homogenization. Here the conditional-product and field-germ argument of Lemma 8.3 supplies that step.

Lemma 8.7 (Conformal rigidity). *Every homeomorphic limit in Lemma 8.6, when obtained from the local flag-surface arrays above, is conformal or anticonformal.*

Proof. Let $f_n = \phi_n \circ H_n^{-1} \rightarrow f$ uniformly. The image of a local quadrilateral R has the same modulus as its actual flag-surface preimage. Strict inner and outer port versions therefore determine the modulus of $f(R)$: uniform convergence of maps and inverses places the four sides between arbitrarily small collars, with their marked sides in the correct order. Conformal rectangle modulus is continuous under these Jordan collar approximations. The two transverse comparisons in (8.3)

supply the two sides of this determination. Thus all these local moduli are readouts of the local EL arrays. Global three-point normalization does not enter them, since Möbius postcomposition preserves modulus.

The distortion bound (8.14) implies that f is differentiable almost everywhere [10, Lemma 2]. Its quasiconformality also gives a nonzero Jacobian almost everywhere [4, Theorem 3.1.2]. At a point z satisfying both conclusions put

$$T(z) = \frac{Df(z)^\top Df(z)}{|\det Df(z)|}. \quad (8.15)$$

This positive symmetric tensor has determinant one and bounded eccentricity. It is a germ readout of the local moduli. Here is an explicit way to recover it. For a unit vector v and $t > 0$, let $R_{v,t}$ have length one in direction v and width t . Differentiability implies uniform convergence of the rescaled images of $z + rR_{v,t}$ to $Df(z)R_{v,t}$, for fixed t . If L is a nonsingular linear map, constant density and straight-line flows give

$$\frac{(|Lv| - t|Lv^\perp|)_+^2}{t|\det L|} \leq \text{EL}_{\leftrightarrow}(LR_{v,t}) \leq \frac{|Lv|^2}{t|\det L|}.$$

Multiplying by t and letting $t \downarrow 0$ recovers $v^\top T(z)v$. Three fixed directions, with countable sequences of r, t and strict covers, determine the tensor. This proves both local readability and spatial measurability.

Lemma 8.3 makes $T(z)$ deterministic for almost every deterministic z . A coordinate rotation conjugates it by the rotation matrix; the possible orientation sign of f has disappeared from (8.15). Applying the rotation part of that lemma to a quarter-turn gives $T = R^\top T R$. For a symmetric two-by-two matrix this sets the off-diagonal entries to zero and equates the diagonal entries. Since $\det T = 1$, we obtain $T = I$ almost everywhere. The quasiconformal differential is therefore a similarity almost everywhere, of one fixed orientation. The analytic characterization of quasiconformal maps [4, Theorem 2.5.4] makes f conformal or anticonformal. Isolated marked points are removable. $\square$

8.9 Actual marks, uniformization, and area

The remaining normalization uses the three fresh area samples of the finite model. We first record how these samples behave under the reference homeomorphisms.

Lemma 8.8 (Area marks under a converging coordinate). *Suppose $H_n : X_n \rightarrow \widehat{\mathbb{C}}$ are homeomorphisms and $(H_n)_* m_n \Rightarrow \mu$ weakly, where μ is diffuse. Conditional on the surface and exploration data used to choose H_n , let x_n^1, x_n^2, x_n^3 be fresh independent samples from m_n . In particular H_n is chosen before these three marks. Then their images converge jointly, together with the surface data, to three conditionally independent μ samples. Almost surely these limiting points are distinct. The Möbius map sending the ordered triple to $0, 1, \infty$ varies continuously with the triple as long as the three points are distinct.*

Proof. For continuous functions f_1, f_2, f_3 on $\widehat{\mathbb{C}}$, conditional expectation of their product is

$$\mathbb{E} \left[\prod_{j=1}^3 f_j(H_n(x_n^j)) \middle| \text{surface data} \right] = \prod_{j=1}^3 \int f_j d(H_n)_* m_n.$$

Weak convergence passes the right side to $\prod_j \int f_j d\mu$. Products of continuous test functions determine the law on the compact product sphere, proving the joint conditional assertion. Diffuseness gives distinctness. In a chart avoiding the limiting points at infinity, the normalized Möbius map is the usual cross-ratio expression; its coefficients depend continuously on the distinct triple. The resulting maps and their inverses converge uniformly in spherical distance by compactness. Other chart choices give the same conclusion when one point is infinite. $\square$

Theorem 8.9 (Canonical uniformization comparison). *Keep any joint extraction of the finite-map encoding and its local observations. Let x_n^1, x_n^2, x_n^3 be the actual independent m_n -samples used to define ϕ_n . In the reference coordinate these marks converge jointly to three conditionally independent samples z^1, z^2, z^3 of μ_h . The maps $\phi_n \circ H_n^{-1}$ have uniformly convergent subsequences, and every limit is the Möbius or anti-Möbius map taking (z^1, z^2, z^3) to $(0, 1, \infty)$. All retained local observations use this same limiting coordinate and the same possible orientation sign.*

Proof. An original edge contributes four equal-mass flags and one quadrangle. Choosing a uniform quadrangle and then one of its two matched tour steps gives a uniform step of the exploration. By Proposition 4.2 and Theorem 4.7, every point in any of its four flags and either step representative have vanishing discrepancy in the reference coordinate, uniformly over the map. Hence, for every continuous test function g ,

$$\int g d(H_n)_* m_n - \frac{1}{2n} \sum_{j=1}^{2n} g(\eta(j/(2n))) \longrightarrow 0.$$

The continuity of η and its quantum-area parameterization identify the Riemann-sum limit as $\int g d\mu_h$. Repeated incidence labels cause no change: the four flags are counted as occurrences. Thus $(H_n)_* m_n \Rightarrow \mu_h$.

The maps H_n were chosen before the fresh samples. Apply Lemma 8.8 to this reference area convergence. The limiting marks have conditional law $\mu_h^{\otimes 3}$, jointly with the retained map and local observations, and are almost surely distinct. Thus both the source marks and their prescribed target images satisfy the normalization hypothesis of the ring criterion.

Conformal invariance identifies the through EL of $(\phi_n \circ H_n^{-1})(A)$ with the actual surface EL for the corresponding annulus. Proposition 8.5, with strict collars, gives the hypothesis of Lemma 8.6 simultaneously for the countable annulus library. That lemma gives uniform subsequential convergence to a quasiconformal homeomorphism. Lemma 8.7 makes it conformal or anticonformal, hence Möbius or anti-Möbius on the sphere. Uniform convergence and convergence of the moving marks preserve their prescribed images. These three images specify the map once its orientation is fixed. All extractions retained the same local arrays and topological correspondence, so no observation acquires a different coordinate. $\square$

Corollary 8.10 (Canonical area and maximum mesh). *In the joint coupling of Theorem 8.9, with limiting normalization f ,*

$$(\phi_n)_* m_n \Rightarrow f_* \mu_h, \quad \max_T \text{diam}_\rho \phi_n(T) \longrightarrow 0.$$

The measure on the right is the area of the quantum sphere in the same three-marked coordinate; a common reflection is retained when the correspondence reverses orientation. In particular, the maximum-mesh conclusion holds in probability along the full sequence.

Proof. Write $f_n = \phi_n \circ H_n^{-1}$. Uniform convergence and the reference area convergence prove the measure assertion. If ω_f is a modulus of continuity of f on the sphere, then

$$\max_T \text{diam}_\rho \phi_n(T) \leq 2\|f_n - f\|_\infty + \omega_f\left(\max_T \text{diam}_\rho H_n(T)\right) \rightarrow 0$$

by (8.1). Every sequence has a further joint extraction with these conclusions, which proves the stated in-probability assertion about mesh. The three samples and ϕ_n throughout are the prescribed finite ones. $\square$

Corollary 8.11 (Area-typical root position). *The exploration root has diffuse limiting area position in the canonical coordinate. This conditional sampling statement remains valid jointly with any root-free map observations retained in the same extraction.*

Proof. Use the exact conditional rerooting statement in Lemma 2.2. A uniform oriented-edge tail can be coupled to a uniform flag by choosing one of its two sides, and then to an independent uniform point of that flag. The two canonical positions differ by at most the maximum flag diameter. Conditional on the root-free map data and the normalization samples, this gives the same limiting continuous-test integrals as sampling from $(\phi_n)_* m_n$. Apply Corollary 8.10. The resulting area measure is nonatomic. $\square$

9 The complete collection of interfaces

We identify every macroscopic interface, including its traversal order and its place in the nested collection. Write $H_n : X_n \rightarrow \widehat{\mathbb{C}}$ for the homeomorphisms of the topological correspondence and $\eta : [0, 1] \rightarrow \widehat{\mathbb{C}}$ for the limiting space-filling exploration, in the common coupling with the surface and all retained observations. In particular, $\eta(0) = \eta(1) = p_*$, the images under H_n of the flag triangles have maximum diameter tending to zero, and the spatial representatives of the discrete tour converge uniformly to η . All the results of this section concern that correspondence; the final coordinate change is made in Proposition 9.12.

There are two steps. Exact surgery on the finite exploration tour represents each FK loop by a face of a noncrossing chord system. Convergence of all these faces gives a complete collection of ordered limiting curves, with no extra macroscopic loops. We then identify these curves with CLE. Exploration switches first identify each curve as a closed initial portion of a CLE traversal; a two-sided winding test rules out any nonconstant omitted portion.

The loop readout follows Sheffield's exploration-tree construction of CLE [41] and its realization through imaginary-geometry flow and counterflow lines [35]. The alternating nested exploration is also used in [3, Section 2.1.6]. Finite-volume contour and cone-time convergence comes from Gwynne and Sun [21]; the argument here upgrades it to simultaneous ordered-curve convergence in the canonical conformal embedding.

9.1 Flexible orders as a closed chord system

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, identified with the boundary of the closed unit disk $\overline{\mathbb{D}}$ by $t \mapsto e^{2\pi it}$. For $a, b \in \mathbb{T}$, write $[a, b]_{\mathbb{d}}$ for the straight chord joining their images. This notation distinguishes a disk chord from an interval in exploration time. A *chord system* is a closed set of unordered endpoint pairs, including all diagonal pairs. Its underlying lamination is the union of its chords and $\partial\mathbb{D}$. Noncrossing means that two distinct chords have disjoint relative interiors. A *face* is a connected component of the complement of the lamination in $\mathbb{D}$; faces are open convex sets.

For the length- $2n$ word, fix the distinct occurrence times

$$t_k = \frac{k - 1/2}{2n}, \quad 1 \leq k \leq 2n.$$

Here t_k labels the visit in the interior of the k th step of the abstract medial tour. It does not assert an exact scalar-chord identity at the interpolated contour height. For a flexible order at position j , with matched burger at $i < j$, use the chord $[t_i, t_j]_d$, labelled by that burger's type. Adjoin the diagonal pairs to obtain $\mathcal{E}_n$. Every letter belongs to at most one flexible pair, so these chords have distinct endpoints, even for consecutive matched letters. This convention is needed for the exact face surgery below. The adjacent-cut times used in the contour limit differ by at most $1/(2n)$; their replacement by t_i, t_j preserves all limiting endpoint assertions. Spatial tracking at these occurrence times follows from tracking at adjacent cuts, continuity of η , and the vanishing image diameter of the intervening flags.

For the limiting excursion $Z = (L, R)$, a backward cone interval is

$$[v(t), t], \quad v(t) = \inf\{s < t : L(u) \geq L(t), R(u) \geq R(t) \text{ for } u \in [s, t]\}.$$

Here t is required to admit at least one such interval of positive length. Away from the end convention, precisely one coordinate agrees at its two endpoints; this coordinate labels the entrance wall. We use the corresponding burger-type label. Let $\mathcal{E}$ consist of the circular endpoint pairs of these intervals and all diagonal pairs.

Lemma 9.1 (Joint convergence of the chord systems). *The coupling may be chosen so that $\mathcal{E}_n \rightarrow \mathcal{E}$ in the Hausdorff topology on unordered pairs of points of $\mathbb{T}$, jointly with all previously retained observations. Both systems are noncrossing. For every convergent sequence of nondegenerate discrete chords whose limiting endpoints have representatives $0 < a < b < 1$, the matched entrance times converge to $v(b) = a$ and the labels eventually agree. Every chord of $\mathcal{E}$ is approximated by discrete chords. The system $\mathcal{E}$ is a measurable function of Z .*

Proof. If flexible matches interlaced, say $i < i' < j < j'$, the burger at i' would still be available when the flexible order at j takes the older burger at i . This contradicts freshness. Thus the discrete chords are noncrossing, and the same holds for their limits.

We use the full statement of Gwynne–Sun [21, Theorem 1.11], with its time interval $[0, 2]$ rescaled to $[0, 1]$. The theorem is for the empty-reduced word at each positive integer n and each fixed inventory parameter in $(0, 1/2)$. Its second assertion identifies all subsequential limits of flexible orders with nonvanishing interval span; its third assertion gives convergence of both entrance quantities and eventual agreement of the wall label. Its fourth assertion simultaneously approximates maximal cone intervals in every rational-endpoint open interval containing a specified rational interior mark. These statements are stronger than uniform contour convergence.

Apply these assertions first at the adjacent-cut times of the cited theorem. The deterministic one-mesh bound just noted gives the same endpoint limits and wall labels at our occurrence times. In particular, it gives the upper endpoint-pair limit. To check the lower limit explicitly, fix $[a, b] = [v(b), b] \subseteq (0, 1)$. Choose open intervals I_k with rational endpoints, containing $[a, b]$, whose closures decrease to $[a, b]$, and fix a rational point of (a, b) . The maximal cone interval in I_k containing that point contains $[a, b]$: cone intervals containing the same interior point are ordered by inclusion. Its endpoints therefore tend to a, b . For each k the distinguished discrete maximal intervals converge by the cited theorem. A diagonal choice approximates $[a, b]$. Equivalently, first apply the theorem simultaneously to the countable family of rational tests and then take

their closure. Compactness of the endpoint-pair space turns the upper and lower conditions into Hausdorff convergence.

There is no additional nondegenerate circular chord at an excursion end. For example, a sequence with one endpoint tending to 0 and the other tending to an interior time would force an interior coordinate value to be zero. A sequence ending at 1 with its entrance bounded away from 0 would do the same, using equality in the entrance-wall coordinate. Both contradict strict interior positivity. The sole possible full-duration interval has equal endpoints on $\mathbb{T}$ and is already included among the diagonal pairs. The resulting closed chord system is determined by the excursion. Hence retaining it causes no ambiguity in the joint subsequential couplings: its limiting conditional law given the excursion is a point mass. $\square$

9.2 The exact finite surgery

Let $\mathcal{L}_n$ be the lamination underlying $\mathcal{E}_n$. A face boundary can be read cyclically by following the time circle on its exposed arcs and using its chord sides to skip the intervening intervals. The circle here records the resolved-tree medial tour, not a second conformal embedding of the map.

The resolved spanning tree has a disk as a small regular neighborhood. Its boundary is the simple medial tour C_n ; the complementary disk is the dual-tree side. The two marked visits associated with a flexible pair lie in the two exploration triangles of one edge quadrangle. The occurrence times fixed above place each mark inside its own visit. Sliding a mark within that visit leaves the four-port pairing unchanged.

Lemma 9.2 (Flag drawing and face surgery). *The faces of $\mathcal{L}_n$ are in bijection with the polygonal FK loops specified in the model. Under this bijection, the exposed time-circle arcs give their cyclic tour order. A transit along a chord changes the spatial route only within its matched edge quadrangle, with an adjacent tour-step error at its endpoints.*

Suppose a face is entered across an outer chord with interval $[a, b]$ and has an immediate child chord with interval $[c, d] \subset (a, b)$. A test point on the tour strictly inside (c, d) lies on the opposite side of the face loop from a test point on the outside time arc if and only if the outer and child chord labels differ. This assertion is made with test points off the surgery quadrangles, or after taking arbitrarily small strict spatial margins.

Proof. In an unoccupied flag triangle the retained corner is cut off by $v = 2/3$. In an occupied triangle that corner is already contained in the strip $f \leq 1/3$, and the boundary is $f = 1/3$. On a common vertex–face-center side these two rules meet at the same point. On the other two side types the occurrences are paired within the same edge quadrangle and have the same occupation status. Consequently the four ports of an edge quadrangle have exactly the usual two FK pairings. They are shown in Figure 2. This verification is occurrence by occurrence, so loops and bridges do not identify distinct ports accidentally.

Starting from C_n , restore the FK pairing at each flexible quadrangle. Cut its two marked visits and reconnect the four cut ends by the other noncrossing pairing. In the abstract disk this is a cut along $[t_i, t_j]_d$. Noninterlacing ensures that the chord lies in one current face, and its distinct endpoints divide that face into two. Induction therefore identifies the final cyclic routes with all disk faces, giving $1 + N_F$ of each. Each reconnection stays in its own quadrangle. The polygonal flag drawing has the same four-port pairings, so it has exactly these cyclic routes in the exposed time-circle order. This argument retains even the smallest finite loops; it does not replace a chord between consecutive occurrences by a diagonal pair.

For the side assertion, realize each switch by a narrow bridge in the disk on the side of its resolved-tree edge. The two burger types use the two opposite disks of C_n . Within either disk the

bridges may be chosen disjoint because their endpoints do not interlace. Fix an outer bridge with interval $[a, b]$, its child bridge $[c, d]$, and a reference point on $C_n(b, a)$. If the bridges use the same disk, the face route bounds the strip between them. The skipped arc $C_n(c, d)$ and the reference arc $C_n(b, a)$ lie on the same side of this route. If the child bridge uses the other disk, the skipped arc $C_n(c, d)$ lies between that bridge and the outer bridge, while the reference arc remains on the other side. Thus the two test points are separated exactly when the bridge types differ; see Figure 5. Subsequent reconnections are disjoint from this face route and cannot change either test point's component. This proves the criterion. The same conclusion can be checked directly from the detailed component identification in [23, Lemmas 2.7 and 2.9]. $\square$

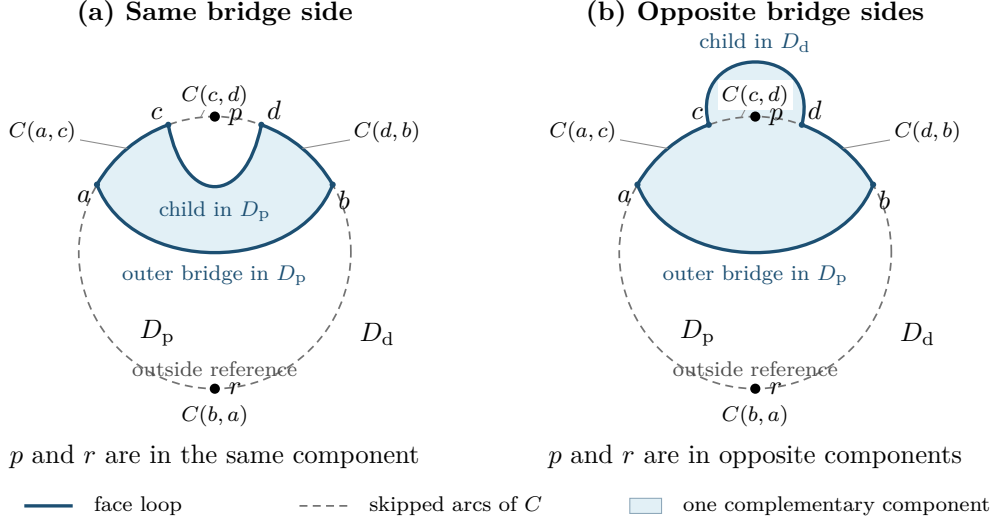


Figure 5: The side test for an outer switch $[a, b]$ and its child $[c, d]$. The circle is the spatial resolved-tree tour C , separating the primal and dual disks D_p, D_d . A point p on the skipped child arc lies with the exterior reference point r exactly when both bridges use the same disk. The bridge curves represent narrow surgery strips in a topological drawing; their lengths do not represent distances in the flag triangles.

Remark 9.3. The finite drawing is a regular neighborhood of $(V(M_n), A_n)$. If it has $k(A_n)$ components and b_n boundary loops, its Euler characteristic is both $|V(M_n)| - |A_n|$ and $2k(A_n) - b_n$. Hence $b_n = 2k(A_n) + |A_n| - |V(M_n)| = \ell(M_n, A_n)$. The surgery description likewise gives one plus the number of flexible orders. Thus the face bijection counts the actual interfaces of the specified ensemble.

9.3 From chord convergence to complete loop matching

Let $\mathcal{L}$ be the lamination underlying $\mathcal{E}$. For a chord with endpoints a, b , the peanosphere identification gives $\eta(a) = \eta(b)$. Thus there is a map

$$Q : \mathcal{L} \longrightarrow \widehat{\mathbb{C}}, \quad Q(e^{2\pi it}) = \eta(t), \quad Q|_{[a, b]_d} \equiv \eta(a).$$

It is well defined and continuous. To see continuity, consider a convergent sequence of points on chords and extract convergent endpoint pairs. At a limit point in the open disk the limiting chord fixes the value. At a limit point of the time circle, strict convexity of the disk forces that point

to be an endpoint of the limiting chord, so the same conclusion follows from $\eta(a) = \eta(b)$ and continuity of η . For a face F of $\mathcal{L}$, define ℓ_F to be the cyclic curve $Q|_{\partial F}$. Constant curves will be discarded.

We state the compactness argument in a form that also controls loops whose indexing face varies with n .

Lemma 9.4 (Convex cells and cyclic curves). *In the common spatial correspondence the following assertions hold.*

1. *For every face F of $\mathcal{L}$ and every $x \in F$, eventually there is a unique face F_n containing x , and $\overline{F}_n \rightarrow \overline{F}$ in Hausdorff distance. The associated spatial FK loops converge to ℓ_F in uniform distance modulo cyclic increasing reparameterization.*
2. *If F_{n_j} is any sequence of faces, then after further extraction $\overline{F}_{n_j}$ has a compact convex limit K . If K has interior, it is the closure of one face of $\mathcal{L}$ and the corresponding spatial loops converge cyclically to that face loop. If K has no interior, the spatial loop diameters tend to zero.*
3. *The multiset of nonconstant ℓ_F is locally finite on the sphere. For every $\varepsilon > 0$, eventually there is a partial bijection between the discrete and limiting face loops which covers every loop of spherical diameter greater than ε , and each matched pair is at individual-loop distance at most ε .*

Proof. Chord endpoint convergence implies Hausdorff convergence of the underlying laminations. A compact connected subset of F has positive distance from $\mathcal{L}$ and therefore lies in one discrete face eventually. Exhausting F by such compact subsets gives $\overline{F} \subset \liminf \overline{F}_n$. Every boundary chord of F is approximated by discrete chords, and F_n lies on the side of each approximating chord selected by x . The intersection of these closed half-disks is $\overline{F}$. This gives the reverse inclusion and proves the first cell assertion.

For a varying sequence, compactness of the space of compact convex subsets of $\overline{\mathbb{D}}$ gives a limit K . If K has interior, no limiting chord can cut that interior: an approximating chord would cut the corresponding discrete face. Choose an interior point of K outside the lamination. The first part then identifies K with a face closure. Such a point exists, since otherwise the lamination would contain an open disk in K ; a chord through that disk would again cut K , which is impossible.

For the curve assertion in this case, parameterize ∂F_n and ∂F by radial bearings from a fixed interior mark. Their radial functions converge uniformly. Indeed an interior disk about the mark is contained in all sufficiently large F_n , and convexity then bounds the radial modulus uniformly; pointwise convergence follows from the cell convergence. These boundary parameterizations preserve cyclic order. On exposed circle arcs, the spatial routes converge to η uniformly. On a chord side, Lemma 9.2 confines the route to a matched quadrangle, whose image diameter tends to zero. Every subsequential limit of such a route is therefore the common endpoint value of Q . The compactness argument used above to prove continuity of Q makes this convergence uniform on the parameter circle. Pauses inserted at chord endpoints are harmless: continuous nondecreasing changes of time are uniform limits of strictly increasing changes of time and give the same zero-distance class in the stated loop topology.

Suppose instead that K has no interior. It is a point or a line segment. A closed convex cell is the convex hull of its contacts with $\partial \mathbb{D}$. Extracting those compact contact sets shows that K is the convex hull of its limiting circle contacts. There are at most two such contacts, since three distinct points of the circle are noncollinear. In the two-contact case, a boundary chord of a discrete cell converges to their connecting segment. For clarity, take a point strictly between the

two contacts and boundary points on either side of it tending to the segment; these points are on chord sides, and any limiting side through an interior point of the segment has exactly those two circle endpoints. The endpoints consequently have equal Q values. All exposed-circle route values and all chord-side route values have that same subsequential limit. Thus the spatial diameters tend to zero.

To prove local finiteness, suppose distinct limiting faces had loop diameters bounded below. Their closures have a subsequential convex limit. The preceding collapsed-cell argument, applied with the fixed lamination, rules out an empty interior. If the limit has interior, an interior disk is eventually contained in every face in that subsequence, contradicting disjointness of distinct faces. Thus only finitely many limiting face loops have diameter above any positive threshold.

Fix $\varepsilon > 0$. Match all limiting face loops of diameter greater than $\varepsilon/2$ to the discrete faces selected by their interior marks. There are finitely many, their selected faces are distinct for large n , and their cyclic errors tend to zero. If unmatched discrete loops of diameter greater than ε remained along a subsequence, Part 2 would produce a noncollapsed limiting face loop of diameter at least ε . Its interior mark would eventually belong to the unmatched discrete face, contrary to the selected matching. This proves the final assertion, and a diagonal sequence of ε gives the claimed loop-ensemble topology. $\square$

9.4 The alternating exploration convention

We now identify the limiting curves. Put the exploration root at ∞ , and adjoin independent uniform times $U_1, U_2, \dots$. Their images $z_j = \eta(U_j)$ are area-typical targets. Almost surely these times are dense, the targets are dense in the sphere, and they avoid all CLE loop traces. We work on one event on which the assertions below hold for all these targets and all rationally truncated exploration windows. Indeed the countable construction of the CLE loops and the frontiers at rational times in a smooth spherical-area parameterization are determined before the independent quantum field is sampled. Their traces have zero Lebesgue area and consequently zero quantum area almost surely. A multiply visited point lies on such a frontier between two of its visits. Thus typical targets also have unique visit times.

The exploration tree is the one determined by the imaginary field in Section 5. Its branch toward z_j is obtained from the space-filling curve by erasing the fillings disconnected from z_j . After a strict initial cut it is radial $\text{SLE}_\kappa(\kappa - 6)$ in its target component. The marked boundary prime end and the growing tip have angular separation $\theta \in [0, 2\pi]$. We use the following convention for extracting *nested* CLE from this tree: traverse a full excursion of θ from one endpoint of $[0, 2\pi]$ to the other, complete its loop by the branch aimed at the excursion's starting point, and repeat in the remaining target component. The completing branch approaches the later prime-end lift of that point from the component separated from the target at the end of the excursion. This is the approach which makes the starting point its terminal second visit [3, Section 2.1.6 and Figure 4]. The directions alternate. This precise iteration is described in [3, Remark 2.3 and Sections 2.1.6–2.1.7]. The whole-plane branching construction is that of [25, Section 2.3.1]; its whole-plane and local radial laws are those of [35, Section 2.1.3].

Lemma 9.5 (The unmarked law of the alternating construction). *The ensemble obtained by this alternating construction from the whole-plane tree has the law of whole-plane CLE_κ . It is locally finite, its loops do not trace themselves or each other, and its unmarked law is invariant under Möbius transformations.*

Proof. First consider a radial branching tree in a proper simply connected domain D , with force point immediately on either side of its boundary root. The alternating construction is the nested-

domain construction of [3, Section 2.1.6]. Its first full crossing, completed through the separated component, gives an outermost loop. The component of its complement containing a target is exactly the radial target component at the crossing time. Conditional on the explored branches and completions, the unexplored trees in the components are independent radial trees. At the next generation the force side is reversed, as follows from $\theta \mapsto 2\pi - \theta$ and the strong Markov property [3, Remark 2.3]. Reflection preserves the unmarked domain CLE law. Induction for finitely many targets and generations, followed by the countable union, therefore gives the independent-domain iteration defining nested CLE_κ . This also gives consistency at a discovered interior bubble: its descendant loops are exactly those of the continued tree in that bubble, and other loops of the domain ensemble cannot enter its interior. Thus the two crossing directions give consecutive generations in this construction.

We pass to the whole plane using stopping domains. Choose a deterministic target, put it at 0, and use the bi-infinite Loewner time of its branch from ∞ . Define

$$S_m = \inf\{t \geq -m : \theta(t) \in \{0, 2\pi\}\}, \quad T_m = \inf\{t \geq S_m : \theta(t) = 2\pi - \theta(S_m)\}.$$

These are stopping times. Full angular crossings occur arbitrarily far toward the initial end [25, Lemma 2.7], so $T_m \rightarrow -\infty$ almost surely. Explicitly, fix any earlier full crossing. Once $-m$ precedes its starting endpoint visit, T_m is no later than its ending visit: whichever endpoint is hit first after $-m$, both endpoints have been visited by then.

Let D_m be the component containing 0 after T_m . The tip and force prime end coincide in the endpoint state $\theta(T_m)$. The whole-plane strong Markov property and conditional independence of branches after target separation imply that the unexplored tree in D_m , conditionally on the exposed branch, is a radial branching tree with the indicated endpoint force side [35, Proposition 2.2 and Section 2.1.3]. The preceding domain argument shows that its future ensemble $\Gamma^{(m)}$ has conditional law CLE_κ in D_m . Its curves lie in $\overline{D}_m$ and may touch ∂D_m .

We check consistency with the global readout using only these domain ensembles. Fix a globally generated loop G with target w and a finite starting crossing. For sufficiently large k , the cut T_k precedes both the separation of w from 0 and that crossing. Then $w \in D_k$, its branch shares the exposed prefix, and G is a loop of $\Gamma^{(k)}$: every subsequent full crossing and its completion are computed from the same restricted tree. Increase k so that $T_k < T_m$ as well. In the domain construction for $\Gamma^{(k)}$, T_m is a later full-crossing time toward 0, and D_m is the corresponding interior bubble. Domain consistency therefore says that G , if its trace enters D_m , is a descendant loop generated by the continued tree there, hence belongs to $\Gamma^{(m)}$. Conversely every loop of $\Gamma^{(m)}$ is a global crossing loop. There are countably many cuts and generated loops, so these statements hold simultaneously. In particular the global and future readouts agree on all loop tests in a compact subset of D_m , including loops which touch ∂D_m elsewhere.

The branch tends to ∞ at its initial end. Thus every fixed disk lies in D_m eventually: for sufficiently negative T_m the entire exposed past lies outside that disk, whose connected interior belongs to the 0-component. The domain limit theorem [37, Theorem A.1] identifies the limit of CLE_κ in such domains with the unique whole-plane law. Its conformal matching estimate is uniform over all proper simply connected domains, with error tending to zero as their boundary recedes from a fixed disk. It applies conditionally to D_m : bounded continuous tests of the loop configurations in a fixed compact set converge to the fixed whole-plane expectation, and dominated convergence removes the conditioning. The local agreement just proved identifies the global alternating readout with this limit. It also transfers local finiteness and the absence of shared arcs directly from a domain ensemble, since every compact set is eventually inside some D_m .

The whole-plane construction in [25, Section 2.3.2] uses a positive-crossing subsequence. Its

agreement in law with this same large-domain limit is stated in Remark 2.6 of that paper. Consequently its Möbius-invariance theorem [25, Theorem 1.1] applies to the present unmarked ensemble. $\square$

We use no tracing in the strong sense of [25, Definitions 2.4–2.5]: no nonconstant open portion of a traversal has its image contained in the union of the complementary trace and the traces of finitely many other loops. Reference CLE traversals are parameterized without constant intervals; pauses will be allowed when comparing them with face contours. This image-containment statement, rather than merely the absence of an identical parameterized subarc, will be needed below.

The next dictionary relates three orders. The interval $[a, b]$ uses the quantum-area time of the space-filling curve: a enters a filled bubble and b completes it. Along a target branch, larger enclosing intervals are encountered first. Its moving clock erases the off-target fillings while retaining the ordered branch trace. The two endpoints of a full angular crossing in branch time will be denoted by σ, τ ; they need not be the area times a, b .

Lemma 9.6 (Cone intervals and exploration switches). *For each typical target time u , the cone intervals containing u describe the successively filled target bubbles. Their order along the branch toward $\eta(u)$ is reverse inclusion. Their entrance-wall labels are the two endpoint states of θ , up to a fixed interchange. Consequently a last closure of one type before a closure of the other type starts the between-tips portion of a nested CLE loop. The tree completes that loop inside the larger interval's filling.*

These switches recover all loops surrounding the target, once each. Every typical target surrounded by such a completed loop belongs to its outer interval's filling. Enclosing cone intervals approach the full time interval at the root end, and arbitrarily small such intervals contain u .

For every fixed strict interior horizon, ancestor-free erasure has a continuous moving trace. In every nonempty interval of its moving clock, erased intervals of both entrance-wall types occur.

Proof. We give the frontier dictionary in the cut surface, where different prime ends over one spatial point remain distinct. Color the left side of the branch red and its right side blue. The boundary of its current target component consists, in prime-end order, of a red arc and a blue arc between the tip W and the marked junction O . These are the boundary colors and exploration order of [3, Section 2.1.7 and Figure 9]. The force prime end is precisely this junction O : it is transported by the Loewner map until a collision, and the same boundary-color rule continues it after the collision. The two prime-end arcs have strictly positive quantum length when they are nontrivial. Therefore one color is absent exactly when O and W coincide on the corresponding side, that is, when $\theta = 0$ or $\theta = 2\pi$.

We use the precise bubble readout of [20, Section 2.3.4, Definition 2.2 and Figure 4; Lemma 3.1]: a filled bubble has interval $[v(b), b]$, its boundary length is $|Z(b) - Z(v(b))|$, and its side is specified by the coordinate equal at the two endpoints. This statement uses backward cone intervals, with the same entrance convention as here. The following frontier verification explains its use on the sphere. When a monocolored component is entered at time a , the space-filling order visits all its descendants before returning to a different component; write b for its completion time. Its interior is visited only during this block, although its boundary points may be visited again. The frontier outside this component is unchanged throughout its filling. Subtracting its terminal heights therefore gives the two remaining frontier lengths

$$(L(s) - L(b), R(s) - R(b)), \quad a \leq s \leq b. \quad (9.1)$$

They are nonnegative, both are zero at b , and exactly one is zero at a . The first entry a is the entrance $v(b)$: in the running-minimum frontier construction of Theorem 3.4, extending the block farther backwards crosses one of its two terminal frontier levels.

The converse uses the same frontier construction. For a cone interval $[v(b), b]$, cut the two frontier trees at heights $L(b)$ and $R(b)$ and retain its filling block. The horizontal identifications inside the block stay above those cuts. Its unfilled boundary arcs are parameterized by the height intervals in (9.1); at its first time the arc for the entrance wall is empty. Maximality of $v(b)$ prevents a preceding block from remaining on these same two sides of the cuts. Thus this is exactly the monocolored component entered there. If the interval contains u , it is the branch's target component, since its interior contains $\eta(u)$; if it precedes u , it is an erased off-target filling. This verifies both directions of the readout without imposing a new law on the field inside the component. Erasing earlier off-target fillings does not change these two boundary arcs in the target component. The argument concerns the restricted quotient and its length coordinates on $[a, b] \subseteq (0, 1)$, which are the same in the sphere realization of Theorem 3.4; no infinite-volume surface law is being assigned to that block.

Combining (9.1) with the prime-end description identifies the entrance-wall labels with the two endpoint states of θ , up to interchanging the colors. Target components shrink along a branch, and their contiguous filling blocks are consequently ordered by reverse inclusion. The exact backward-cone erasure for a target branch is also stated in [3, Section 4.2 and Definition 4.14]; the ancestor-free construction underlying it is [16, Proposition 1.13 and Lemma 10.4].

Starting at either state, the last visit to it before reaching the other is the beginning of a full crossing of $[0, 2\pi]$. The alternating constructor therefore gives precisely the stated switches and their between-tips portions. Conditional on that portion, the remaining piece is chordal SLE in the specified prime-end component, as in [3, Section 2.1.6] and [25, Section 2.3.2]. More precisely, let σ, τ be the starting and ending branch times of the full crossing, and view the target bubble at σ in its prime-end disk. The tip and force point coincide at its designated boundary root x . The completing branch approaches the later lift of this same x through the component separated from the target at τ . The excursion and completion therefore give a closed continuous curve β in the closed prime-end disk, starting and ending at x . Both pieces lie in the larger cone interval's filling.

Let f map that disk to the outer bubble. Its continuous extension exists because the boundary is locally connected. The closed curves $f((1 - \varepsilon)\beta)$ lie in the bubble and converge uniformly to the completed loop as $\varepsilon \downarrow 0$. Simple connectedness gives zero winding about every point outside the bubble for each approximant. Winding stability gives the same assertion for the completed loop at every off-trace point outside the bubble. Thus every surrounded typical target is in that bubble.

Iteration supplies exactly the successive nested loops about a target. Their domains decrease to that target. In the whole-plane version they also exhaust toward the initial point. Equivalently, the full crossings of the recurrent angular diffusion occur at all scales; the whole-plane branch is transient at its initial end and tends to its target at its terminal end. For any fixed compact subinterval of $(0, 1)$, large enclosing bubbles contain its image. Their fillings therefore contain that time subinterval, and their endpoints tend to 0, 1.

Finally, for the unconditioned Brownian contour, the two boundary processes after ancestor-free erasure, read in forward branch order, have negative jumps and stable index $\kappa/4 \in (1, 2)$, one for each side [16, Proposition 1.13]. This reverses the time direction of the positive-jump processes in that statement. Each Lévy measure has infinite mass near zero, so each coordinate has jumps in every nonempty clock interval. The jumps are precisely the erased cone intervals of the indicated wall type. The time-changed trace is continuous directly: every inverse-clock jump skips an interval $[a, b]$ with $\eta(a) = \eta(b)$, and η is continuous. This is the inverse-clock argument in [16,

Proposition 10.6]. The erasure retains the full ordered branch traversal; the skipped intervals are space-filling visits to off-target bubbles, not moving pieces of that branch. Thus a portion on which the branch changes position occupies a nonempty interval of its moving clock. For the excursion, exhaust by strict interior rational windows containing the full Brownian-time span of the tested clock interval, including all its erased intervals. Local absolute continuity transfers the null event that one label is missing there. Taking the countable intersection gives the assertion for rational and independent uniform horizons. Only this almost-sure property is transferred; the excursion's time-changed coordinates need not be Lévy processes. $\square$

9.5 Ordered prefixes of the limiting face curves

We now return to the cyclic curves supplied by Lemma 9.4. The exploration dictionary will identify successively longer prefixes of each such curve with one CLE traversal. First, contour genericity makes the order of a parent and its child unambiguous.

Lemma 9.7 (Strict endpoints of nested cone intervals). *If two nondegenerate cone intervals in $(0, 1)$ are distinct and one contains the other, then both containments of their endpoints are strict.*

Proof. Two intervals with the same terminal time have the same entrance $v(t)$, so are equal. Suppose instead that $[a, d]$ is properly contained in $[a, b]$ and has the same entrance. Interchange the two coordinates if necessary so that the outer wall is R : $R(a) = R(b)$ and $R \geq R(b)$ on $[a, b]$. If the inner wall is L , its cone inequality and the outer inequality give $R(a) \geq R(d) \geq R(b) = R(a)$. Hence $[a, d]$ is both an L -chord and an R -chord, contrary to Lemma 5.4(i). If the inner wall is also R , then $R(a) = R(d) = R(b)$, and d is a scalar local minimum at that level, strict by Brownian scalar genericity. But b is a backward joint record with positive witness, so its R -height cannot be a scalar branch level by Lemma 5.4(ii). This rules out a shared entrance as well. $\square$

The *outer chord* of a face means its boundary chord which separates it from the time-circle point 0. If its interval is $[a, b]$, its *children* are the other boundary chords, with intervals inside (a, b) . They are the maximal proper cone intervals skipped by that face. Write $K_F = \partial F \cap \partial \mathbb{D}$ for its circle contacts, read in increasing time from a to b . Extending $\eta|_{K_F}$ constantly across each child interval gives the ordered face contour $\alpha_F : [a, b] \rightarrow \widehat{\mathbb{C}}$. Its two endpoint values agree.

Lemma 9.8 (Ordered prefixes from children). *Every face with interior has an outer chord with endpoints in $(0, 1)$. If an immediate child has the opposite label, the portion of α_F from a to that child's entrance is a nonconstant initial portion of the CLE loop indexed by their switch.*

Every switch toward a typical target has exactly one incident face with this prefix. For every nonconstant face contour there is a sequence of opposite-type children whose identified prefixes extend one another and whose remaining contour diameters tend to zero. In particular, α_F is a closed initial portion of one oriented CLE traversal, up to pauses.

Proof. The enclosing chords of Lemma 9.6 approach the full time interval. An interior point of a face eventually lies between one of those chords and the opposite circle arc. The maximal circle gap of that face containing time 0 therefore has endpoints in $(0, 1)$. Its bounding chord is the asserted outer chord. Almost every time in (a, b) is inside a child: typical times have arbitrarily small enclosing cone intervals, whereas no cone chord can cut the interior of F . We henceforth impose this fact for the independent uniform targets as well as for almost every time.

Let $[c, d]$ be a child and take a target horizon $r \in (c, d)$. Ancestor-free erasure toward r , restricted to $[a, c]$, is exactly the face contour. Here is the elementary nesting check. Every child

interval before c is a filling disconnected from that target and is erased. Conversely a deleted cone interval meeting a retained circle contact would have to cross the outer chord or a child chord, unless it were contained in one of the skipped children. Crossing is impossible, and the latter case deletes no further face contact. Endpoint differences are constant pauses. Thus the two procedures retain the same points in the same order.

For an opposite-type child, there is no intermediate containing interval between it and $[a, b]$, by maximality of the child. Lemma 9.7 gives $a < c < d < b$. Their closures are consequently the two endpoints of a full-winding excursion in Lemma 9.6. The preceding erasure check identifies the face prefix with its between-tips branch segment. That segment is nonconstant: a full crossing of the angular interval makes a nontrivial Loewner hull. Its traversal order is the branch order.

Conversely, take the two consecutive nested intervals of a target switch. A chord from the target time on the circle to time 0, outside the larger interval, crosses both bounding chords. The open segment between those crossings meets no lamination chord. A chord meeting it would separate these two circle points and therefore give an intermediate interval containing the target. The endpoints of the segment are distinct, since the two bounding chords are distinct and their intersections are in their relative interiors. Closedness of the lamination puts this open segment in one complementary face. That face has the two specified chords on its boundary. This proves existence; the side of the outer chord fixes uniqueness. It also proves that the face has interior even when chords accumulate on either side of the branch.

We next make the exhaustion assertion precise. On a varying compact subarc of α_F before a later varying portion, choose a child strictly after it and a rational horizon inside that child. Such a child exists because almost every intervening time is in a child, and a child cannot skip two retained contacts with distinct images. The preceding erasure check identifies the earlier subarc with an interval of this horizon's ancestor-free trace. By Lemma 9.6, this is the full ordered moving trace. Its portion between distinct spatial values therefore occupies a nonempty interval of the moving clock: an interval collapsed by that clock is a skipped filling whose two endpoint images agree. On this compact earlier part of the trace, the erasure check identifies its retained set with the face contacts, up to endpoint pauses. The jumps of the inverse clock are the maximal open gaps of this retained set, hence precisely the immediate child intervals there. An interval inside an already skipped child is not another inverse-clock jump. The compact restriction before the later child's entrance removes any cutoff at the target horizon. The last assertion of Lemma 9.6 therefore gives immediate children of both labels on the chosen varying portion. This use of the stable processes is simultaneous for all rational horizons and all interior windows; the face and subarc may therefore be chosen afterwards.

Set $x_0 = \alpha_F(b)$. Given a sufficiently small $\delta > 0$, let t_δ be the last time with $\rho(\alpha_F(t_\delta), x_0) \geq \delta$. There is a varying portion after t_δ and before the contour becomes constant at x_0 . Apply the preceding paragraph to a compact part of that variation, leaving some variation after it. An opposite-type child entrance $c_\delta > t_\delta$ occurs there, and

$$\text{diam}_\rho \alpha_F([c_\delta, b]) \leq 2\delta.$$

Choosing successively smaller δ and later varying subarcs makes the entrances nondecreasing. If a constant terminal interval is present, the entrances approach its first point and the same bound holds. This produces nested nonconstant prefixes with vanishing remaining diameter.

All these prefixes belong to the same CLE loop: each contains the first nonconstant prefix, and distinct CLE loops share no nonconstant arc. They also follow the same oriented traversal from the same starting prime end. No self-tracing prevents switching to a different visit of that initial arc. Parameterize this traversal on $[0, 1]$; the prefix endpoints are nondecreasing parameters,

bounded by 1, and hence converge to a parameter s . Continuity of that traversal and the vanishing remainder diameter identify the face contour with the traversal up to s , allowing pauses. Since the face contour is closed, it is a closed initial portion, as asserted. $\square$

9.6 A winding test for an omitted strand

The prefix lemma leaves a specific possibility: a face contour might close before its CLE traversal has finished. Equality of traces would not settle this question of order. We instead show that any omitted moving portion changes winding around an open set, which the finite side test can detect.

For an oriented closed curve A and a point z outside its trace, $\text{Ind}(A, z)$ denotes its integer winding number. We use the orientation inherited from the exploration; a CLE loop has index zero or its designated sign. In particular its nonzero-index components are exactly its surrounded components. No rectifiability is needed: the index is the degree of the map from the parameter circle to the unit circle viewed from z .

Lemma 9.9 (An omitted moving portion changes a winding test). *Almost surely, simultaneously for every loop L of the exploration ensemble, the following holds. Let P be a closed initial portion of its oriented traversal. If the omitted portion is nonconstant, there is an open disk V disjoint from the trace of L such that*

$$\text{Ind}(P, z) \neq \text{Ind}(L, z), \quad z \in V.$$

Consequently equality of these indices on a dense off-loop target set forces P to be the whole traversal, up to pauses.

Proof. The deterministic step is a local index comparison. Suppose that a closed curve C and L share an oriented compact segment A , and write $L = A * B$, $C = A * E$. If the complementary arcs B, E miss a connected ball U , then additivity of degree gives

$$\text{Ind}(L, z) - \text{Ind}(C, z) = \text{Ind}(B * \overleftarrow{E}, z), \quad z \in U \setminus L. \quad (9.2)$$

The right side is constant on U . Consequently, if two off-trace points of U have different indices for C , they also have different indices for L . If P misses U , its index is constant there and cannot agree with L at both points. Local constancy off compact traces then gives the disk V in the statement.

We will apply this comparison with C a chordal SLE closed along a boundary side. To cover an initial portion chosen after seeing the loop, we first construct one simultaneous family of comparison paths. Every moving open portion of a loop contains an interior segment with law absolutely continuous with respect to a segment of ordinary chordal SLE_κ . Indeed each loop is its full-winding branch piece followed by its conditional chordal completion. In the first piece use countably many windows stopped while the driving and force points are separated by at least $1/m$, in a fixed bounded time window after a strict initial cut. The force drift is bounded there, and coordinate change followed by Girsanov gives the asserted local absolute continuity [40, Theorem 3 and Section 5]. Force collisions have no time interval. In the completion use its conditional chordal law in the cut domain. Each moving interval contains an interior excursion: a continuous Loewner trace spends zero capacity time on the domain boundary [44, Proposition 1.7]. Restrict to a compact subinterval of such an excursion. Enumerate the countably many branches, completions, and stopped windows, and refine them by all rational compact subintervals in these interior excursions. The resulting countable family of comparison segments contains a member inside every moving open portion, including those selected subsequently.

An ordinary chordal SLE in a Jordan disk has open pockets of both passage sides arbitrarily close to each moving interior trace point. To verify the needed two-sided assertion, decorate a quantum wedge by an independent chordal SLE. Its left and right boundary-length processes in quantum natural time are independent spectrally negative stable processes of index $\kappa/4$ [16, Theorem 10.1]. Both have jumps in every nonempty time interval, and each jump cuts off an open pocket on its designated side. We need this clock to describe the full ordered traversal, as well as to be continuous. The proof of [16, Theorem 1.16], with both force weights zero, obtains ordinary chordal SLE by welding two wedges of weight $\gamma^2 - 2$ to a middle wedge of weight $2 - \gamma^2/2$; the ambient weight is $3\gamma^2/2 - 2$. In the middle wedge the interface is a concatenation of $\text{SLE}_\kappa(\kappa/2 - 4; \kappa/2 - 4)$ traces. Duplantier et al. [16, Lemma 10.4 and Proposition 10.6] identify its entire ordered counterflow traversal and its continuous quantum-natural parameterization. The inverse-clock jumps skip space-filling visits to bubbles, not moving portions of that counterflow traversal.

The exterior welding in the proof of Theorem 1.16 retains this same interface and the ordered quantum bubbles it encloses. The intrinsic bubble description of quantum natural time in [16, Theorem 6.22, Definition 6.23 and Remark 10.5] therefore retains its clock, with the fixed deterministic normalization. This construction gives a continuous parameterization of the full ordinary chordal traversal. Every moving portion thus occupies a positive clock interval. Jumps of each stable coordinate in every rational clock interval, together with continuity, imply that closing tips of pockets of both sides approach every moving interior trace point simultaneously. Interior points of these pockets may be chosen arbitrarily close to their tips and remain off the entire chordal trace. These nearby interior points supply the two-sided test. Integrating out the auxiliary field gives the property for the unlocked chordal law.

Close the reference chordal path by one specified boundary side of the disk. Its winding indices on pockets of the two passage sides differ by one. One way to see this without a smoothness assumption is ordered approximation of a continuous Loewner trace by simple traces [44, Theorem 1.1]. Truncate away from its terminal point, complete within a shrinking terminal neighborhood, and close on the same boundary side. The Jordan curve theorem gives the side test for each approximant, and uniform convergence preserves winding about any fixed off-trace points. For a cut domain perform this argument in its prime-end disk, using only interior test points. Hence the assertion survives conformal coordinates and the stopped local changes of measure above.

The full ordinary chordal reference law also has strong no self-tracing. For a fixed indexed full crossing, the conditional completion given the explored trunk is ordinary chordal SLE in its remaining component, and its concatenation with the trunk is a CLE loop [25, Section 2.3.2]. Normalize this component to a prime-end disk. An open portion whose image there lay in its complementary trace would give the same containment in the physical CLE loop. It has an interior moving subinterval by [44, Proposition 1.7]; conformal injectivity there prevents its physical image from being constant. This contradicts the strong CLE property stated above. The normalized completion has the fixed ordinary chordal law conditionally on the trunk, so the deduction gives a probability-one property of that law. The same interior argument transfers it to the cut-domain reference paths used here.

Before choosing any initial portion, retain a complete reference path for every member of the countable comparison family. For one member, disintegrate its reference chordal law given the compared segment and weight by the Radon–Nikodym derivative of the actual segment law. Sampling from this conditional extension kernel couples the actual segment with a complete chordal path which agrees with it; the full path marginal is absolutely continuous with respect to

chordal SLE. These kernels exist on the continuous-path spaces. Use auxiliary randomness for the countably many extensions, and intersect all their no-tracing and two-sided-pocket events, together with the corresponding events for the original loops. This gives a single probability-one event on the enlarged space.

On that event fix arbitrary L and P . The omitted moving portion cannot have its entire trace in P , by no self-tracing. It therefore contains a moving open interval whose trace misses the compact trace of P . Choose an already enumerated comparison segment inside it, and let c be its already retained complete reference path. Thus c and L agree on an open parameter segment, and the required properties of both full curves hold before this choice.

For any continuous curve with no self-tracing, the set of repeated parameter times is meagre. To check this, for disjoint compact rational parameter intervals J, K , the set of times in J whose images lie in the image of K is closed. It has empty relative interior: otherwise an open portion would have its image contained in the complementary trace, violating the strong no-tracing property. The countable union of these sets covers all repeated times, with the cyclic endpoint convention for L . Apply this to both L and the complete path c . On their common open segment choose, by the Baire theorem, a time visited only once by each full curve. Let its position be x .

Choose a smaller common compact segment A around that time. Both complementary arcs are compact and miss x . There is consequently a ball U about x which misses these two complementary arcs, P , and the cut-domain boundary. Close c by its boundary-side arc to obtain C . Inside U , the only portions of L and C are their identical oriented segment A . The two passage pockets of c supply points of U at which the indices of C differ by one. The deterministic comparison (9.2) now supplies the required open disk V .

All subsequent choices are deterministic on the single event just constructed. The conclusion therefore holds for every closed initial portion. Since this conclusion concerns only the original loops, integrating out the auxiliary reference paths gives the asserted probability-one event on the original space. $\square$

9.7 Identification with all nested CLE loops

Theorem 9.10 (Complete CLE readout). *The nonconstant curves ℓ_F , with their cyclic orders and with orientations forgotten, are exactly the whole-plane nested CLE_κ ensemble determined by the independent space-filling exploration. Each CLE loop occurs once.*

Proof. We first transfer the finite side test to a fixed face. For each fixed boundary chord of F , the corresponding sides of its approximating faces have the same limiting endpoints. To verify this, take a point in the relative interior of that chord and a transverse line through it meeting F . The endpoints of the resulting convex-cell sections converge. Their supporting chord sides have subsequential endpoint limits on the circle; a limiting side through the chosen interior point must be the full bounding chord, since its line has only two circle intersections. Lemma 9.1 then gives eventual agreement of their labels.

Fix a typical target in one child and a test time outside the outer interval, using time 0 as the reference point at infinity. These locations are off the limiting face contour: by Lemma 9.8, that contour is an initial portion of a CLE loop, and the targets avoid all CLE traces. For a target time u strictly inside a child, choose discrete tour times $u_n \rightarrow u$. Convergence of the two child endpoints puts u_n inside the corresponding discrete child for all large n , and the spatial correspondence sends its tour point to $\eta(u)$ in the limit. The analogous assertion holds for the exterior reference point. Their positive distance from the compact limiting contour and uniform

loop convergence preserve the finite winding test. Lemma 9.2 therefore implies

$$\text{Ind}(\alpha_F, \eta(u)) \neq 0 \iff u \text{ is in an opposite-type child of } F. \quad (9.3)$$

Targets outside the outer interval have index zero. The nonzero index has the orientation sign of the prefix. The statement holds simultaneously for all faces and all the countably many targets; there are only countably many faces, since each contains a rational disk point.

By Lemma 9.8, a nonconstant face is a closed initial portion P of one CLE loop L . Conversely every CLE loop has a surrounded open component, hence contains a typical target. Its switch supplies a face with a nonconstant prefix, so no CLE loop is absent from this assignment.

We show that targets surrounded by one fixed L give the same outer interval. Let I_1, I_2 be their switch outer intervals. Every typical target surrounded by L lies in each outer bubble by Lemma 9.6. Typical points have unique visit times, so both intervals contain both target times and are nested. Suppose, for contradiction, that I_2 is a proper subinterval of I_1 . Let J_1 be the child of I_1 used by its switch. The first target time lies in the interiors of both J_1 and I_2 . These two intervals are therefore comparable, and maximality of J_1 among proper subintervals of I_1 gives $I_2 \subseteq J_1$. Thus the second target also sees the switch with outer interval I_1 . The between-tips prefix is identical for both targets in that child, by target invariance and the erasure description. It identifies L for both targets, since it is a nonconstant shared arc. But the second target already sees the switch with outer interval I_2 producing L , contradicting the once-only nested construction. Hence $I_1 = I_2$. The face on its inner side is unique, so every target surrounded by L is tested by an opposite-type child of the same face.

It follows from (9.3) that P and L have the same index on every typical target surrounded by L . They also agree on every typical target outside L . Indeed targets outside the outer interval have index zero. A target inside it is almost surely in a child; if it were in an opposite-type child, that switch would produce a loop surrounding the target and sharing the nonconstant prefix of L . It would therefore be L , a contradiction. A same-type child has index zero by the side test. The indices of P and L thus agree on the entire dense typical target set. Lemma 9.9 excludes any nonconstant omitted portion. Hence the face traverses the complete loop in its order.

Finally, two nonconstant faces assigned to the same L would, by their opposite-type children, supply two outer intervals for targets surrounded by L . We just proved that these intervals coincide; their unique inner face is then the same. This proves multiplicity one and completes the identification. $\square$

Corollary 9.11 (Joint matching in the reference sphere). *On every common contour extraction, the collections $H_n(\hat{\Gamma}_n)$ converge to the complete nested ensemble Γ of Theorem 9.10, in the loop topology of the theorem. The convergence is joint with the surface, area, and all additional observations retained in the common extraction. Every macroscopic loop on either side is included in the matchings.*

Proof. Combine Lemmas 9.1 and 9.4 with Theorem 9.10. The topological correspondence is the one of Theorem 4.7. All limiting chord and face data are measurable functions of the retained excursion, so the identification holds in its existing joint coupling. In particular, it is not necessary to recouple an individual loop separately from the other observations. $\square$

Proposition 9.12 (Canonical coordinates and independence). *Before three-mark normalization, conditional on the reference quantum surface, Γ has the fixed unmarked whole-plane CLE_κ law. Let T be the Möbius map taking three fresh independent quantum-area marks to 0, 1, ∞ . Then $T\Gamma$ has that same conditional law given the normalized embedded quantum sphere.*

On an extraction for which $\phi_n \circ H_n^{-1} \rightarrow f$ uniformly, the canonical loop collections $\Gamma_n = \phi_n(\widehat{\Gamma}_n)$ converge to $f\Gamma$. Here $f = T$ or $f = \mathfrak{r} \circ T$, where $\mathfrak{r}(z) = \bar{z}$; this is the same orientation alternative as for all other retained observations.

Proof. The reference surface is decorated by an independent unclocked space-filling SLE in Theorem 3.4. Its exploration tree and unmarked CLE are functions of that unclocked curve; changing to the quantum-area clock changes neither. Thus the conditional loop law is fixed. Given the surface and three fresh area marks, T is deterministic and the loop law remains unchanged by its action: whole-plane CLE_κ is invariant under every Möbius transformation [25, Theorem 1.1]. Taking conditional expectations and then forgetting the unnormalized surface and marks proves the claimed conditional independence in the normalized embedding.

For the convergence assertion, a uniform individual-loop matching of error δ becomes a matching of error at most

$$2\|\phi_n \circ H_n^{-1} - f\|_\infty + \omega_f(\delta),$$

where ω_f is a modulus of continuity of f on the compact sphere. Uniform continuity of f^{-1} ensures that every loop of prescribed positive diameter in the new coordinate had diameter bounded below in the old one. Hence the complete matchings of Corollary 9.11 remain complete after this coordinate change. Theorem 8.9 identifies f and its single orientation sign, proving the last assertion. $\square$

10 Crossing scales and rigidity of local passage costs

The metric argument is first carried out for the bilateral word. We use the joint local observations of Theorem 6.1, their re-expression under changes of time unit, and the simultaneous annulus estimate of Section 7. All costs in this section are numbers of original primal edges. Put

$$p = \frac{1}{d_\gamma}, \quad \xi = \gamma p, \quad Q = \frac{2}{\gamma} + \frac{\gamma}{2} > 2.$$

The reference metric is the intrinsic LQG metric $D = D_h$, with the normalization fixed in the statement of the theorem. Theorems 10.18 and 10.21 identify the local passage law and its deterministic normalization.

The comparison strategy follows Gwynne and Miller's LQG metric uniqueness proof: optimal constants, repeated annular opportunities and local finite-energy field changes bring a geodesic near a shortcut and improve the comparison [19, Section 1.5 and Sections 3–6]. The continuum metric theory rests on LFPP tightness [12], weak-metric estimates [14], and that existence and uniqueness theorem. Here the objects being compared are port passages. Their local kernels, selected changes of law and eventual endpoint completion require the additional arguments below. We continue to work in the original oriented peanosphere coordinate, with its real and imaginary field laws. The canonical coordinate and loop identifications already established do not change those laws. This section identifies local costs up to a deterministic subsequential scalar; Section 11 uses the intrinsic GH input to determine that scalar and reach every vertex.

10.1 Reference metric facts and annular cutoffs

The reference metric is already defined on the continuum surface. We recall the estimates used to compare it with discrete passages; none of these statements identifies a graph-distance limit.

Write $D_h(\cdot, \cdot; U)$ for the internal metric on U , and fix one deterministic normalization of the γ -LQG metric. Set $p = 1/d_\gamma$ and $\xi = \gamma p$. For a length metric D and a continuous function f , the Weyl weighted metric means

$$(e^{\xi f} \cdot D)(z, w) = \inf_P \int_0^{\text{len}(P; D)} e^{\xi f(P(t))} dt,$$

where the infimum is over D -rectifiable paths from z to w , parameterized by D -arclength [19]. Write $h_r(0)$ for the circle average of h on $\partial B_r(0)$.

Theorem 10.1 (LQG metric inputs). *The subcritical LQG metric is a field-measurable continuous length metric inducing the surface topology. Its internal metric on U is determined by $h|_U$. It obeys*

$$D_{h+f} = e^{\xi f} \cdot D_h, \quad \mu_{h+f} = e^{\gamma f} \mu_h, \quad (10.1)$$

$$D_{h \circ \psi + Q \log |\psi'|}(z, w) = D_h(\psi(z), \psi(w)) \quad (10.2)$$

for continuous f and conformal coordinate maps ψ . The whole-plane GFF metric is proper and geodesic, and the geodesic between each fixed pair of distinct points is almost surely unique.

For every compact planar set and every $0 < \beta < \xi(Q - 2)$, there are almost surely finite C and positive r_0 such that, for $|z - w| < r_0$ in that set,

$$D_h(z, w; B_{2|z-w|}(z)) \leq C|z - w|^\beta. \quad (10.3)$$

The corresponding estimates before almost-sure truncation have polynomially high probability uniformly over the reference scale, with the normalization $r^{\xi Q} e^{\xi h_r(0)}$.

Proof. Existence, uniqueness, locality, and Weyl scaling are [19]; conformal covariance is [18, Theorem 1.3]. Properness is [14, Lemma 3.8]; a proper length space is geodesic, as also noted at the start of Section 4 of that paper. Fixed-pair uniqueness is [34, Theorem 1.2]. The precise internal estimate, including its probability and normalization assertions, is [14, Lemma 3.20, equations (3.66)–(3.67)]. Apply it along dyadic radii and Borel–Cantelli to obtain (10.3). The scale constant is $r^{\xi Q}$ for the uniquely normalized LQG metric. On a fixed compact chart, local field absolute continuity and Weyl scaling transfer the almost-sure assertion to our other real-field laws away from their insertions. Uniform probability bounds under a changed law require the density control established when that law is used. $\square$

Lemma 10.2 (Deterministic geodesic anchors). *Fix a deterministic countable dense subset A of the plane. Almost surely every nonempty open subarc of every nonconstant whole-plane D_h -geodesic contains a nondegenerate segment of the unique geodesic between some two points of A .*

Proof. Intersect fixed-pair uniqueness over $A \times A$ with the simultaneous strong-confluence event of [7, Theorem 2 and Proposition 6]. In a given open geodesic subarc choose a strictly interior interval $[s, t]$ and $0 < \varepsilon < (t - s)/2$. Proposition 6 supplies metric neighborhoods of its two endpoints such that all geodesics between those neighborhoods contain the original segment $[s + \varepsilon, t - \varepsilon]$. Since A is also metric dense, choose distinct anchors in these neighborhoods. Their unique geodesic contains the required segment. The event is simultaneous over all starting geodesics; only the anchor pair is selected afterward. For later local applications, one first confines a sufficiently short interior piece below the positive cost of exiting a surrounding buffer. Locality and local absolute continuity then transfer this whole-plane statement to that piece. $\square$

The simultaneous annulus theorem can retain the following metric regularity events together with a passage test. Its field and spatial alignment are unchanged.

Corollary 10.3 (Additional local cutoffs). *The good events in Theorem 7.4 may include finitely many local internal D_h across and around regularity tests, and bounds for the Gaussian pairings associated with any fixed finite family of smooth Cameron–Martin recipes supported strictly inside E . Their cutoffs can be chosen before applying the theorem so that their combined marginal failure is arbitrarily small, uniformly in the label.*

Proof. By Theorem 10.1, in each fixed buffered geometry the internal metric induces the Euclidean topology, positive separations between disjoint compact ports are positive, and the required confined connection costs are finite. Finite maxima, positive minima, and moduli of continuity therefore admit high-probability finite cutoffs and positive lower cutoffs. Only internal metrics in the stated dependency buffer are used. Under (7.5), the coordinate and constant identities for this reference metric give

$$D_{h^{c,r,v}}(u, u') = v^{-p} D_h(c + ru, c + ru')$$

with the corresponding internal domains. These are the ordinary unit-coordinate regularity events at every label.

For $f \in C_c^\infty(E^\circ)$, the pairing $(h, f)_\nabla$ is a centered Gaussian of variance $\|f\|_\nabla^2$ under the real reference law. Constants pair to zero; Dirichlet norms and pairings are invariant under affine coordinate rescaling. The same holds for the centered imaginary field. Gaussian tails and a finite union bound give any required common probability cutoff. On $|(h, f)_\nabla| \leq T$, the Cameron–Martin factor $\exp((h, f)_\nabla - \|f\|_\nabla^2/2)$, and its reciprocal, have fixed finite bounds. These are field events and are adjoined to the local output tests without changing their kernels. $\square$

10.2 Port comparisons and deterministic chaining

We first specify the comparisons that will be proved. A passage certificate includes its ordered endpoint and intermediate ports, a compact confinement, its upper cost, and the upper costs of its ordered subpassages. A certificate is a retained finite ordered path record, with arbitrary positive losses in its port, confinement and cost requirements; it is not merely the spatial range of a path. Lemma 5.10 and Theorem 6.1 retain these records jointly, including their consistent refinements and restrictions. Statements about an open confinement are read using compact subsets of that confinement; endpoint and intermediate ports can be made arbitrarily small.

Definition 10.4 (Local passage comparisons). Fix a joint limiting law of the field, its reference internal metrics, and the compatible ordered passage records. An *upper comparison with constant C* means the following simultaneous assertion. For every D -rectifiable path P compactly contained in an open set U , every finite ordered subdivision of P , and every positive endpoint, tube, and cost tolerance, there is an attainable certificate which follows the subdivided path through those tolerances, with each marked subpassage costing at most C times the corresponding D -length plus the assigned cost tolerance. An *upper constant* is a deterministic C for which this holds almost surely in every ordinary chart.

A *lower comparison with constant c* means that every finite-cost certificate, with limiting endpoints x, y , has cost at least $cD(x, y)$. It applies to every ordered subcertificate as well. A *lower constant* is a deterministic $c \geq 0$ for which these assertions hold almost surely in every ordinary chart.

For $x \neq y$ in U , let $F_U(x, y)$ be the infimum of the upper costs of certificates from x to y compactly confined in U , with the endpoint ports shrinking to x, y . Set $F_U(x, x) = 0$.

The upper comparison permits the endpoints of a discrete path to be chosen within the prescribed ports. It gives no bound on the cost of reaching a prescribed lattice vertex. This distinction persists until Section 11.

Here is the deterministic use of a simultaneous annulus cover. The annuli have fixed shape parameters: a central hole, a surrounding annular band, and a strictly larger buffer. Their centers range over fixed fine meshes at finitely many radii. A positive number w_i is attached to each annulus; in applications it is v_i^p . All shape parameters are held fixed while a discrete approximation tends to its limit.

Lemma 10.5 (Chaining through an annulus cover). *Suppose every point of a compact set K belongs to a central hole of a finite family of buffered annuli contained in an open set U . Their largest outer radius is at most r_* , their holes have a fixed positive clearance from their bands, and their reference weights are $w_i > 0$. Assume that a reference path from a hole to its surrounding separator has D -length at least aw_i , where $a > 0$.*

1. *If the i th annulus contains an attainable surrounding primal circuit of cost at most Lw_i , then every D -rectifiable path in K can be followed, with endpoint errors tending to zero with r_* , at cost at most*

$$\frac{L}{a} \text{len}_D(P) + L \max_i w_i.$$

For any fixed finite ordered subdivision of the reference path, the following walk can also retain its intermediate ports. Each marked subpassage has cost at most (L/a) times the corresponding reference length plus $2L \max_i w_i$. All spatial errors are $O(r_)$, and arbitrary fixed open tube losses are allowed.*

2. *Suppose instead that every primal traversal from the hole side to the outside of the band costs at least ℓw_i . Suppose also that a fixed reference circle lies strictly outside the tested band and strictly inside its buffer, with internal D -diameter at most Aw_i in that buffer. Every bounded-cost certificate from x to y whose range lies in K then satisfies*

$$\text{cost} \geq \frac{\ell}{A} D(x, y) - o_{r_*}(1).$$

The error depends only on the reference metric modulus on the fixed compact neighborhood of K .

Both conclusions hold for limits of discrete paths when the annular tests have strict margins and all selected ordered subrecords are retained.

Proof. Fix the annulus cover before taking a discrete approximation. Use the strictly buffered upper tests to choose all its simple primal circuits simultaneously on that approximation, with costs at most $Lw_i + o_n(1)$, and denote their filled sides by Ω_i . Transfer them to the reference sphere by H_n . For large n , each specified central hole lies strictly inside its Ω_i , and Ω_i lies in the corresponding outer disk. The errors of H_n are smaller than the fixed clearances. In the estimates below we may first use any $a' < a$ to absorb those errors, and let $a' \uparrow a$ after taking the limit.

Starting at $P(0)$, select a hole containing that point and follow P until its first encounter with the boundary of the selected filled circuit. At that encounter select another hole containing the point, and repeat, stopping when the endpoint is reached. If an index is chosen several times, retain its occurrences separately. Write $\Omega_1, \dots, \Omega_m$ for this ordered list and $[s_j, s_{j+1}]$ for the corresponding path intervals. Each interval except possibly the last has D -length at least $a'w_j$,

and its image lies in Ω_j . These intervals have disjoint interiors, so

$$\sum_{j < m} w_j \leq (a')^{-1} \text{len}_D(P).$$

The finite cover has a positive minimum weight, so there are only finitely many complete intervals, bounded uniformly in the approximation by $1 + \text{len}_D(P)/(a' \min_i w_i)$. Thus all summed $o_n(1)$ cost errors vanish at this fixed cover. The point $P(s_{j+1})$ belongs to $\partial\Omega_j \cap \text{int}(\Omega_{j+1})$. Consequently Ω_{j+1} cannot be contained in Ω_j .

We reduce this list in its original order. Keep an ordered list of retained disks. When a new disk contains the last retained disk, delete that last disk, assigning its path interval to the new disk, and repeat until no such containment remains. The new disk cannot be contained in the preceding retained disk: otherwise the last disk just deleted would have been contained there as well, contrary to the defining property of the retained list. The new disk also overlaps the preceding disk in its interior, since it contains the deleted disk whose interior overlapped that preceding disk. Thus consecutive retained disks have intersecting interiors and neither contains the other. Their Jordan boundaries therefore intersect. These are intersections of actual primal circuits, so they contain common primal vertices. No transverse-crossing claim about limiting circuit traces is needed.

Choose a common vertex for each consecutive pair. On each retained circuit follow one arc between its two selected vertices; on the first and last circuit choose the free endpoints arbitrarily. If there is only one retained circuit, both endpoints may be chosen there. The resulting primal walk uses at most the full cost of each retained circuit, and hence at most

$$L \sum_{j=1}^m w_j \leq \frac{L}{a'} \text{len}_D(P) + L \max_i w_i.$$

Each retained disk has spatial diameter $O(r_*)$, contains all path intervals assigned to it, and the assignments remain consecutive in path order. Thus the walk's endpoints are within $O(r_*)$ of the prescribed endpoints, and its circuits lie in any fixed open tube around P once r_* is sufficiently small.

This ordered reduction also retains finitely many intermediate ports. Assign a marked reference time to the retained disk containing its path interval. The entire corresponding circuit is within $O(r_*)$ of its marked reference point. Choose each marked vertex to be the incoming intersection on its circuit occurrence, or the free initial vertex on the first one. Several marks assigned to one disk may use the same vertex; their order is then preserved without an additional traversal. For an individual subpassage, all retained circuit intervals strictly between its two end marks are charged to disjoint reference intervals within that subpassage. At most the two circuits carrying the end marks are not so charged. Its cost is therefore at most

$$\frac{L}{a'} \text{len}_D(P|_{[t,t']}) + 2L \max_i w_i.$$

The full-path bound above still has its single terminal term. In applications the number of prescribed marks is fixed and the covers are refined with $\max_i w_i \rightarrow 0$, so these boundary errors fit any assigned positive subpassage tolerances. All intersections and joins were made on one discrete surface. Pass to the retained finite records only after this construction, then let $a' \uparrow a$ and refine the covers.

For the lower bound, apply the same stopping construction to a discrete path, stopping on the fixed circles beyond the tested bands. Thus every complete step contains a full band traversal.

Mark in every complete step its traversal of the tested band. The marks occur on disjoint ordered path intervals. A budget T allows at most $T/(\ell \min_i w_i)$ complete steps; thus they can be retained under subextraction. If I is the set of selected complete steps, then

$$\text{cost} \geq \ell \sum_{i \in I} w_i.$$

Apply the same chronological deletion to the filled intermediate reference circles, retaining a chain of intersecting boundaries. Connect the successive circle-intersection points within their respective buffers, using the internal-diameter bounds. This gives a reference route of length at most $A \sum_{i \in I} w_i$. The first and last reference connections have Euclidean displacement at most a fixed multiple of r_* ; their costs are $o_{r_*}(1)$ by uniform continuity of D on the compact neighborhood. Hence

$$D(x, y) \leq A \sum_{i \in I} w_i + o_{r_*}(1).$$

The last, possibly incomplete, disk contributes only an endpoint error: its reference circle is approached at the last complete step, and its remaining Euclidean displacement is bounded by its diameter.

For discrete approximations, keep the finite cover, the strict band margins, and all stopping marks fixed first. The number of marks is bounded by the displayed budget. The subrecord property transfers the traversal lower tests, and the attained-witness property transfers the circuit upper tests. Robust separation and intersection hold before passing to a limit. Finally refine the annulus covers. This order of limits does not require any estimate for costs from arbitrary lattice endpoints. $\square$

Lemma 10.6 (Confined reference connections). *Almost surely, on each compact subset of an ordinary chart there are $\chi > 0$, $H < \infty$, and $r_0 > 0$ such that*

$$D(x, y; B_{2|x-y|}(x)) \leq H|x-y|^\chi, \quad |x-y| \leq r_0. \quad (10.4)$$

For rectangles of bounded aspect ratio this gives a uniform internal connection modulus up to all sides and corners. The same bound, with a constant depending only on the aspect ratio and the compact cutoff, holds in every open neighborhood of the closed rectangle, uniformly as that neighborhood decreases to the rectangle.

Proof. For a whole-plane field, Lemma 3.20 of [14], equations (3.66)–(3.67), gives the internal-ball bound with any $0 < \chi < \xi(Q-2)$, with polynomially high probability along dyadic scales. Borel–Cantelli gives (10.4); local field absolute continuity transfers its almost-sure version to the ordinary charts used here. Quantitative cutoffs are taken in the aligned reference units before this transfer.

For completeness, let R be a closed rectangle with bounded aspect ratio, $z \in R$, and let $F_t(z)$ clamp the two coordinates of z to the rectangle whose sides are shifted inward by t . Put $t_n = (31/32)^n t$. Then

$$|F_{t_n}(z) - F_{t_{n+1}}(z)| \leq \frac{\sqrt{2}}{32} t_n, \quad \text{dist}(F_{t_n}(z), \partial R) \geq t_n.$$

The ball used in (10.4) therefore lies strictly inside R . Connecting successive clamped points and summing a geometric series gives an interior path from z to $F_t(z)$ of length at most $H_1 t^\chi$, including a continuous extension at z . For $x, y \in R$, use t equal to the smaller of $|x-y|$ and one quarter of

the shorter side. The segment between $F_t(x)$ and $F_t(y)$ lies in the t -interior of R . Divide it into steps of size at most $t/8$ and use (10.4) on each step. Bounded aspect ratio bounds the number of steps when $|x - y|$ is comparable to the side length; otherwise it is bounded absolutely. Adding the two inward paths proves the asserted modulus. Every constructed path lies in the closed rectangle, so enlarging the rectangle by an arbitrarily thin open collar cannot increase its cost. $\square$

Lemma 10.7 (Vanishing joins and stability). *A finite upper comparison implies the following assertions, simultaneously on each compact ordinary chart.*

1. *Every point has arbitrarily small surrounding primal circuits whose limiting costs tend to zero, uniformly over the compact set.*
2. *The functions F_U are symmetric, satisfy the triangle inequality, and are locally continuous in their endpoints. They obey $F_U(x, y) \leq CD(x, y; U)$.*
3. *Nonconstant certificates can be concatenated at a common limiting endpoint with arbitrary open and cost loss. Comparisons with fixed constants, strict shortcut witnesses, and saturation on compact geodesic arcs pass to joint tangent extractions that retain the reference internal metrics on their collars.*

Finite connected systems of reference ports and rectifiable paths can in particular be followed by primal paths with all prescribed robust intersections. These assertions concern free endpoint ports.

Proof. Choose a fixed sufficiently fine angular subdivision of a small Euclidean circle. Lemma 10.6 connects consecutive vertices inside balls contained in its annular neighborhood. The result is a rectifiable reference closed path of winding one, with length at most $H_2 r^\chi$. The same construction works uniformly for centers on fine meshes and, by strict hole margins, surrounds every point of the compact set. Follow its successive pieces using the upper comparison, with transverse overlapping ports. The topology comparison of Corollaries 4.9 and 4.10 gives a surrounding primal walk and hence a simple surrounding subcircuit, with cost at most $CH_2 r^\chi + o(1)$. Take countably many gridded covers and then let r decrease.

To join two nonconstant certificates ending at z , choose a small surrounding circuit which both certificates must meet. Cut them at their encounters with the circuit and join along that circuit. The added cost tends to zero. This proves the triangle inequality; cases with a constant subpassage use $F_U(z, z) = 0$. Symmetry comes from reversing primal paths. Following internal reference paths proves $F_U \leq CD(\cdot, \cdot; U)$. Consequently, for endpoints in a compact subdomain,

$$|F_U(x, y) - F_U(x', y')| \leq C(D(x, x'; U) + D(y, y'; U)).$$

Lemma 10.6 gives the local continuity asserted here.

In a tangent extraction the reference field and its internal metric are retained with their fixed joint law. Their compact moduli are therefore tight. A strict low-cost witness uses finitely many compact ports and retains its cost slack. Conversely, a strictly cheaper certificate in a proposed saturated limit has a finite attained record in a strict collar. In preceding laws, correct its endpoint errors by the circuits just constructed. Their costs tend uniformly to zero on compact reference-modulus cutoffs, contradicting the preceding saturation inequalities. First fix the strict collar and finite record, next take the extraction limit, and finally decrease the endpoint tolerances. This proves the stated stability without requiring costs to prescribed lattice endpoints. $\square$

Lemma 10.8 (Marked strict improvement). *Consider a sequence of increasingly fine simultaneous annulus covers with fixed reference regularity cutoffs $a, A > 0$. Include the tests of Lemma 10.5, a*

lower bound aw_i for reference advance across a fixed thinner middle band, and lower bounds aw_i for the internal reference distance between opposite face ports of the finite rectangle library constructed below. The ports have strict disjoint thickenings and the rectangles have strict surrounding collars. Fix this finite library and then these cutoffs before applying the simultaneous annulus theorem. Corollary 10.3 permits their combined failure probability to be arbitrarily small; decreasing a and increasing A incorporates the original chaining cutoffs.

1. Suppose C is an upper constant. If every marked reference geodesic traversal in the good bands admits an alternative certificate at ratio at most $C - \delta$, then $C - \varepsilon_+$ is an upper constant for some $\varepsilon_+ > 0$ depending only on δ, a, A and the fixed band geometry.
2. Suppose c is a lower constant. If every marked rectangular subtraversal in the good bands costs at least c_1 times its internal reference endpoint distance, where $c_1 > c$, then $c + \varepsilon_-$ is a lower constant for some $\varepsilon_- > 0$ depending only on $c_1 - c, a, A$ and the band geometry. The second assertion permits $c = 0$.

It is enough that the indicated good annulus events have the high probability and positive label density required by the simultaneous annulus theorem.

Proof. Use the ordered stopping construction of Lemma 10.5 and mark a crossing of a thinner middle band in each complete step. Its reference advance is at least $a_1 w_i$ for a fixed $a_1 > 0$. The marked intervals are disjoint. Connecting the outer reference circles as in that lemma gives

$$D(x, y) \leq A_1 \sum_i w_i + o(1) \quad (10.5)$$

with a fixed $A_1 < \infty$.

For the upper assertion, run the construction along a reference geodesic. Replace each marked interval by its improved certificate, and retain the upper comparison C on the intervening pieces. At this fixed cover there are only finitely many gaps. The complete ordered upper comparison follows them with assigned cost tolerances whose sum is arbitrarily small; Lemma 10.7 supplies the joins with equally small total error. These tolerances are chosen after fixing the marked cover, so no error proportional to its possibly growing number of marks is left when the cover is refined. The saving is at least $\delta a_1 \sum_i w_i$. Use (10.5) and then refine the covers. For instance any $0 < \varepsilon_+ < \delta a_1 / A_1$ is admissible after the strict losses. For a general rectifiable path in a prescribed open tube, its compact range has positive reference distance from the complement of a slightly smaller buffered tube. Partition it into pieces shorter than that distance. Reference geodesics joining the consecutive endpoints then stay in the tube and have total length no greater than the original path. Apply the preceding construction to those pieces, keeping the specified ordered ports. This proves the complete upper-following assertion.

For the lower assertion, fix a normalized spatial length $h > 0$ much smaller than the thinner band's remaining crossing distance and its clearance from the retained band collar. Use squares of half-side h centered on a grid of mesh $h/4$. The starting point of each marked crossing lies within $h/8$ in each coordinate of one such center. Every square so selected stays in the retained band collar, and the crossing must exit it. Suppose its first exit is on the right side $x = c_x + h$. Retain the subpassage after its last crossing of $x = c_x + h/2$ before that exit. It stays in the exact rectangle

$$[c_x + h/2, c_x + h] \times [c_y - h, c_y + h]$$

and joins its opposite vertical faces. The other three exit sides give the corresponding rectangles. These form a finite library of aspect-four rectangles; corner exits are allowed. On discrete

approximations, round the stopping locations to nearby vertices and use the strict face-port and collar margins, then retain their limits. The ordered subpassages remain disjoint. The stipulated finite-library cutoffs give reference endpoint distance at least $a_2 w_i$ in the retained collar, with fixed $a_2 > 0$. Apply the lower comparison c on every gap and the improved comparison on each marked rectangle. The reference triangle inequality for these finitely many ordered endpoints yields

$$\text{cost} \geq cD(x, y) + (c_1 - c)a_2 \sum_i w_i.$$

If $c = 0$, the strictly positive rectangle charge itself bounds the number of complete steps at a fixed mesh. If $c > 0$, the same bound holds and the lower comparison can also be subdivided intrinsically. Thus the required marked subextractions exist in either case. Equation (10.5) gives any $0 < \varepsilon_- < (c_1 - c)a_2/A_1$ after the strict losses.

Finally, the simultaneous annulus theorem makes these constructions available about every path location with probability tending to one along the specified covers. Countable exhaustion of paths by ports, then of chart compacts and tolerances, makes the improved comparison an almost-sure statement. No rate of discrete convergence is used: the finite cover is always fixed before its discrete approximation. $\square$

10.3 Quantile scales and a first-exit lower bound

For the normalization experiment represent the cone in its circle-average embedding: its marked points are $0, \infty$ and

$$\sup\{r > 0 : h_r(0) + Q \log r = 0\} = 1.$$

This is the convention of [16, Definition 4.10]. Fix three strictly nested versions of one round annular circuit test, with one fixed dependency buffer strictly inside $\mathbb{D}$ and away from 0 . At every microscopic time unit use this same embedded cone input law, the same band geometries, and the same reference mark convention. In particular the field's additive height is fixed by this convention; it is not reset within a selected local sample. At time unit 4^j , let X_j be the least unscaled cost of the middle test, using the contour correspondence and short-arc interpolation to read winding. On the event that the correspondence does not yet resolve the strict buffers, use any positive finite fallback value. That event has probability tending to zero. Fix a quantile level $\tau < 1$ sufficiently close to one, and define

$$A_j = \inf\{t > 0 : \mathbb{P}[X_j \leq t] \geq \tau\}. \quad (10.6)$$

Changing finitely many initial values makes every A_j positive and finite. The topological comparison makes the circuit tests available with probability tending to one; hence the fallback convention has no effect on any conclusion below. At atoms we use inequalities with strict cost and probability slack. In particular,

$$\mathbb{P}[X_j < A_j] \leq \tau, \quad \mathbb{P}[X_j \leq A_j] \geq \tau.$$

The narrower and wider bands allow the corresponding implications to pass through the local certificate topology.

We shall use first exits from finitely many strict probability tests. Here is their precise convention. A bad traversal test consists of a finite compact endpoint/confinement specification and a bounded cost threshold. Insert strict larger endpoint ports, a strict larger confinement, and a strict larger cost threshold. In the compactified observation space, a continuous function with values in $[0, 1]$ can be chosen to equal one on the smaller bad test and zero outside the larger bad test. Work first on compact reference-data cutoffs and then exhaust those cutoffs. Tests of the

expectation of this function are continuous in the joint law. A weak inequality is consequently preserved under extraction, whereas a limiting strict inequality is eventually preserved in the approximants. The attained-witness and subrecord assertions in Theorem 6.1 are what make the two buffered tests correspond to the required passage statements. This convention will also be used for circuit tests and numerical ratio tests.

Lemma 10.9 (Initial growth). *The quantile scales satisfy*

$$\limsup_{j \rightarrow \infty} \frac{\log_4 A_j}{j} \geq p.$$

Proof. Fix $\varepsilon \in (0, p)$ and a sufficiently high-probability lower traversal test in a buffered band. Start at j_0 with cost unit one and, at level $j \geq j_0$, use the denominator

$$t_{j,j_0} = 4^{(p-\varepsilon)(j-j_0)}.$$

At each fixed offset $j - j_0$, a bounded number of primal edges makes vanishing spatial advance as $j_0 \rightarrow \infty$. Thus the traversal lower test passes, with any fixed cost and probability slack, on arbitrarily long initial runs.

Suppose first failures existed for arbitrarily large starts. Their distance from their starts tends to infinity. Extract the joint local observations at those failures, retaining every fixed preceding offset, with denominator t_{j,j_0} . At offset i the passing lower test gives a barrier proportional to $4^{-(p-\varepsilon)i}$ in the current units. The reference metric unit there is 4^{-pi} . Their ratio is therefore $4^{\varepsilon i}$. On labels $i \in [N, 2N]$, the simultaneous annulus theorem and the lower part of Lemma 10.5 give a lower comparison at least $c_0 4^{\varepsilon N} D$, with $c_0 > 0$ independent of N . Letting N tend to infinity says that every fixed separated traversal has infinite cost in the extracted law. It strictly passes the original lower test. The continuous buffered-test convention then contradicts that these scales were failures.

Consequently a sufficiently late start has no later failure. A circuit in the normalization band contains a traversal of one of the strict lower test bands. The lower probability cutoff was chosen stronger than needed for the quantile level in (10.6); hence, for all sufficiently large j ,

$$A_j \geq c_\varepsilon 4^{(p-\varepsilon)j}$$

with some $c_\varepsilon > 0$. This eventual lower bound will also ensure that a single edge has vanishing cost in A_j units. Letting ε decrease to zero proves the asserted limsup. $\square$

Lemma 10.10 (A finite comparison from preceding scales). *Consider a joint extraction in units A_j for which*

$$\limsup \frac{A_{j-i}}{A_j} \leq K 4^{-pi} \quad \text{for every fixed } i \geq 0 \quad (10.7)$$

with one finite K . Its limiting passage law has a finite, positive optimal upper constant C . It has an optimal lower constant $c \in [0, C]$. Both constants are deterministic and attained in the sense of Definition 10.4. Their inequalities persist after every fixed change of volume unit and under further joint tangent extractions retaining the reference internal metrics.

Proof. The quantile test at offset i , transported to its ordinary field chart, supplies a high-probability circuit bound proportional to $A_{j-i}/A_j \leq K 4^{-pi}$. Take the quantile sufficiently high that the simultaneous annulus theorem applies after the fixed field density changes and reference regularity cutoffs. Its weights are $w_i = 4^{-pi}$. The upper part of Lemma 10.5 gives a finite upper

constant. If the optimal upper constant were zero, follow a reference circuit in the narrower normalization band at arbitrarily small cost and stitch its finitely many transverse pieces. The reference circuit length is finite with arbitrarily high probability. The middle normalization test would then cost less than, say, $1/2$ in A_j units with probability greater than τ , contradicting (10.6). Thus $C > 0$.

Define C as the infimum of deterministic almost-sure upper constants and c as the supremum of deterministic almost-sure lower constants. Use countably many rational ports, chart compacts and tolerances in these definitions. Intersect the probability-one events for constants decreasing to C and increasing to c ; the compact certificate extraction and arbitrary positive cost losses show that the limiting constants themselves hold. Symmetry and the existence of finite nonconstant upper passages imply $c \leq C$.

For a volume factor v , the same discrete map with time unit $v4^j$ and denominator $A_j v^p$ has field input $h - \gamma^{-1} \log v$ and costs v^{-p} times the original costs. This is the joint array re-expression of Proposition 6.8. It does not identify two differently indexed kernels. The ordinary field laws on a strict patch are equivalent under the fixed shift, so the probability-one comparisons retain the same constants in these units. Further tangent extractions retain their inequalities by Lemma 10.7 and the ordered lower subrecords. $\square$

10.4 Optimal constants and a saturated tangent

Fix a deterministic disk containing the compact annular geometries and their collars used in this argument. We now use a new ordinary comparison experiment: the real field is a whole-plane GFF pinned by its circle average on a deterministic circle outside that disk. The imaginary field keeps its independent whole-plane law and uniform phase, and the local outputs use the same kernels. Mutual local field equivalence transfers the upper and lower comparisons on each strict patch; a countable atlas makes them simultaneous. Their optimal constants are therefore the same C, c . This choice precedes all witness and hit selections. No realized field is repinned. The later neighborhood U and, after discarding finitely many coarse windows, all modification supports will lie inside the fixed disk. Consequently every modifying function vanishes on the chosen pin.

We also fix the distinction between the law, its indexed coordinates, and a further tangent. A unit test at label v evaluates its indexed kernel under this one ordinary normalized input law, the same for every v . Thus h_v in (6.14) has that fixed law; $h = h_v + \gamma^{-1} \log v$ expresses the old quantum units. We do not extract the diverging fields $h - \gamma^{-1} \log v$ for one fixed unnormalized h . Fixed finer offsets are retained jointly from this normalized input. The almost-sure comparison constants transfer to each such test by the indexed identity and fixed-label field equivalence. The uniform witness probabilities below are proved directly under this common ordinary input law, using the annulus theorem and optimality of the constants. A varying-label tangent is a new joint limit of those unconditioned indexed laws, formed with the normalized input held fixed as in Proposition 6.9. Auxiliary shortcut draws will be sampled only after that tangent has been fixed; they do not replace its original passage system.

We will prove that the constants in Lemma 10.10 coincide. For the moment suppose $c < C$. A nearly saturated witness at volume label v is a reference geodesic traversal of a fixed closed annular band for which every competing certificate in a fixed strict open enlargement costs at least $(C - \delta)$ times its length. A shortcut witness is a certificate in a closed rectangle R , whose endpoints lie on opposite faces, with cost less than c_1 times their reference internal distance in a fixed open collar Ω of R . The rectangle comes from a fixed gridded library in the annular band. Endpoint separation, collar clearance and strict cost losses are included in the observations.

We use the absence of these witnesses only with a reserved ratio slack. For a fixed geodesic traversal, failure of the lower bound on all competing certificates means that one finite attained record has cost below that bound; rationally enlarged ports and confinement and a rational cost threshold still witness the strict inequality. Conversely, absence of all such shortcut records means that every retained rectangle subcertificate obeys the asserted lower bound. The reference internal metric and the ordered record sets are both part of the local observation. On fixed length and modulus cutoffs, the reference geodesic family is compact in uniform parametrization; its geodesic identities are checked at rational times. The endpoint continuity in Lemma 10.7 and countable port refinement make the corresponding witness events measurable. In applying their negations below, first reserve an intermediate ratio, then fix the port and collar losses, and only afterward take an extraction or refine an annulus cover. Thus the negations supply the finite alternatives and universal lower subrecords used in Lemma 10.8.

Lemma 10.11 (Many labels carry extremal witnesses). *There are fixed band geometries and $\eta > 0$ such that the following holds. For every $\delta > 0$ and all sufficiently large N , more than three quarters of $i \in [N, 2N]$ have a nearly saturated witness at $v = 4^{-i}$ with probability at least η . For each fixed $c_1 \in (c, C)$, the analogous statement holds for shortcut witnesses. The lower probability and spatial separation for nearly saturated witnesses can be chosen independently of δ .*

Proof. Choose the band geometry and high-probability reference regularity cutoffs first. Choose η below the failure threshold in the simultaneous annulus theorem for a label set of density one quarter. If the first assertion failed along arbitrarily large windows, at least one quarter of their labels would have no nearly saturated witness with probability greater than $1 - \eta$. On these labels every marked reference geodesic traversal has an alternative at ratio below $C - \delta$, with the retained strict open loss. The first part of Lemma 10.8 decreases the upper constant by a positive amount, contradicting its optimality. The probability threshold used in this argument depends on the annular geometry and its reference regularity, not on δ ; only the required fineness and cost losses depend on δ .

If the shortcut assertion failed, a quarter of the labels would, with the requisite high probability, have no low-ratio rectangle witness. Every marked rectangular subpassage would then cost at least $c_1 D(\cdot, \cdot; \Omega)$. The second part of Lemma 10.8 strictly increases the lower constant. This is a contradiction also when $c = 0$. Start with an intermediate ratio between c and c_1 if a strict collar or cost loss must be absorbed. The finite rectangle construction in that lemma supplies the fixed positive spatial separation. Reference diameter and modulus cutoffs, taken with probability loss less than $\eta/2$, make all retained lengths finite and separated from zero in the normalized units. $\square$

Lemma 10.12 (Saturation with deterministic anchors). *If $c < C$, a further tangent law has upper constant exactly C , admits strict shortcuts with ratio bounded away from C , and has the following property with positive probability. There are fixed distinct deterministic anchors, rational $0 < t_1 < t_2 < 1$, and a fixed bounded open set U such that the unique reference geodesic P between those anchors, parametrized on $[0, 1]$ proportionally to D -length, satisfies*

$$F_U(P(t_1), P(t_2)) = CD(P(t_1), P(t_2)), \quad P([t_1, t_2]) \Subset U. \quad (10.8)$$

The event can be restricted to deterministic positive lower and finite upper bounds on the total and interval lengths.

Proof. Let $\delta_k \downarrow 0$. The two sets of labels in Lemma 10.11 intersect. Choose indices $i_k \rightarrow \infty$ in these intersections and put $v_k = 4^{-i_k}$. This choice is deterministic, since it uses probabilities under the

common normalized input law. Extract the unconditioned joint laws, including every fixed finer offset and the reference internal metrics, by the restricted diagonal construction of Proposition 6.9. The discrete indices are chosen so that the effective time unit $n_k v_k$, and every retained fixed multiple of it, tend to infinity. Only after this joint extraction do we retain the positive-probability witness events. On a fixed reference-modulus and length cutoff, the nearly saturated geodesics have equicontinuous arclength parametrizations, lie in the fixed compact band, and have separated endpoints. Extract their paths and the internal metric on their fixed open collar. A strictly cheaper limiting certificate would be attained with strict ports in the preceding laws, and its endpoint errors could be corrected by Lemma 10.7. This contradicts the near-saturation inequality. The limiting arc is therefore saturated. The same extraction keeps the fixed-gap shortcut witness by its strict cost and spatial margins. Its upper optimal constant is C , since the inherited upper bound holds and saturation forbids a smaller one. Its lower optimal constant is strictly below C ; applying Lemma 10.11 in this law supplies a fixed positive density of still finer labels with shortcut ratio at most $C - \delta_0$, for one $\delta_0 > 0$.

Take a short strictly interior subarc of the saturated arc. It is an ambient reference geodesic: when an internal collar was used to take the extraction, shorten the arc until its length is below the positive cost of leaving a still larger collar. We are in the exterior-pinned whole-plane experiment fixed above. Proposition 6 of [7] applies simultaneously to every reference geodesic. It gives endpoint neighborhoods such that geodesics joining points in those neighborhoods contain a nondegenerate interior segment of our arc. Shrink these endpoint neighborhoods so that short internal connections to the arc, followed by the arc, cost less than the positive cost of leaving the larger collar. Every joining geodesic then stays in that collar. Thus the anchor selection uses only the protected internal metric, as required for the local field-law transfer. Choose the endpoints from a fixed countable Euclidean dense set. They are metric dense as well. Fixed-pair uniqueness holds simultaneously for the countably many pairs by [34, Theorem 1.2]. Thus the saturated arc contains a nondegenerate segment of one of these unique anchor geodesics.

Saturation passes to its subarcs: otherwise splice a cheaper subpassage with the upper comparison on the two remaining pieces and contradict saturation of the larger arc. Choose rational fractional times inside this segment and a rational open neighborhood with compact clearance. Countably many anchor pairs, rational intervals, neighborhoods and length cutoffs cover the positive-probability saturation event. One fixed choice has positive probability, giving (10.8). $\square$

10.5 Amplifying and preserving a shortcut

We work in the tangent of Lemma 10.12. Its finer favorable labels have a shortcut ratio at most $C - \delta_0$, where $\delta_0 > 0$ is fixed. Here $v = 4^{-i}$ is a favorable shortcut label, $\hat{v} = Mv$ is the displaced annular label in Theorem 7.4, and $u = v^p$. By the reference cutoffs in Lemma 10.11, its denominator $d = D_h(z_0, z_1; \Omega)$ can be required to satisfy

$$a'u \leq d \leq A'u, \quad \text{cost}(\mathcal{S}) \leq (C - \delta_0)d \quad (10.9)$$

for constants $0 < a' < A' < \infty$. The certificate $\mathcal{S}$ lies in a closed rectangle R and has endpoints on opposite faces. A preservation patch V contains R with strict margin and satisfies $\bar{V} \subset \Omega$. We also require

$$D_h(z_0, z_1; V) \leq A'u \quad (10.10)$$

uniformly as the open rectangular collar V decreases to R . Lemma 10.6 bounds the internal diameter of R in every such collar. Its normalized cutoff has arbitrarily high probability, uniformly

in the label. Increase A' and restrict the unit shortcut test by this cutoff, losing less than half its fixed positive probability. Thus (10.10) and the small terminal-cap moduli are fixed before M is chosen. After a real shift the same requirements are read for D_{h+b} .

Lemma 10.13 (Many compartments). *Fix a desired shell failure probability $\varepsilon > 0$ and a sufficiently large fixed M . Put $w = (Mv)^p$. There are finitely many disjoint buffered compartments inside a fixed annular shell and a finite deterministic set of real shifts $\mathcal{B}$ with the following property, uniformly over the favorable labels. With probability at least $1 - \varepsilon$, one compartment and one $b \in \mathcal{B}$ have an auxiliary passage draw satisfying (10.9) for D_{h+b} , in actual cost units u , with the specified preservation patch, the bound (10.10) for D_{h+b} , and the reference regularity cutoffs. The event is local in the union of the compartment buffers. The necessary traversal order can be required to be determined in their fixed larger imaginary-field buffers.*

Proof. First restrict the positive-probability unit shortcut test so that its traversal order is determined inside a fixed finite imaginary buffer. The finite-order localization and whole-plane flow-tree determination in Lemma 6.6 allow the omitted probability to be less than half the original success probability. Fix this buffer before packing its copies. The resulting unit test has probability at least some $\eta_0 > 0$.

Inside the chosen shell, pack k smaller copies with mutually disjoint full buffers. Their relative shape is fixed; their spatial scale decreases when k increases. Decompose both fields into fresh Dirichlet parts and exterior harmonic parts on those buffers. After subtracting the coarse real value and the imaginary phase, the harmonic oscillation on a strict inner buffer has a scale-uniform Gaussian bound. Choose a bound H so that, with probability at least $1 - \varepsilon/4$, at least $k/2$ compartments have the required harmonic bound. Indeed the expected fraction failing any one bound is uniformly small, so Markov's inequality gives this conclusion; independence of the harmonic parts is not used.

Here is the exact coordinate and volume compensation. If the parent annulus has center c , radius r , and label Mv , its normalized real input is

$$g(x) = h(c + rx) + Q \log r - \gamma^{-1} \log(Mv).$$

For a compartment $x = d + az$, with $a > 0$, its input at label v after the preservation shift b is

$$g(d + az) + Q \log a + \gamma^{-1} \log M + b. \tag{10.11}$$

Write H_c for its exterior-measurable coarse value in $g(d + a \cdot)$. Round $-H_c - Q \log a - \gamma^{-1} \log M$ to the fixed mesh for b . The residual is bounded uniformly; the absolute label v has canceled. This chooses the field of a new auxiliary draw. By (6.14), $O_v = v^{-p}O$ on the same incidence array, and the reference metric has the identical scalar re-expression. Thus a unit-index shortcut has actual cost u times its unit cost, with $u = v^p$, and its preservation kernel is exactly the kernel for $h + b$. No change-of-field rule for a fixed passage cost is used.

On a compartment whose harmonic profile is bounded by H , bounded Cameron–Martin extensions and the bounded residual shift give a uniform conditional lower probability $p_0 > 0$ of success, as in Lemma 7.1. To see the direction of this estimate, restrict the reference harmonic profile to a fixed bounded set still carrying positive success probability. Between that restriction and any allowed target profile the Gaussian likelihood ratio has a uniformly bounded second moment. If L is the likelihood ratio and E the retained unit event,

$$\mathbb{P}_{\text{ref}}(E)^2 \leq \mathbb{P}_{\text{target}}(E) \mathbb{E}_{\text{target}}[L^2].$$

This bounds the target probability below uniformly. The same calculation applies to the imaginary harmonic part modulo its phase. Conditional passage kernels do not change the likelihood ratio. The constants depend on η_0, H , the fixed buffer shapes and the residual bound, and not on the absolute size of the compensating shift or on k .

Now choose k so large that $(1 - p_0)^{k/2} < \varepsilon/4$. Conditional on the exterior fields, the fresh fields and the passage draws in these disjoint buffers have product laws. Thus all good compartments fail with probability at most this amount. With this finite geometry fixed, truncate the coarse compensation values to a bounded range, losing probability less than $\varepsilon/4$. The possible mesh values now form the finite set $\mathcal{B}$. For each b , sample its full joint kernel coordinate, retaining the passage records, reference metrics, all required volume offsets, and feasible-order outputs under the shifted local input. Given the fields, different prospective b coordinates may be sampled independently; their within-coordinate consistency is preserved. The disjoint-compartment arrays have product conditional laws. All these draws precede selection of a successful coordinate. The finite-order data and their directed local traversals are retained at the same time. Additional reference regularity cutoffs on these finitely many fixed geometries can be chosen to lose the remaining $\varepsilon/4$.

This selection order is essential: the profile bound fixes p_0 before the number of compartments is chosen; the finite shift menu is truncated afterward. Uniformity comes from the bounded residual and the exact array re-expression, not from continuity of passage costs in a real-field shift. $\square$

The next lemma is deterministic once the reference metric and shortcut have been given. For each proposed smooth function f , suppose a passage system has upper comparison C with D_{h+f} and contains the retained shortcut on V , where $f = b$. These are its only required properties. Write F^f for its infimum certificate cost with compact confinement in the shell. The lemma constructs f and obtains a strict gain in that system. Lemma 10.15 subsequently constructs such an output system with the required dominated law.

Lemma 10.14 (A finite family of two-gate modifications). *Fix the cutoff data of Lemma 10.13, including its finite exact shift menu. Work in shell coordinates in which all modifications are supported in $B_2(0) \setminus \overline{B_{1/8}(0)}$. Let the reference anchor geodesic enter $B_{1/4}(0)$ and let both anchors lie outside $B_2(0)$. Write l_-, l_+ for the least reference costs from the anchors to $\partial B_2(0)$. On the reference shell cutoff suppose*

$$D(z_-, z_+) - l_- - l_+ \geq aw. \quad (10.12)$$

Choose M so that $A'u < aw/8$. There is a finite deterministic family of smooth functions, compactly supported in the permitted shell, with uniformly bounded amplitudes and Dirichlet norms, such that whenever a shifted shortcut from Lemma 10.13 is available, one of these functions has the following effect.

It equals the shortcut's exact shift b on its preservation patch V . For the modified reference metric $D^f = D_{h+f} = e^{\xi f} \cdot D_h$, every geodesic between the anchors contains a segment compactly within the shell for which that passage system admits a competing certificate confined in the shell, of cost strictly less than C times its reference length. The saving is at least $\delta_0 a' u/2$. The certificate uses the preserved shortcut on V and the upper comparison for its two approaches. All inequalities admit strict spatial and cost margins. The probability lost in imposing the required reference moduli and finite-domain cutoff bounds can be made arbitrarily small before using the simultaneous annulus theorem.

Proof. The proof concerns the reference metric and a retained passage certificate. A Weyl identity

is used only for D^f . Choose

$$0 < e < \min \left\{ \frac{A'u}{100}, \frac{\delta_0 a'u}{16(C+1)} \right\}, \quad B_0 = 2A'u.$$

All individual approximate-path and gluing errors below are allocated smaller fixed fractions of e .

Landing and terminal caps. Let $x_-, x_+ \in \partial B_2(0)$ minimize the two exterior landing costs. One can equivalently use approximate minimizers within the reserved error. For landing sets within τ of those minima, the reference triangle inequality and (10.12) give mutual distance at least $aw - 2\tau$. The reference upper modulus therefore separates these sets by a uniform positive Euclidean amount on the cutoff event. Choose two disjoint outer caps around x_-, x_+ , with all original connection errors less than e . Around the shortcut endpoints z_0, z_1 choose disjoint terminal caps whose internal D_{h+b} connection errors to the respective endpoints are less than e in Ω . Choose outer stops just inside $\partial B_2(0)$ and terminal stops just outside V . Decrease the margin of V below the terminal-cap scale. Lemma 10.6 provides the uniform connection estimates needed for this operation, including when an endpoint lies at a corner.

The allowed regions; see Figure 6. Join the outer stops to their corresponding terminal stops by two disjoint simple polygonal arcs outside V , avoiding the central omitted disk and the opposite caps. To construct them, first join the outer boundary component to the rectangular boundary component. Cutting along this arc leaves the planar domain connected, so the second prescribed joining arc can be drawn there. Fix their short end approaches before thickening the arcs.

Use two nested thickenings to obtain a smaller allowed set E_0 and a larger set

$$E_1 = O \cup T_0 \cup S \cup T_1.$$

Here O contains the exterior and a thin inner collar of $\partial B_2(0)$; the chamber S contains V with $\bar{S} \subset \Omega$; the tubes T_0, T_1 are disjoint; $O \cap S = \emptyset$; and the only overlaps are

$$O \cap T_j \subset \mathcal{A}_j, \quad S \cap T_j \subset \mathcal{B}_j, \quad j = 0, 1,$$

where $\mathcal{A}_j$ and $\mathcal{B}_j$ are the outer and terminal caps. The relevant overlaps have closures in slightly larger caps with the same error estimates. The smaller set has positive clearance from the complement of the larger set along every interior boundary. An outer wall layer lies between them, compactly in the permitted support, so exiting E_1 from the intended route requires crossing a fixed positive-width wall.

Inside each tube take a closed core K_j , a finite union of overlapping rectangles of positive width, connecting its two stops. It is separated from O, V , and the wall layer. Its old internal diameter is bounded by Kw on a fixed-domain cutoff. A transverse marker slab spans the whole tube away from the caps. Its two longitudinal faces have a fixed positive separation. Every complete tube transit crosses this slab.

A smooth function with the required plateaux. After fixing this geometry, let $m_*w > 0$ be a lower cutoff for a crossing of any wall or marker slab. Choose $H \geq \max\{0, b\}$ and $J \geq \max\{0, -b\}$ such that

$$e^{\xi H} m_*w > 2B_0, \quad e^{-\xi J} Kw < e/3.$$

Next choose $\rho > 0$ so small that a point within ρ of a core can access that core at old cost less than $ee^{-\xi H}/3$ in a slightly larger local collar. Keep these core collars disjoint from O, V , and the wall. The choice order is H , then J , then ρ ; the lower marker cutoff uses its fixed longitudinal faces and does not depend on ρ .

Choose a smooth baseline g which is nonnegative on O , at least b on S , exactly b on V and the smaller terminal caps, zero on the smaller outer caps and near both support boundaries, and exactly H on the walls and markers. It can be required to lie between $\min\{0, b\}$ and H . These closed plateau requirements have positive clearance, so smooth cutoff functions construct g . Let χ_j be smooth, equal to one on K_j , supported in its ρ -collar, with the two supports disjoint. Set

$$f = (1 - \chi_0 - \chi_1)g - J(\chi_0 + \chi_1). \quad (10.13)$$

Then $f = b$ on V , $f = 0$ outside the permitted shell, and $f = -J$ on each core. It is nonnegative on O , at least b on S outside the tubes, and equals H on the wall. On a marker it equals H except within the ρ -collar of the core. Outer cap hops have multiplier at most one, and terminal cap hops have multiplier at most $e^{\xi b}$.

Reference forcing. Follow the original exterior minimizing paths, the cheap cores and their cap hops, and an internal connection across V . This gives a modified competitor of length at most

$$l_- + l_+ + A'u + 2e < l_- + l_+ + B_0. \quad (10.14)$$

For any modified anchor geodesic, the portion between its first and last visits to $\partial B_2(0)$ has length at most B_0 , since the two exterior portions cost at least l_-, l_+ . This local portion cannot cross the high wall, and hence stays in E_1 . Every complete tube transit must approach its core within ρ while crossing the marker; otherwise it pays more than B_0 at height H . At such a marker point p_j , use the access to the core, its cheap internal connection, and the terminal hop. These give

$$D^f(p_j, z_j; \text{local collar}) < e. \quad (10.15)$$

The first outer cap visited must be $\mathcal{A}_0$. If it were $\mathcal{A}_1$, the prefix lies in O , where its old length is no larger than its modified length. Joining its last point to x_+ inside that cap and following the old exterior path to z_+ would give an old anchor connection of length at most $l_- + l_+ + B_0 + e$, contrary to (10.12). The last outer cap is similarly $\mathcal{A}_1$.

Take the first visit to $\mathcal{A}_1$ after the first visit to $\mathcal{A}_0$, and then the last visit to $\mathcal{A}_0$ before it. The intervening open path avoids both outer caps. The allowed-set overlaps imply that it lies wholly in O or wholly in $T_0 \cup S \cup T_1$. The first possibility again gives an old competitor, now with at most two cap errors, of length $l_- + l_+ + B_0 + 2e < l_- + l_+ + aw$. Thus there is an inner transition from T_0 through S to T_1 .

On that transition, take the first terminal-cap-1 visit and the last terminal-cap-0 visit before it. Between them the path lies in S outside both tubes. Its shift is at least b , and the cap endpoints are each within internal D_{h+b} distance e of their respective z_j . The central segment therefore has modified length at least $d - 2e$. Select marker points p_0, p_1 on the complete tube transits before and after this segment. As they belong to a geodesic,

$$D^f(p_0, p_1) \geq d - 2e.$$

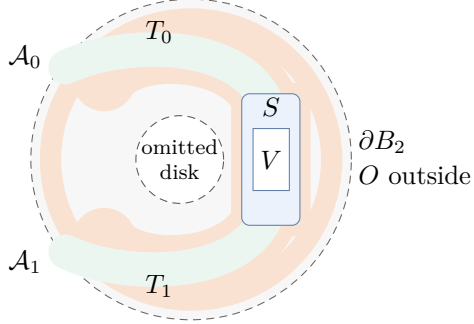
This last/first selection incurs at most one error per cap regardless of repeated excursions.

Passage gain and finiteness. Use (10.15), the output upper comparison, and the retained shortcut certificate. Concatenating with its strict margins gives

$$F^f(p_0, p_1) \leq (C - \delta_0)d + 2Ce \leq CD^f(p_0, p_1) - (\delta_0 d - 4Ce).$$

The final parenthesis is at least $\delta_0 a' u / 2$, allowing the reserved open losses.

(a) Spatial placement inside the permitted shell

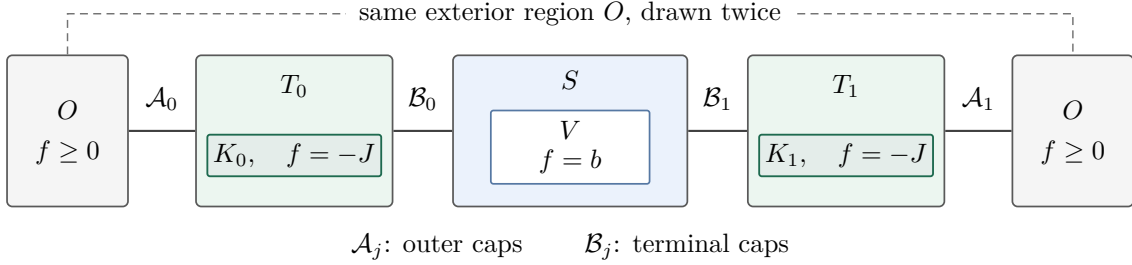


The two tubes join distinct outer caps to the chamber S and avoid the central omitted disk. The orange collar represents the high wall around the interior route.

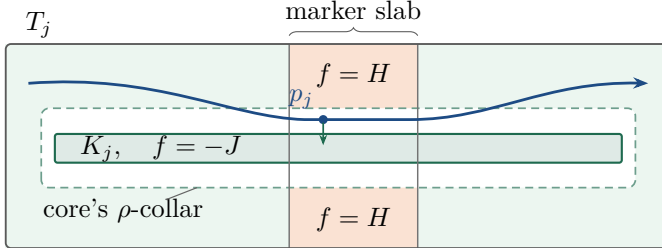
The wall and the field modification have positive clearance from both support boundaries. The outer region O also includes a thin inner collar of ∂B_2 .

This panel shows placement only; the adjacency and marker constraints are detailed below.

(b) Allowed-region adjacency



(c) A transverse marker forces access to the core



A complete transit with D^f -length at most B_0 must enter the ρ -collar, since

$$e^{\xi H} m_{*} w > 2B_0.$$

From such a point p_j , the cheap core and terminal hop give

$$D^f(p_j, z_j; \text{local collar}) < e.$$

Figure 6: The gate construction in Lemma 10.14. (a) Schematic placement of the chamber and two tubes away from the central omitted disk. The orange collar indicates the high wall; its exact shape and width are not prescribed by the drawing. (b) Adjacency of the allowed regions; the two O boxes denote the same exterior region. The only overlaps are in the outer caps $\mathcal{A}_j$ and terminal caps $\mathcal{B}_j$. The preservation patch $V \subset S$ has $f = b$, and the tube cores K_j have $f = -J$. (c) On a transverse marker, $f = H$ outside the core's ρ -collar. The crossing-cost cutoff forces a transit of D^f -length at most B_0 into that collar; the cheap core and terminal hop then give access to z_j . The high wall confines the route to E_1 .

Here is the order that makes the recipe family deterministic. Fixing M and the amplification geometry fixes e/w and B_0/w . The initial reference-modulus cutoffs give deterministic cap radii and positive endpoint separations. All endpoint and rectangle parameters therefore lie in a deterministic compact family with fixed aspect and boundary-clearance bounds. Each valid polygonal construction persists on a neighborhood of its parameters. Choose a finite subcover of these parameter families; equivalently use sufficiently fine deterministic endpoint and rectangle bins. First add uniform reference-modulus cutoffs for this finite library, then its wall and core cutoffs, and finally choose H, J, ρ uniformly over it. The resulting finite list of smooth functions has bounded amplitude and bounded Dirichlet norm. These later wall and core cutoffs change the plateau amplitudes; they do not change the earlier spatial parameter family, gap or budget. Translation and Euclidean dilation preserve the Dirichlet energy in two dimensions. All cutoff losses can be made arbitrarily small without changing the earlier gap aw or the budget B_0 : the new wall and core constants are absorbed by H, J . This proves the asserted finite, strict construction. $\square$

10.6 The weighted change-of-measure contradiction

Fix the anchor pair, interval and neighborhood in (10.8). Restrict the saturation event to total reference length in a fixed compact subinterval of $(0, \infty)$ and to a fixed positive clearance of the saturated interval in U . All modifications below are centered near its middle third. For a favorable label v , use the shell event of Lemmas 10.13 and 10.14, aligned to Mv . Its weight is $w = (Mv)^p$. The original geodesic's passage through the central hole supplies (10.12). Require also the reference outer-circle diameter bound Aw and the local lower crossing cutoffs used in the gate construction.

For the finite list of smooth functions, add a cutoff on every Gaussian pairing appearing in its Cameron–Martin likelihood. The functions have uniformly bounded Dirichlet norms after spatial rescaling, so these finitely many pairing cutoffs have arbitrarily high marginal probability, uniformly in the label. The simultaneous annulus theorem applies to their intersection with the other shell events.

Lemma 10.15 (Transfer before selection). *For each fixed ball, label and gate recipe, let μ be any subprobability measure obtained by restricting the source joint law to saturation, a successful hit, that recipe, and its likelihood cutoff. Its image under the one-ball modification can be completed to a measure ν satisfying*

$$\nu \leq K\mathbb{P}, \tag{10.16}$$

where $\mathbb{P}$ is one common unmodified joint law of the tangent field, reference metric and passage observations. The finite K is independent of the ball, label, window length and selected hit.

Proof. Fix the recipe before imposing a hit or saturation condition. Condition on both fields. On its preservation patch retain the auxiliary draw for the exact field $h + b$, and outside a strictly larger modification buffer retain the original exterior observations. Their joint law is the product-kernel marginal in Lemma 6.11. Under $h' = h + f$, these are exactly the two required output kernels, since $f = b$ on the first patch and $f = 0$ on the second. Complete all other output observations with their correct conditional law. The imaginary field is unchanged.

The only density change is now that of the real field. The Cameron–Martin density, evaluated on the source field, is

$$L_f(h) = \exp \left((h, f)_{\nabla} + \frac{1}{2} \|f\|_{\nabla}^2 \right).$$

On the imposed likelihood cutoff it is at most K . Therefore the unselected image restricted to that cutoff is bounded by $K\mathbb{P}$. Restricting further to any event determined by the old observations, the

auxiliary shortcut, the hit or the chosen recipe gives a submeasure and preserves this domination. Thus no conditional independence after selecting a hit is required. The maximum cutoff over the fixed finite recipe list is uniform under spatial dilation and translation, giving the claimed single K . $\square$

Lemma 10.16 (Strict loss approaches equality). *There are deterministic $\epsilon_N \downarrow 0$ and source events whose probabilities tend to one such that every selected modification in the N th annular window satisfies, in its output,*

$$0 < C - \frac{F_U(P'(t_1), P'(t_2))}{D_{h'}(P'(t_1), P'(t_2))} \leq \epsilon_N. \quad (10.17)$$

Here P' is the output reference geodesic between the same anchors. The denominator remains bounded above and away from zero.

Proof. All supports have radius at most a deterministic number tending to zero, their centers range over a fixed compact set, and their amplitudes are bounded by a fixed constant. The old D -diameters of these supports tend uniformly to zero. On either an old or a modified geodesic, the part between the first and last support visits can be replaced by a reference connection between two points of that support, at cost bounded by a fixed amplitude factor times its old diameter. Outside the support the two length rules agree. Consequently all modified anchor paths have old length tending to the old anchor distance. Global comparison by the bounded multiplier keeps them in a fixed proper metric ball. Their arclength parametrizations are equicontinuous; every subsequential limit is an old geodesic. Uniqueness identifies it with P . The collapsed support-spanning portions have vanishing arclength, so fractional times converge as well. This reasoning is uniform over the finite recipe geometries and all their centers in the compact set: a failure of uniformity would give a contradicting sequence of shrinking supports.

In particular, the gate segment lies inside the same fixed interval $[t_1, t_2]$ of P' for all sufficiently fine windows, with compact clearance in U . Apply the output upper comparison to the two remaining pieces of this interval and use the strict gain from Lemma 10.14. This proves the strict left inequality in (10.17).

For the other inequality, take a proposed output competitor in U with bounded cost. Cut it at its first and last encounters with a slightly larger ball than the observation buffer. Its exterior subcertificates have the original costs and can be retained in the old law. Replace the whole intervening portion by one old short connection, and correct its two moved endpoints by old short connections. Lemma 10.7 bounds all added costs by a deterministic function of the old compact reference modulus, the largest support radius, and the endpoint displacements, tending to zero. Repeated visits cost no additional error because only the first and last encounters are used. If the competitor never enters the ball, it transfers directly. Strict larger buffers ensure compact confinement of the retained exterior records even when an individual competing path has only its own positive clearance in U .

An output competitor whose cost was below the old saturated value by a fixed positive amount would therefore contradict (10.8). This proves the required lower bound on its cost up to an error tending uniformly to zero on old reference-data cutoffs. The reference endpoint distances converge, and the fixed interval length has a positive lower bound. The resulting ratio error tends to zero in probability. Choose a deterministic sequence $\epsilon_N \downarrow 0$ by successive probability cutoffs; retain the source event on which all errors are at most ϵ_N . Its probability tends to one, proving the assertion. $\square$

Lemma 10.17 (Weighted image-count contradiction). *The saturated tangent of Lemma 10.12 cannot admit the fixed-gap shortcuts described there.*

Proof. Let $\mathcal{X}$ denote the measurable space of the complete joint observations, and let $\mathcal{I}_N$ be the finite set of centers, scales, volume labels and recipes in the simultaneous annulus construction, and write $w_i = (Mv_i)^p$. These weights are deterministic. Under the one common unmodified output law $\mathbb{P}$, define

$$R = \frac{F_U(P(t_1), P(t_2))}{D_h(P(t_1), P(t_2))}, \quad E_N = \{0 < C - R \leq \epsilon_N\}.$$

Set $R = +\infty$ unless $P([t_1, t_2]) \subseteq U$ and the denominator is positive. Thus R is a single random variable, independent of the choice of ball or recipe.

For an index i , let G_i be the following event of the output alone. The output anchor geodesic satisfies the fixed total-length cutoff and makes a complete transit of the recipe's marker slab inside its enlarged shell, with subpath length at least $a_3 w_i$. Also require the alignment test, expressed using the output field minus that recipe's fixed function f_i . Choose $a_3 > 0$ below the source marker-crossing cutoff times the minimum bounded Weyl factor. Every modified good hit has these properties. All marker faces, shells and functions are deterministic for index i ; the reference geodesic is measurable and almost surely unique. Hitting times, with the retained strict spatial margins, therefore make G_i measurable from the output field and geodesic. It refers to no discarded source path or auxiliary draw.

On the saturation event, intersected with the high-probability cover and error events, every point of the middle third of the interval has at least cN good covering holes. The time spent by a reference geodesic in any one ball is at most Aw_i : its first and last outer-circle visits can be connected within the surrounding buffer at this cost. Integrating the covering multiplicity along an interval of length bounded below gives

$$\sum_{i \in \mathcal{I}_N} w_i \mu_i(\mathcal{X}) \geq c_1 N \tag{10.18}$$

for $c_1 > 0$, where μ_i is the source submeasure for the corresponding good hit and a selected valid recipe. Choose one valid recipe measurably from the fixed finite list for each hit. The source probability of the retained saturation event stays bounded below, since its original probability was positive.

Push each submeasure separately through its modification, obtaining ν_i . By Lemma 10.15, $\nu_i \leq K\mathbb{P}$ for the same $\mathbb{P}$ and K . By Lemma 10.14 and Lemma 10.16, it is supported on the output event $G_i \cap E_N$ defined above.

The output events satisfy

$$\sum_{i \in \mathcal{I}_N} w_i \mathbf{1}_{G_i} \leq c_2 N \tag{10.19}$$

on the fixed output total-length cutoff. At a fixed spatial radius, choose the first qualifying marker transit specified by each G_i . Its length is at least $a_3 w_i$ and its whole parameter interval stays in that enlarged shell. The mesh balls have bounded overlap, so integration along the output geodesic bounds the sum of these subpath lengths by a fixed multiple of its total length. There are $O(N)$ spatial radii. The alignment coarse value is measured outside the modification support; alternatively, its change is bounded by the fixed amplitude. Consequently only boundedly many volume labels can be aligned with one spatial ball. The finite number of recipes contributes another fixed factor. This proves (10.19).

Combining the two weight bounds and the common-law domination gives

$$c_1 N \leq \sum_i w_i \nu_i(\mathcal{X}) \leq K \int \mathbf{1}_{E_N} \sum_i w_i \mathbf{1}_{G_i} d\mathbb{P} \leq K c_2 N \mathbb{P}(E_N).$$

For a fixed random variable R , the probability $\mathbb{P}(0 < C - R \leq \epsilon_N)$ tends to zero. This is a contradiction. All modifications were applied separately; neither independence of selected hits nor a simultaneous modification of overlapping balls was used. $\square$

Theorem 10.18 (Rigidity of a finite port comparison). *Every limiting law satisfying (10.7) has coincident optimal constants $c = C = \lambda$, where $0 < \lambda < \infty$ is deterministic. In every ordinary open chart,*

$$F_U(x, y) = \lambda D_h(x, y; U).$$

The lower comparison applies to all certificates; the upper comparison has the complete ordered-port meaning of Definition 10.4.

Proof. If $c < C$, Lemma 10.12 constructs a saturated tangent with fixed-gap shortcuts. Lemma 10.17 excludes it, so $c = C$. Positivity and finiteness follow from Lemma 10.10.

The upper inequality for the displayed internal distance follows by following internal reference paths. For the reverse inequality, start with a bounded-cost certificate compactly confined in U and its actual discrete approximating paths. Before taking limits, add ordered marks whenever their accumulated normalized cost first crosses a multiple of 2^{-k} , retaining the previously specified marks. The single-edge cost tends to zero, since the initial-growth argument gives $A_j \rightarrow \infty$. For each fixed k the number of marks is bounded by the cost budget. Lemma 5.10 lets us retain all these finite refinements consistently along a diagonal.

Apply the lower comparison to the subrecord between any two retained marks. Their D -distance is at most $1/c$ times their intervening cost. This is the modulus that was unavailable before $c > 0$. The marked points therefore extend to a continuous path parametrized by accumulated cost; flat cost intervals have identical spatial endpoints. Its image stays in the retained compact confinement. For every finite subdivision, the sum of its reference endpoint distances is at most the certificate cost divided by c . Taking the supremum over subdivisions bounds its D -length by that value. Thus the certificate costs at least $cD(x, y; U)$, and infimizing gives the reverse inequality. This construction uses consistent finite refinements and the positive lower bound; no continuous path is inferred from a spatial range alone. Exhaust compact confinements and retain the arbitrary positive open losses to obtain the assertion for every open U . $\square$

10.7 Propagation of the deterministic normalizers

Lemma 10.19 (Quantile control in a rigid law). *There are deterministic $0 < k_- < k_+ < \infty$, depending only on the fixed quantile and buffered test geometries, with the following property. If a joint extraction in A_j units has comparison λD , then every fixed offset $t \in \mathbb{Z}$ has*

$$k_- \lambda 4^{pt} \leq \liminf \frac{A_{j+t}}{A_j} \leq \limsup \frac{A_{j+t}}{A_j} \leq k_+ \lambda 4^{pt}. \quad (10.20)$$

In particular $\lambda \in [k_+^{-1}, k_-^{-1}]$. The statement includes forward offsets whose ratios were first extracted in $[0, \infty]$.

Proof. Fix $t \in \mathbb{Z}$ and put $v = 4^t$. In the joint extraction retain the raw observations at effective time unit $v4^j = 4^{j+t}$ and cost denominator $A_j v^p$; denote their local kernel array by K_t . Retain the shifted-base-coordinate description and the prescribed circle-average cone description jointly, as spatial relabellings of the same time-labelled raw records in the common aggregate reference extraction, before changing any field law. This is the effective-time array supplied by Theorem 6.1(1),(4). Evaluating K_t with the fixed embedded cone input and band convention used in (10.6) therefore gives the limiting buffered observations of $X_{j+t}/(A_j v^p)$. The time-labelled raw records are those of that same word model at time 4^{j+t} , and their spatial relabelling uses this fixed input. No identification of K_t with the kernel at time 4^j is asserted.

The indexed unit identity of Proposition 6.8 instead identifies K_t at the shifted input $h_v = h - \gamma^{-1} \log v$: its passage costs and reference metric are both v^{-p} times those at the original input h . The comparison constant in this shifted-input experiment is thus λ . On the fixed dependency buffer, the shifted cone restriction and the unshifted circle-average cone restriction are mutually absolutely continuous. The same kernel K_t is used under these two input laws. Hence their probability-one local comparison with constant λ transfers to the fixed unshifted cone input. This step uses only an almost-sure transfer for each fixed t ; it does not require a density bound uniform in t . Internal metrics and all circuit and traversal tests here are confined to the retained buffer. Countably many fixed offsets may be retained together.

Now take the reference cutoffs under that one fixed cone input law. A circuit in the middle band contains a reference advance across a strict smaller band. Its positive lower cutoff can have probability sufficient to contradict the upper quantile if the ratio in (10.20) were too small. The all-certificate lower comparison gives the left bound. Conversely a rectifiable reference winding path in the narrower band has a finite-length cutoff with probability greater than τ , with the needed probability slack. Following and stitching it gives a middle-band circuit and hence the right bound. Both cutoffs depend only on this fixed input law and the test geometries, so the constants k_-, k_+ are independent of t and of the extracted law. Strict buffered tests and the two quantile inequalities after (10.6) justify these transfers, including possible quantile atoms. Setting $t = 0$ gives the stated compact interval for λ . The same bounds exclude zero or infinite forward ratios before renormalizing at that offset. $\square$

Lemma 10.20 (Comparable preceding power ratios). *Suppose a joint extraction at scales $j \rightarrow \infty$ satisfies the following tests at every fixed preceding offset: adjacent A-ratios lie in a fixed compact subinterval of $(0, \infty)$, there is the normalized upper circuit quantile, and a fixed sufficiently high-probability normalized lower traversal test holds. Then*

$$r_i := 4^{pi} \lim \frac{A_{j-i}}{A_j}, \quad i \geq 0,$$

can be extracted with $r_i \in (0, \infty)$, and one constant L_0 satisfies $L_0^{-2} \leq r_i \leq L_0^2$ for all i .

Proof. The adjacent bounds give positive finite limits for every fixed offset after a diagonal extraction. Fix $\theta < 1/4$. Choose a large L_0 , uniformly in the starting offset i . If, for arbitrarily large N , more than θN values of $k \in [N, 2N]$ satisfy $r_{i+k} > L_0 r_i$, the preceding lower tests at these deterministic labels give barriers of size at least a fixed constant times $L_0 4^{-pk}$ in the start- i units. The simultaneous annulus theorem and Lemma 10.5 force a lower comparison proportional to $L_0 D$ at the starting scale. For sufficiently large L_0 this contradicts its upper circuit quantile, because a fixed reference band crossing has positive distance with the required probability. All constants are uniform in i : at the starting scale use its own A_{j-i} and evaluate its effective-time kernels with the same fixed cone input and band convention as in (10.6). The ordinary local laws

used by the simultaneous annulus theorem are likewise fixed at every label. Thus the reference probability cutoffs in this contradiction are chosen once; no uniform absolute-continuity bound for an unbounded family of additive shifts is being used.

Likewise, if more than θN labels have $r_{i+k} < r_i/L_0$, their upper circuit quantiles and upper chaining give a following constant proportional to L_0^{-1} . A reference traversal with a fixed finite-length cutoff then contradicts the starting normalized lower traversal test. Increase L_0 once to handle both contradictions. Consequently, for every fixed i and all sufficiently large N , more than $(1 - 2\theta)N$ indices $\ell \in i + [N, 2N]$ satisfy

$$r_i/L_0 \leq r_\ell \leq L_0 r_i. \quad (10.21)$$

For two fixed starts i, i' , the shifted intervals differ in at most $2|i - i'|$ indices. Since $1 - 2\theta > 1/2$, the two sets of indices in (10.21) intersect for sufficiently large N . At an intersection index, $L_0^{-2} \leq r_i/r_{i'} \leq L_0^2$. Take $i' = 0$, for which $r_0 = 1$. The constant is independent of i , as required. $\square$

Theorem 10.21 (All-scale port normalization). *There is a deterministic positive function $B(n)$, tending to infinity, and constants $s > 0$, $K < \infty$ such that*

$$\frac{B(n')}{B(n)} \leq K \left(\frac{n'}{n} \right)^s, \quad 1 \leq n' \leq n. \quad (10.22)$$

One can take $s = p/2$ and $B(n) = A_{\lfloor \log_4 n \rfloor}$, after changing finitely many initial values.

Along every deterministic sequence of diverging time units, joint crossing-data extractions in $B(n)$ units have local comparison λD with a deterministic λ in a fixed compact subinterval of $(0, \infty)$, after further extraction. The comparison includes all finite-cost lower certificates, complete ordered-port upper following, the simultaneous small surrounding circuits of Lemma 10.7, and finite connected systems of paths with prescribed transverse intersections. The same comparison holds on every ordinary compact patch of a conditioned finite-sphere extraction using these deterministic units and the common local kernels.

Proof. We propagate a finite list of tests. Choose a broad adjacent-ratio interval (L^{-1}, L) which is strictly passed by the ratios of Lemma 10.19. Put $s = p/2$ and choose an integer m so large that those rigid ratios strictly pass

$$\frac{A_{j-m}}{A_j} < 4^{-sm}.$$

Finally choose a sufficiently small fixed normalized lower traversal threshold so that every rigid law with $\lambda \in [k_+^{-1}, k_-^{-1}]$ strictly passes its high-probability lower test. Use the continuous buffered-test convention above, with enough probability slack that limits of passing tests still suffice for the simultaneous annulus theorem. The upper circuit quantile is already supplied at every index by (10.6). The remaining finite test family consists of the adjacent-ratio bounds, the displayed m -step bound, and the buffered lower traversal test. We use two different implications: limits of passing tests retain the closed inequalities needed for Lemma 10.20, whereas every rigid law passes the tests with strict margins and therefore forces nearby approximants to pass. The first-failure argument below uses these implications in that order.

There are arbitrarily long success runs at arbitrarily large indices. Indeed, by Lemma 10.9, for every $\varepsilon > 0$ the sequence $4^{-(p-\varepsilon)j} A_j$ has arbitrarily late record maxima. Take record indices with $\varepsilon \downarrow 0$. For every fixed preceding offset i their ratios satisfy

$$\frac{A_{j-i}}{A_j} \leq 4^{-(p-\varepsilon)i}.$$

Every joint limit therefore satisfies (10.7). Theorem 10.18 and Lemma 10.19 imply that all the chosen tests strictly pass at every fixed backward and forward offset. If some bounded window of tests failed along these record indices, another joint extraction would contradict that strict passage. A diagonal choice yields success runs whose lengths tend to infinity.

Suppose there were infinitely many later failures. Starting from the success runs just obtained, extract at the level immediately before the first later failure. Every fixed preceding offset passes the tests. Lemma 10.20 supplies the uniform exact-power bound (10.7). The extracted law is rigid, and Lemma 10.19 includes the first forward offset. Thus all the tests strictly pass at the alleged failure as well. The continuity and strict-loss convention for the tests gives a contradiction. Hence all sufficiently large indices pass.

In particular, for those indices, $A_{j-m}/A_j \leq 4^{-sm}$ and the adjacent ratios are bounded. Write $j - j' = km + r$, with $0 \leq r < m$. Iterating gives

$$\frac{A_{j'}}{A_j} \leq L^r 4^{-smk} \leq L^m 4^{sm} 4^{-s(j-j')}.$$

Changing the constant handles the finite initial indices. Passing from j, j' to $\lfloor \log_4 n \rfloor, \lfloor \log_4 n' \rfloor$ changes this bound by at most another fixed factor, proving (10.22). Taking $n' = 1$ proves $B(n) \rightarrow \infty$.

For an arbitrary sequence of powers of four, all fixed preceding tests now pass. Extract their ratios and apply Lemma 10.20 again. Theorem 10.18 identifies the local port law, and Lemma 10.19 puts its deterministic factor in the fixed positive compact interval. For a general time unit n , extract also $n/4^{\lfloor \log_4 n \rfloor} \in [1, 4]$. The bounded time-unit re-expression of the contour, field and certificate arrays absorbs this converging factor into λ ; its possible p th powers remain in a fixed compact interval. No convergence rate is needed for this extraction.

The simultaneous circuit and finite-system assertions follow from Lemma 10.7 and the robust topological intersection statements. Finally, the finite-volume part of Theorem 6.1 uses the same kernels and the same deterministic cost units. Lemma 6.10 dominates the actual normalized sphere field, with its actual additive height, by the ordinary field law on each such patch. Together with Proposition 6.8, it transfers the probability-one comparisons there. A countable chart exhaustion makes them simultaneous off the exploration root.

For the sphere with n edges the duration-one letter clock has unit $2n$, so this first gives the comparison in $B(2n)$ units. The propagated adjacent-ratio bounds imply that $B(2n)/B(n)$ lies in a fixed positive compact interval: the two base-four integer parts differ by at most one. Retain a deterministic subsequential limit r of this ratio and absorb it into the comparison constant. Thus the $B(n)$ convention used in Section 11 has factor $r\lambda$, which we again denote by λ , with all passage and circuit conclusions scaling together. Neither an exact ratio limit nor an additional time-scaling factor is required. Uniform costs to prescribed lattice endpoints are supplied separately in Section 11. $\square$

11 Completion in the canonical coordinate

The local passage comparison allows endpoints to be chosen within small ports. We now identify distances between arbitrary vertices, including vertices approaching the exploration root. The intrinsic GH input supplies a compact metric sphere; the task is to identify its points with their limiting conformal locations.

Fix $q \in (0, 4)$. We use the deterministic normalization a_n in Theorem 1.2, with the continuum metric normalization specified there. Put

$$V_n = V(M_n), \quad \delta_n = a_n d_n, \quad z_n(v) = \phi_n(v), \quad \mu_n = (\phi_n)_* m_n.$$

The metric d_n here uses every original primal edge. In particular, shortest paths used below are paths in the original primal graph.

We keep two coordinates distinct. The original oriented peanosphere coordinate has field h° , metric D° , area μ° and exploration root $p_*^\circ = \infty$. It has the quantum-sphere and independent imaginary-field laws used in Sections 3 and 6. Write

$$x_n(v) = H_n(v), \quad f_n = \phi_n \circ H_n^{-1} \longrightarrow f$$

along a joint extraction of Theorem 8.9. The limit f may be Möbius or anti-Möbius, and its orientation may depend on all the retained data. Define the canonical geometric objects by

$$\mu = f_* \mu^\circ, \quad p_* = f(p_*^\circ), \quad D(z, w) = D^\circ(f^{-1}(z), f^{-1}(w)). \quad (11.1)$$

These definitions transport geometry sample by sample. Field-density and local-kernel arguments use the oriented peanosphere field and, when constructed below, its orientation-preserving reembedding by fresh area marks. We verify the field law in that reembedding explicitly; no field-law assertion is inferred from the possibly adaptive orientation of f . Port and circuit conclusions in the canonical coordinate are the pathwise transports of the already proved conclusions for x_n ; uniform convergence of f_n and f_n^{-1} preserves their strict margins.

Let T be the orientation-preserving Möbius map that takes the three actual limiting normalization samples in the original coordinate to 0, 1, ∞ , and set

$$\mu^+ = T_* \mu^\circ, \quad D^+(z, w) = D^\circ(T^{-1}(z), T^{-1}(w)).$$

This pair has the prescribed canonically marked quantum-sphere law. Write $\mathfrak{r}(z) = \bar{z}$ on $\widehat{\mathbb{C}}$, with $\mathfrak{r}(\infty) = \infty$. It acts on a measure by pushforward, on a distance graph by $(z, w, r) \mapsto (\mathfrak{r}z, \mathfrak{r}w, r)$, and on a loop collection by composing every loop with $\mathfrak{r}$. These actions are continuous in the topologies of the main theorem; the graph and loop distances are unchanged. The map f is either T or $\mathfrak{r} \circ T$. We will remove this common reflection at the level of the joint law after identifying the metric graph.

Theorem 11.1 (Completion of the metric graph). *In every joint subsequential extraction supplied by Theorems 8.9 and 10.21, one may further extract the intrinsic spaces (V_n, δ_n) and their spatial distance graphs. Almost surely, every resulting limit of the latter is*

$$\mathcal{G}_D = \{(z, w, D(z, w)) : z, w \in \widehat{\mathbb{C}}\}. \quad (11.2)$$

Consequently

$$(\mu_n, K_n(a_n)) \implies (\mu^+, \mathcal{G}_{D^+})$$

through all positive integers. In any convergent joint extraction with additional reference or local observations, its metric component is the same graph (11.2). The Hausdorff metric for the second component is that induced by $\rho(z, z') + \rho(w, w') + |r - r'|$.

The proof compares the passage scale $B(n)$ of Theorem 10.21 with the intrinsic scale a_n . We first show that $a_n B(n)$ is bounded and retain a deterministic subsequential limit $\alpha \geq 0$; let λ be the passage comparison constant on the same extraction. There are two possibilities. If $\alpha > 0$, the lower passage comparison and small primal separators identify the location relation as a homeomorphism, with metric factor $\alpha\lambda$. If $\alpha = 0$, those separators force every intrinsic point except one to have the same spatial location. Two separated local escape tests will rule out that concentration. Once $\alpha > 0$, the intrinsic diameter law fixes $\alpha\lambda = 1$.

We begin by retaining the full location relation. The intrinsic input identifies each marginal limit of (V_n, δ_n) as a nontrivial compact metric sphere, but gives no identification with the coordinate z_n .

11.1 Location relations and separators

Lemma 11.2 (Retaining every vertex). *Suppose that, on a coupled subsequence, (V_n, δ_n) converges in Gromov–Hausdorff distance to a compact metric space (Y, d_Y) , and that $z_n(V_n)$ becomes dense in $\widehat{\mathbb{C}}$. There are maps $j_n : V_n \rightarrow Y$ and numbers $e_n \downarrow 0$ such that*

$$|d_Y(j_n(u), j_n(v)) - \delta_n(u, v)| \leq e_n, \quad \sup_{y \in Y} \text{dist}_{d_Y}(y, j_n(V_n)) \leq e_n. \quad (11.3)$$

After a further subsequence, the finite sets

$$R_n = \{(j_n(v), z_n(v)) : v \in V_n\}$$

converge in Hausdorff distance in $Y \times \widehat{\mathbb{C}}$ to a compact relation R that projects onto both factors. On the same subsequence,

$$K_n(a_n) \longrightarrow \mathcal{K}(R) := \{(z, w, d_Y(y, y')) : (y, z), (y', w) \in R\}. \quad (11.4)$$

Proof. A correspondence of distortion tending to zero in a Gromov–Hausdorff realization gives (11.3): choose one related point of Y for each vertex. The image is a net with error tending to zero. The space of nonempty compact subsets of the compact space $Y \times \widehat{\mathbb{C}}$ is compact, so a subsequence of R_n converges. Net density in the two factors proves the two surjectivity assertions.

For the upper inclusion in (11.4), take any converging sequence of triples from $K_n(a_n)$ and extract the two Y -coordinates of its vertices. They give two pairs in R , and (11.3) identifies the third coordinate. For the reverse inclusion, approximate any two pairs of R by pairs of R_n at the same indices. Their distances again converge by (11.3). This proves both Hausdorff inclusions. $\square$

The mesh assertion of Corollary 8.10 supplies the spatial density needed in this lemma: every flag contains an original vertex, and the maximum spherical flag diameter tends to zero. The maps j_n need not be continuous or agree with the spatial coordinate.

Here is the elementary use of the topology of the intrinsic limit. For a set of vertices S_n , write $G_n \setminus S_n$ for the graph obtained by deleting those vertices and their incident edges.

Lemma 11.3 (A collapsing separator). *Assume (11.3). Let $y_0 \in Y$, and suppose that $Y \setminus \{y_0\}$ is path connected. If*

$$\sup_{v \in S_n} d_Y(j_n(v), y_0) \longrightarrow 0,$$

then, for any vertices u_n, v_n with $j_n(u_n) \rightarrow y$ and $j_n(v_n) \rightarrow y'$ in $Y \setminus \{y_0\}$, the two vertices are in the same component of $G_n \setminus S_n$ for all sufficiently large n . The conclusion also holds when $y = y'$.

Proof. Choose a continuous path P from y to y' in $Y \setminus \{y_0\}$. If the endpoints coincide, take the constant path. Its compact image has distance $b > 0$ from y_0 . By uniform continuity, choose a finite partition of P into pieces whose endpoints are at d_Y -distance less than $b/10$. Approximate the partition points by $j_n(V_n)$, using u_n, v_n at the ends. For large n , consecutive chosen vertices have δ_n -distance less than $b/5$, and each chosen vertex has Y -distance at least $4b/5$ from y_0 .

Join consecutive vertices by graph shortest paths. Every vertex of such a path is at δ_n -distance less than $b/5$ from its initial vertex, so (11.3) keeps its Y -image at distance at least $b/2$ from y_0 . It therefore avoids S_n . Concatenation proves the assertion. The same proof shows the following useful version: for this fixed path P , it is enough that all of $j_n(S_n)$ eventually lies within distance $b/4$ of y_0 . $\square$

Every point of a topological sphere has a path-connected complement. Thus this lemma applies to the intrinsic limit furnished by Theorem 1.2. When S_n is a primal circuit or a connected primal skeleton, a path in $G_n \setminus S_n$ cannot cross it on the embedded surface: primal edges meet only at their incidence vertices. The strict separation assertions in Corollary 4.10 therefore apply to these graph paths, including in maps with loops or multiple edges.

11.2 The comparison scale and the exploration root

Lemma 11.4 (Small bypasses at an area mark). *Almost surely there are rectifiable closed paths L_k in $\widehat{\mathbb{C}} \setminus \{p_*\}$, each separating p_* from the complement of a neighborhood of p_* , whose spatial diameters and D -lengths tend to zero. Consequently, for $z, w \neq p_*$, the internal distance in $\widehat{\mathbb{C}} \setminus \{p_*\}$ equals $D(z, w)$. The metric completion of that punctured space is $(\widehat{\mathbb{C}}, D)$.*

Proof. Work first with D^0 in the original oriented sphere. A fixed orientation-preserving inversion puts $p_*^0 = \infty$ at zero. By Proposition 3.5, near this mark the field is represented by a GFF with a logarithmic insertion of strength γ , a locally continuous term, and the finite random additive constant that normalizes its area; the change to the unit-area law has a positive integrable density. We may first work with

$$h^\gamma = h^0 - \gamma \log |\cdot|,$$

where h^0 is a whole-plane GFF normalized on the unit circle. The local continuous term changes lengths by factors bounded above and below on a fixed smaller disk, by Weyl scaling.

For $0 < r < 1/4$, consider the annulus $r < |z| < 2r$. The around-annulus estimate for h^γ gives a disconnecting closed path of length at most

$$r^{\xi(Q-\gamma)} e^{\xi h_r^0(0)} T_r, \tag{11.5}$$

where the nonnegative variables T_r form a tight family. This is the around-distance estimate, not just a point-to-circle estimate; see Definition 3.7 and the proof of Proposition 3.14 in [14]. It also follows by scaling a unit annulus, using the spatial coordinate rule and the tight unit-annulus internal distances. The normalized field $h^0(r \cdot) - h_r^0(0)$ has the same annular law at every r .

Now

$$Q - \gamma = \frac{4 - \gamma^2}{2\gamma} > 0, \quad \frac{h_r^0(0)}{\log(1/r)} \longrightarrow 0 \quad \text{in probability.}$$

Indeed the numerator is centered Gaussian of variance $\log(1/r)$. Thus (11.5) tends to zero in probability; no independence between its factors is needed. Bounded local continuous perturbations, the finite area-normalizing factor, and an absolutely continuous probability weighting preserve this convergence. Choose deterministic radii $r_k \downarrow 0$ so that the probabilities of a length exceeding

2^{-k} are summable. Borel–Cantelli gives such paths almost surely in the original oriented sphere. Their diameters tend to zero because they stay in the corresponding shrinking annuli. Transport them by f . Equation (11.1) preserves their metric lengths, and continuity of f gives vanishing canonical spatial diameter. This obtains the asserted L_k without changing any field law.

Let $z, w \neq p_*$. Take a D -geodesic between them. If it passes through p_* , replace the portion between its first and last intersections with L_k by a portion of L_k ; both endpoints lie outside the annulus for large k . The replacement avoids p_* and adds at most the length of L_k . A closed path with nonzero winding can, for this purpose, be replaced by a separating simple subcircuit of its trace; equivalently one can use the first and last crossing of its separating trace and either connecting portion of the closed path. The internal distance is therefore at most $D(z, w)$ after letting $k \rightarrow \infty$. The reverse inequality is immediate. Finally the punctured sphere is D -dense by the topology of the LQG metric, and $(\widehat{\mathbb{C}}, D)$ is complete. This identifies its completion. $\square$

Lemma 11.5 (Bounded deterministic comparison scales). *The sequence $a_n B(n)$ is bounded. On every deterministic subsequence one may therefore pass to a further deterministic subsequence on which*

$$a_n B(n) \longrightarrow \alpha \in [0, \infty). \quad (11.6)$$

In a joint extraction on this subsequence, let λ be the deterministic positive multiplier of Theorem 10.21. In δ_n -units, local port following has upper factor $\alpha\lambda$. If $\alpha > 0$, the lower certificate comparison has that same factor.

Almost surely, about every spatial point there are shrinking strict primal separating circuits whose limiting δ_n -diameters tend to zero. Moreover, if $\alpha = 0$, then for every $\varepsilon > 0$ there are connected primal skeletons of δ_n -diameter tending to zero whose complementary spatial components have diameter at most ε for all sufficiently large n .

Proof. Choose a fixed strictly buffered annular test in an ordinary patch of the original oriented coordinate x_n , with vertices on both sides. The positive lower port comparison in $B(n)$ -units implies that, for some $b, \theta > 0$ and all sufficiently large n , every traversal of that annulus has cost at least $bB(n)$ on an event of probability at least θ . To obtain fixed b, θ , use the positive reference distance between its compact boundary sets and the fixed positive lower bound for λ ; then truncate to a positive-probability event. If the initial choice of patch depends on the location of the exploration mark, choose one member of the countable ordinary atlas with positive probability and restrict to its strict containment event.

Every path between the two sides contains an annular traversal. Hence the ambient graph diameter is at least $bB(n)$ on this event. If $a_n B(n)$ were unbounded, a deterministic subsequence tending to infinity would contradict the tightness of $\text{diam}(V_n, \delta_n)$ in Theorem 1.2. This proves boundedness and (11.6). Multiplying the upper and lower comparisons by $a_n B(n)$ gives the stated factors.

Away from the original exploration root, the simultaneous circuit conclusion is Lemma 10.7, multiplied by the bounded factors $a_n B(n)$. Transporting the circuits by $f_n \rightarrow f$ gives the asserted conclusion away from p_* . Its simultaneous quantifier is relevant here: a point chosen from a later metric relation is allowed. The countable strict port library and local internal Hölder bounds in that lemma provide the assertion on every compact set off p_* . At p_* use Lemma 11.4, cover each of its finitely long reference paths by finitely many ordinary patches, and apply port following and Corollaries 4.9 and 4.10. The resulting primal circuit has a strict central hole and lies in a shrinking neighborhood of p_* . Its limiting cost is at most $\alpha\lambda$ times the reference length, with arbitrarily small open losses. These lengths tend to zero.

For the last assertion, choose finitely many strict central holes covering the sphere, each bounded by one of the preceding reference circuits and with filled side of spatial diameter less than ε . Join the circuits by finitely many reference paths off p_* . Such paths have finite D -length by Lemma 11.4. Use transverse ports at all required intersections. Lemma 4.11 realizes this finite construction as a connected primal skeleton. Each complementary component lies on the filled side of one of the covering circuits, so has the stated diameter, allowing the fixed strict margins to be chosen in advance. The total cost in $B(n)$ -units is bounded along the extraction for this finite construction. Multiplication by $a_n B(n) \rightarrow 0$ makes its δ_n -diameter tend to zero. $\square$

11.3 Positive scale and continuity at all endpoints

Fix a joint coupling on which the preceding convergences hold, and retain the relation R of Lemma 11.2. The intrinsic limit Y is almost surely homeomorphic to a sphere. All remaining topological arguments are deterministic on this event.

Proposition 11.6 (Eliminating nontrivial fibers). *If $\alpha > 0$, then R is the graph of a homeomorphism $\pi : Y \rightarrow \widehat{\mathbb{C}}$, and*

$$d_Y(y, y') = \alpha \lambda D(\pi(y), \pi(y')) \quad (y, y' \in Y). \quad (11.7)$$

In particular,

$$\lim_{r \downarrow 0} \limsup_{n \rightarrow \infty} \sup_{\substack{u, v \in V_n \\ \rho(z_n(u), z_n(v)) \leq r}} \delta_n(u, v) = 0. \quad (11.8)$$

Proof. Write $c = \alpha \lambda > 0$. We first prove that every two related pairs satisfy

$$d_Y(y, y') \geq c D(z, z') \quad ((y, z), (y', z') \in R). \quad (11.9)$$

Choose representing vertices and graph shortest paths between them. Their δ_n -lengths are bounded. Parameterize them by this length and keep them constant after their endpoints, on a common bounded time interval. Project the edges by the surface homeomorphisms and the canonical coordinate.

For each $r > 0$, any projected path of spherical diameter at least r makes a fixed positive advance in one member of a finite ordinary annular cover. Such a cover can avoid a ball of radius much smaller than r about p_* : a path with that diameter has a subpath making a comparable advance outside the smaller ball. The compact boundary sets in these finitely many tests have positive D -separation. The lower certificate comparison therefore gives, for all sufficiently large n , a positive lower δ_n -cost for every such advance. If this uniform lower bound failed, a sequence of failing traversals would yield a lower-certificate violation in one of the finitely many tests. The maximum single-edge spatial diameter tends to zero and $a_n \rightarrow 0$, so the length-parameterized projections are equicontinuous. A subsequence converges uniformly to a continuous path P from z to z' .

Every subarc of P compactly off p_* pays at least c times the D -distance of its ends. Indeed a finite ordinary cover and subdivision reduce it to the local lower comparisons, with costs added over disjoint subpaths. For $t > 0$, remove the portion of P between its first entry into and last exit from the D -ball of radius t about p_* . The retained pieces are compactly off the root and their endpoint distances sum to at least $D(z, z') - 2t$. The same statement, with one retained piece, applies if an endpoint is p_* . If the ball is not visited there is no removal. Letting $t \downarrow 0$ proves (11.9).

Taking $y = y'$ in (11.9) shows that R assigns each y a unique spatial point. Its graph is compact, so the resulting surjection $\pi : Y \rightarrow \widehat{\mathbb{C}}$ is continuous. In fact (11.9) gives its continuity directly in the D -topology.

Suppose that a fiber over z contains more than one point. Take the shrinking primal circuits about z from Lemma 11.5. For each fixed circuit test, extract its j_n -image as a compact subset of Y . Choose the tests so that the diameters of these limiting subsets tend to zero. After a further subsequence of tests they converge to one point y_0 . Their spatial diameters tend to zero at z , so $\pi(y_0) = z$. Choose $y \in \pi^{-1}(z) \setminus \{y_0\}$ and choose $x \in Y$ with $\pi(x) \neq z$. Inequality (11.9) keeps x away from y_0 .

There is a path in $Y \setminus \{y_0\}$ from y to x with positive metric clearance from y_0 . Fix a sufficiently fine circuit test that its limiting Y -image lies within one fourth of this clearance from y_0 . Then take n large. Representatives of y lie in its strict central hole, and representatives of x lie on its exterior side. The last version of Lemma 11.3 joins those representatives while avoiding the entire circuit, contradicting its separation. Thus the fibers are singletons. A continuous bijection between compact metric spaces is a homeomorphism.

To obtain (11.8), a contrary sequence would have spatially approaching pairs at distances bounded away from zero. Extract both Y -coordinates. The relation then assigns the same spatial point to two distinct Y -points, contradicting the singleton-fiber conclusion. This argument is uniform over all vertices.

It remains to prove the upper bound in (11.7). For $z, z' \neq p_*$ and any $e > 0$, Lemma 11.4 supplies a rectifiable path compactly in the punctured sphere, from z to z' , of length at most $D(z, z') + e$. A finite cover by ordinary patches and the port upper comparison give discrete following paths with arbitrarily small spatial errors at their ends and limiting cost at most $c(D(z, z') + e)$. Equation (11.8) now connects their ends to any prescribed representing vertices at vanishing additional cost: first make the fixed port errors small, then take n large, and finally let those errors tend to zero. Let $e \downarrow 0$. Endpoints equal to p_* follow by the continuity of D and of π^{-1} . This proves the reverse of (11.9), and hence (11.7). $\square$

11.4 The zero-scale alternative

The topology of a sphere also describes what would happen if the port scale vanished in GH units. This description is stronger than a statement about typical vertices.

Proposition 11.7 (Concentration at one spatial point). *If $\alpha = 0$, there are unique points $y_0 \in Y$ and $z_0 \in \widehat{\mathbb{C}}$ such that*

$$R = (\{y_0\} \times \widehat{\mathbb{C}}) \cup (Y \times \{z_0\}). \quad (11.10)$$

Moreover z_0 is a measurable function of the limiting distance graph $\mathcal{K}(R)$ alone. It is the unique point z for which

$$(z, z, r) \in \mathcal{K}(R) \quad \text{for some } r > 0. \quad (11.11)$$

Proof. For each $k \geq 1$, take a connected skeleton from Lemma 11.5 with complementary spatial mesh at most 2^{-k} . Its δ_n -diameter tends to zero. Diagonally extract all these countably many skeletons, so that their j_n -images converge respectively to points $y_k \in Y$.

Fix k . For any $(y, z), (y', z') \in R$ with $y, y' \neq y_k$, Lemma 11.3 joins representing vertices in the complement of the skeleton. They therefore lie in the same spatial complementary component. Passing to the limit gives

$$\rho(z, z') \leq 2^{-k}. \quad (11.12)$$

This reasoning also applies to two representations of the same point $y \neq y_k$.

Pass to a subsequence of k with $y_k \rightarrow y_0$. Any two fixed points of $Y \setminus \{y_0\}$ differ from y_k for all sufficiently large k . Equation (11.12) shows that all their spatial representations agree with a single point z_0 . There is at least one such point of Y because Y is nontrivial. The surjectivity of R onto $\widehat{\mathbb{C}}$ implies $(y_0, z) \in R$ for every $z \neq z_0$; closure gives it also for $z = z_0$. Surjectivity onto Y and closure likewise give $(y, z_0) \in R$ for every $y \in Y$. This proves (11.10). Its exceptional row and column are unique because both factors contain more than one point.

For $z \neq z_0$, the only Y -point related to z is y_0 , so the only third coordinate above (z, z) in (11.4) is zero. At z_0 the entire space Y is related, and a pair realizing its positive diameter supplies (11.11). The criterion is measurable: for a closed spatial set C , its exceptional point lies in C if and only if, for some integer $m \geq 1$, the compact graph hits the closed set $\{(z, z, r) : z \in C, r \geq 1/m\}$. These are Borel conditions in the compact-set topology. Thus z_0 can be retained without reference to an exploration root or a choice of GH realization. $\square$

Lemma 11.8 (The concentration point avoids the root). *In the setting of Proposition 11.7, almost surely $z_0 \neq p_*$.*

Proof. Keep the original canonical coordinate for this argument. The finite pair $(\mu_n, K_n(a_n))$ is determined by the unrooted decorated map and the three normalization samples. Conditional on those data before forgetting incidences, the initial vertex of a uniform oriented root edge has law

$$\nu_n^{\text{root}} = \sum_{v \in V_n} \frac{\deg(v)}{2n} \delta_{z_n(v)}.$$

Lemma 2.2 couples this distribution to μ_n at spherical error at most the maximum flag diameter. Consequently, for bounded continuous g on $\widehat{\mathbb{C}}$ and bounded continuous F of the root-free pair,

$$\mathbb{E}[F(\mu_n, K_n(a_n))g(z_n(v_*))] - \mathbb{E}\left[F(\mu_n, K_n(a_n)) \int g d\mu_n\right] \rightarrow 0.$$

The difference is bounded by $\|F\|_\infty$ times the modulus of continuity of g at the maximum flag diameter, which tends to zero in probability and is bounded.

Pass to the joint limit and use bounded continuous tests followed by the monotone-class theorem. Conditional on $(\mu, \mathcal{K}(R))$, the point p_* has distribution μ . The point z_0 is measurable from $\mathcal{K}(R)$ and μ has no atoms by Corollary 8.10. Hence

$$\mathbb{P}[p_* = z_0] = \mathbb{E}[\mu(\{z_0\})] = 0.$$

$\square$

We now return to the original oriented peanosphere coordinate. Uniform convergence of f_n^{-1} gives the geometric relation

$$R^\circ = (\text{id}_Y \times f^{-1})(R) = \lim_n \{(j_n(v), x_n(v)) : v \in V_n\}. \quad (11.13)$$

In the zero-scale case it is the cross relation with spatial center $\zeta_0 = f^{-1}(z_0)$ and metric center y_0 . Lemma 11.8 implies $\zeta_0 \neq p_*^\circ = \infty$. This pullback is an identity of locations on each sample; the field remains h° with its original oriented law.

The local-kernel construction retains first-exit observations in this original coordinate. For nested spatial disks $U \Subset V \Subset W$, define

$$E_n^\circ(U, V) = \max_{\substack{u \in V_n \\ x_n(u) \in U}} \delta_n(u, \{v \in V_n : x_n(v) \notin V\}). \quad (11.14)$$

We use a maximum of zero when the inner set is empty. The outer set is nonempty for large n in all tests below. Distances in (11.14) equal the minimum cost up to the first exit from V . With full stars protected in W , the readout uses only that protected graph: the first exiting edge and its endpoint remain in W once the flag mesh is small. Lemma 5.10 retains these quantities, with strict inner/outer domain sandwiches, in $[0, \infty]$ for the present deterministic units a_n . We write $E^\circ(U, V)$ for such a retained limit. All applications use strict containment and strict cost inequalities, so the sandwiches may be fixed before extraction.

Proposition 11.9 (Excluding concentration by two escape tests). *The alternative $\alpha = 0$ is impossible.*

Proof. Suppose that the deterministic subsequence in (11.6) has $\alpha = 0$. The cross relation (11.10) then holds almost surely for the finite-sphere extraction. We will use an orientation-preserving reembedding by fresh area marks and produce two separated positive escape readouts in that reembedded sphere law. The cross relation is preserved by the reembedding, so it must hold in the same experiment.

Finding one local witness. The original oriented root is already at infinity, and $\zeta_0 \neq \infty$ by (11.13). Sample two fresh points s^1, s^2 independently from μ° , conditionally on the existing continuum input and output arrays. Use fresh sampling randomness: their conditional distribution depends only on the real surface. They are finite, distinct and different from ζ_0 almost surely. Form the orientation-preserving affine map

$$g(z) = \frac{z - s^1}{s^2 - s^1}.$$

This map is defined in the original oriented chart, without using f . Put $\hat{x}_n = g \circ x_n$, $\hat{\zeta}_0 = g(\zeta_0)$, $\hat{\mu} = g_*\mu^\circ$, and let $\hat{h}$ be the usual orientation-preserving coordinate transform of h° by g . The geometric relation $\hat{R} = (\text{id}_Y \times g)(R^\circ)$ is still a cross, now at $\hat{\zeta}_0$ and y_0 .

We check the law used for this reembedding. We call it the *oriented sphere law with fresh area marks*, distinguishing it from the original coordinate x_n and from the possibly reflected canonical coordinate. The real field $\hat{h}$ has the ordinary root-and-two-area-mark sphere law of Proposition 3.5. The map g depends only on the real surface and the two fresh sampling variables. Conditional on those variables, the original imaginary field has its unchanged independent whole-plane law. An affine coordinate change preserves that law with its angular convention: translation and dilation use the stated symmetries, and rotation adds a constant to the uniform phase modulo its angular period. Thus the transformed imaginary input retains that same law independently of the transformed real surface. The new marks are auxiliary continuum variables sampled independently of the output arrays given the original continuum input. Theorem 6.1 therefore retains the same intrinsic observation kernels in this reembedding. All field-density applications below are to $\hat{h}$ with this oriented law. No reflected field is used.

Let $\hat{E}_n$ and $\hat{E}$ denote (11.14) and its retained limit with $\hat{x}_n$ in place of x_n . These still use the same graph and the same deterministic distance units a_n .

Choose disks from a countable ordinary atlas such that

$$\hat{\zeta}_0 \in U \Subset V \Subset W, \quad \hat{\mu}(\overline{W}) < \frac{1}{3}, \quad (11.15)$$

and the closed buffer $\overline{W}$ avoids the marks. Such choices exist by diffuseness at $\hat{\zeta}_0$ and the preceding mark avoidance. Since Y is not a point, its maximal distance from y_0 is positive. Countable

selection therefore gives fixed disks and a fixed $e > 0$ for which (11.15) holds and some $y \in Y$ has $d_Y(y, y_0) > 4e$, on an event of positive probability.

Represent this point y by vertices u_n . Their spatial locations in $\hat{x}_n$ tend to $\hat{\zeta}_0$, so they are eventually in U . Uniformly over vertices whose spatial locations are outside V , their Y -images tend to y_0 . Otherwise a subsequence would produce a pair of $\hat{R}$ with spatial entry outside V and metric entry different from y_0 , contradicting its cross form. Equation (11.3) now shows that $\hat{E}_n(U, V) > 3e$ for large n on this event. Using slightly nested disks and the readout sandwiches gives

$$\mathbb{P}\left[\hat{E}(U, V) > 2e, \hat{\mu}(\overline{W}) < \frac{1}{3}\right] > 0. \quad (11.16)$$

Placing two witnesses in the same oriented sphere law. The fixed disks in this paragraph are in the oriented $\hat{x}_n$ -coordinate. The readout in (11.16) is a local observation in the sense of Lemma 5.10. First restrict to a positive-probability subevent on which all imaginary-order data needed by the test are determined in one finite enlarged imaginary buffer; Lemma 6.6 allows this restriction. The domination part of Lemma 6.10 transfers the positive event to an ordinary-field experiment with the same local observation kernel. No distance or clock normalization is changed.

Translate two copies of this experiment into disks $U_i \subseteq V_i \subseteq W_i$, $i = 1, 2$, whose closed real buffers and enlarged imaginary buffers are mutually disjoint and avoid the marks. Use independent restrictions and their observation kernels. Each copy has a positive probability of satisfying the translated version of (11.16); hence the intersection has positive probability. Its two real-buffer masses sum to less than $2/3$.

The positivity part of Lemma 6.10, together with the product assertion of Theorem 6.1, puts this event in the actual joint limiting law of the two finite-sphere readouts. Proposition 6.8 applies with translations, whose derivatives are one, so both readouts still use the original deterministic units a_n . We conclude that, in the oriented sphere law with fresh area marks,

$$\mathbb{P}[\hat{E}(U_1, V_1) > 2e, \hat{E}(U_2, V_2) > 2e] > 0. \quad (11.17)$$

This conclusion uses the joint kernel theorem for the two readouts. It does not follow from their individual marginals.

The cheap exit contradiction. On every sample satisfying the cross relation for $\hat{R}$, at least one of the disjoint closed buffers $\overline{W}_i$ misses $\hat{\zeta}_0$. Fix such an index i . Uniformly over vertices in U_i , their Y -images tend to y_0 . Choose any fixed spatial point of $W_i \setminus \overline{V}_i$ and vertices tending to it; their Y -images also tend to y_0 . Therefore the maximum distance from any inner vertex to this exterior target tends to zero, by (11.3). A shortest graph path to the target has a first exit from V_i and its initial segment costs no more than the whole path. Thus $\hat{E}_n(U_i, V_i) \rightarrow 0$, including all fixed readout sandwiches, and $\hat{E}(U_i, V_i) = 0$. This contradicts (11.17). Both the two positive readouts and the cross relation were assertions in the oriented sphere law with fresh area marks. Since the affine reembedding preserves the cross relation sample by sample, the contradiction also excludes $\alpha = 0$ for the canonical relation R . $\square$

11.5 Deterministic normalization and the full sequence

The final identification first fixes the numerical scale and then removes the common orientation choice. The latter uses the following elementary observation, which also applies to the complete tuple in Section 12.

Lemma 11.10 (One common orientation ambiguity). *Let E be a measurable space with a measurable involution τ . Suppose random elements X, Y satisfy $X \in \{Y, \tau Y\}$ almost surely, and both $\text{Law}(X)$ and $\text{Law}(Y)$ are invariant under τ . Then $\text{Law}(X) = \text{Law}(Y)$. The choice between Y and τY may depend on the entire coupled random object.*

Proof. For any bounded measurable $F : E \rightarrow \mathbb{R}$, the unordered pairs $\{X, \tau X\}$ and $\{Y, \tau Y\}$ coincide. Hence

$$F(X) + F(\tau X) = F(Y) + F(\tau Y)$$

almost surely. Taking expectations and using both invariances gives $2\mathbb{E}F(X) = 2\mathbb{E}F(Y)$. $\square$

Lemma 11.11 (Identifying the scalar). *If X is a finite, strictly positive random variable and $c > 0$ is deterministic, then cX and X have the same law only if $c = 1$.*

Proof. If $c > 1$, equality of laws implies, for every finite $t > 0$,

$$\mathbb{P}[X \leq t] = \mathbb{P}[X \leq t/c^k] \longrightarrow \mathbb{P}[X = 0] = 0.$$

This contradicts the almost sure finiteness of X . If $c < 1$, apply the same argument to $1/c$. $\square$

Proof of Theorem 11.1. Start with an arbitrary deterministic subsequence of positive integers. The preceding sections and Theorem 1.2 give a joint further extraction of the reference data, local readouts and intrinsic spaces. Their marginals are tight; adjoining the compactified local arrays preserves tightness. The full distance graphs are tight as well: their first two coordinates lie in a fixed compact sphere, and their third coordinates are bounded by the tight intrinsic diameters. Take a Skorohod coupling for these retained data, and then apply Lemma 11.2 pathwise.

Lemma 11.5 allows the deterministic subsequence to be chosen so that $a_n B(n) \rightarrow \alpha$. Proposition 11.9 gives $\alpha > 0$, and Proposition 11.6 identifies the location relation as a homeomorphism satisfying

$$(Y, d_Y) \cong (\widehat{\mathbb{C}}, \alpha \lambda D).$$

Both α and λ are deterministic: the first is a limit of numerical normalizations, and the second is the law-wide rigid comparison constant in Theorem 10.21. By (11.1), f is an isometry from $(\widehat{\mathbb{C}}, D^\circ)$ to $(\widehat{\mathbb{C}}, D)$ on each sample, regardless of its orientation or dependence on the data. Therefore the unmarked space $(\widehat{\mathbb{C}}, D)$ has the ordinary unit-area marginal of $(\widehat{\mathbb{C}}, D^\circ)$. This is the same marginal as (Y, d_Y) by Theorem 1.2. Their diameters are finite and strictly positive. Lemma 11.11 gives $\alpha \lambda = 1$; no assertion about the law of an adaptively reembedded field is used here.

Equation (11.4) is now exactly (11.2). In particular, it identifies every vertex pair in the Hausdorff limit, rather than only a countable or area-typical set of pairs. The area convergence is already joint in Corollary 8.10. The resulting canonical pair is

$$X = (\mu, \mathcal{G}_D) \in \{X^+, \tau X^+\}, \quad X^+ = (\mu^+, \mathcal{G}_{D^+}),$$

where reflection acts simultaneously on the measure and both spatial coordinates of the distance graph. We now remove this possible adaptive sign at the level of the pair's law.

The law of each discrete pair $(\mu_n, K_n(a_n))$ is reflection invariant: reflection is a weight-preserving bijection of the oriented finite map ensemble, preserves graph distances and flag areas, and conjugates the normalized coordinate because $0, 1, \infty$ are fixed by conjugation. Reflection is continuous in the weak-measure and compact-graph topologies, so $\text{Law}(X)$ is invariant as well. The law of X^+ is reflection invariant by the reflection symmetry of the ordinary quantum sphere, its area and metric, and its three independently sampled marks. Lemma 11.10, applied to this

pair, gives $\text{Law}(X) = \text{Law}(X^+)$ even though the choice between $f = T$ and $f = \tau \circ T$ may depend on every retained observation.

Every deterministic subsequence therefore has a further subsequence with the prescribed pair law. The subsequence criterion proves the asserted full-sequence convergence. The metric identification itself was made in the joint extraction, so it remains (11.2) with any additional retained observations; the common orientation sign is kept with those observations. The normalization $a_n \rightarrow 0$ is the one from the companion intrinsic theorem. $\square$

12 Joint identification and the full sequence

The preceding identifications were made on a common contour extraction. We now show that their joint law is the one in Theorem 1.1. The only remaining coordinate choice is a single possible reflection of the entire tuple; its sign need not be independent of that tuple.

Proof of Theorem 1.1. Keep the deterministic normalization a_n of Theorem 1.2, and start with an arbitrary deterministic subsequence of positive integers. By Theorems 3.3 and 3.4, pass to a further joint extraction of the contours and their reference curve-decorated sphere. Retain the compatible local arrays used in the conformal, loop and metric arguments. The compactified observation spaces permit this joint extraction, and their conditional-law identifications remain valid with the additional observations. Write h° , μ° and D° for the field, area and metric in the original oriented reference coordinate, and Γ° for its complete nested loop ensemble.

The homeomorphisms H_n of Theorem 4.7 are chosen before the three fresh finite area samples. The reference area convergence $(H_n)_* m_n \Rightarrow \mu^\circ$ is established in Theorem 8.9. By Theorem 8.8, the sample images converge jointly to three points with conditional law $(\mu^\circ)^{\otimes 3}$ given the reference surface and its exploration. Let T be the orientation-preserving Möbius map sending these three limiting samples to $0, 1, \infty$. Theorems 8.9 and 8.10 give, after further extraction,

$$f_n := \phi_n \circ H_n^{-1} \longrightarrow f \quad \text{uniformly}, \quad f \in \{T, \tau \circ T\}, \quad \tau(z) = \bar{z}.$$

Here $\tau(\infty) = \infty$. On this same coupling, $\mu_n := (\phi_n)_* m_n$ converges weakly to $f_* \mu^\circ$, and the maximum spherical diameter of a flag image tends to zero.

Theorems 9.11 and 9.12 give complete loop matchings on this extraction. Every loop of prescribed positive diameter on either side is matched, with uniform error tending to zero up to reparameterization. Thus Γ_n converges to $f\Gamma^\circ$, retaining cyclic order, multiplicity and all nesting. The map f is exactly the map used for the area limit.

The distance graphs are tight jointly with these data. Their first two coordinates lie in the compact sphere, and their heights are bounded by $a_n \text{diam}(V(M_n), d_n)$, which is tight by Theorem 1.2. Below any fixed height L , the nonempty compact subsets of $\widehat{\mathbb{C}}^2 \times [0, L]$ form a compact Hausdorff hyperspace. Retain a convergent further subsequence of these graphs and of the intrinsic spaces. Theorem 11.1 identifies its graph limit, sample by sample, as

$$\{(f(z), f(w), D^\circ(z, w)) : z, w \in \widehat{\mathbb{C}}\}.$$

This identification includes all vertex pairs, including pairs approaching the exploration root. Its scalar is one for the prescribed a_n : the completion argument obtains a deterministic scalar multiple of the reference metric and fixes that scalar using the intrinsic diameter law. It makes no change to the retained area or loop coupling.

Consider first the orientation-preserving target tuple

$$Y = \left(T_*\mu^\circ, \{(T(z), T(w), D^\circ(z, w)) : z, w \in \widehat{\mathbb{C}}\}, T\Gamma^\circ \right).$$

The first two entries are the area and intrinsic metric of the ordinary unit-area sphere in its three-area-mark coordinate. Conditional on the reference surface, Γ° has the fixed whole-plane nested CLE law, by [Theorem 9.12](#). The three marks use fresh sampling randomness, so for every bounded measurable loop functional G ,

$$\mathbb{E}[G(T\Gamma^\circ) \mid \text{reference surface and three marks}] = \int G(T\mathcal{L}) \, d\text{CLE}_\kappa(\mathcal{L}) = \int G(\mathcal{L}) \, d\text{CLE}_\kappa(\mathcal{L}).$$

The second equality is Möbius invariance. Taking conditional expectation given the normalized embedded marked surface proves the independence required by [Theorem 1.1](#). Thus Y has exactly the asserted target law.

Let X be the limiting discrete tuple on this coupling. The three identifications above give $X \in \{Y, \mathfrak{r}Y\}$, with reflection acting simultaneously on the measure, both spatial coordinates of the distance graph, and every loop. Both tuple laws are reflection invariant. For the discrete tuple, reflection is a weight-preserving bijection of the oriented finite-map ensemble. It preserves flag areas and all-edge graph distances and transports the independent area samples. Since $0, 1, \infty$ are fixed by conjugation, the normalized tuple is reflected as well. Reflection is continuous in all three topologies, so this invariance passes to X . The law of Y is invariant by reflection symmetry of the ordinary quantum sphere, its area and metric, and the independent unoriented CLE ensemble. [Theorem 11.10](#) now gives $\text{Law}(X) = \text{Law}(Y)$, even when the sign of f depends on all the retained data.

Every deterministic subsequence therefore has a further jointly convergent subsequence with the same specified law. The subsequence criterion proves (1.3) through all positive integers. On each such extraction the maximum flag diameter also tends to zero. If (1.4) failed in probability, a subsequence would have a fixed positive probability of exceeding a fixed positive threshold, contradicting its further extraction from [Theorem 8.10](#). This proves the mesh conclusion and completes the theorem. $\square$

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